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(54) ETHANE OR ETHANOL INTO 3-HYDROXYPROPIONATE USING AN ENGINEERED MICROORGANISM

(71) Applicant: **INDUSTRIAL MICROBES, INC.**, Alameda, CA (US)

(72) Inventors: **Elizabeth Jane CLARKE**, Alameda, CA (US); **Noah Charles HELMAN**, Alameda, CA (US); **Timothy Brian ROTH**, Alameda, CA (US); **Yuan ZHANG**, Alameda, CA (US)

(73) Assignee: **INDUSTRIAL MICROBES, INC.**, Alameda, CA (US)

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(57) **ABSTRACT**

Provided are synthetic organisms and methods for the conversion of ethane and related substrates into 3-hydroxypropionate and related products.

Specification includes a Sequence Listing.

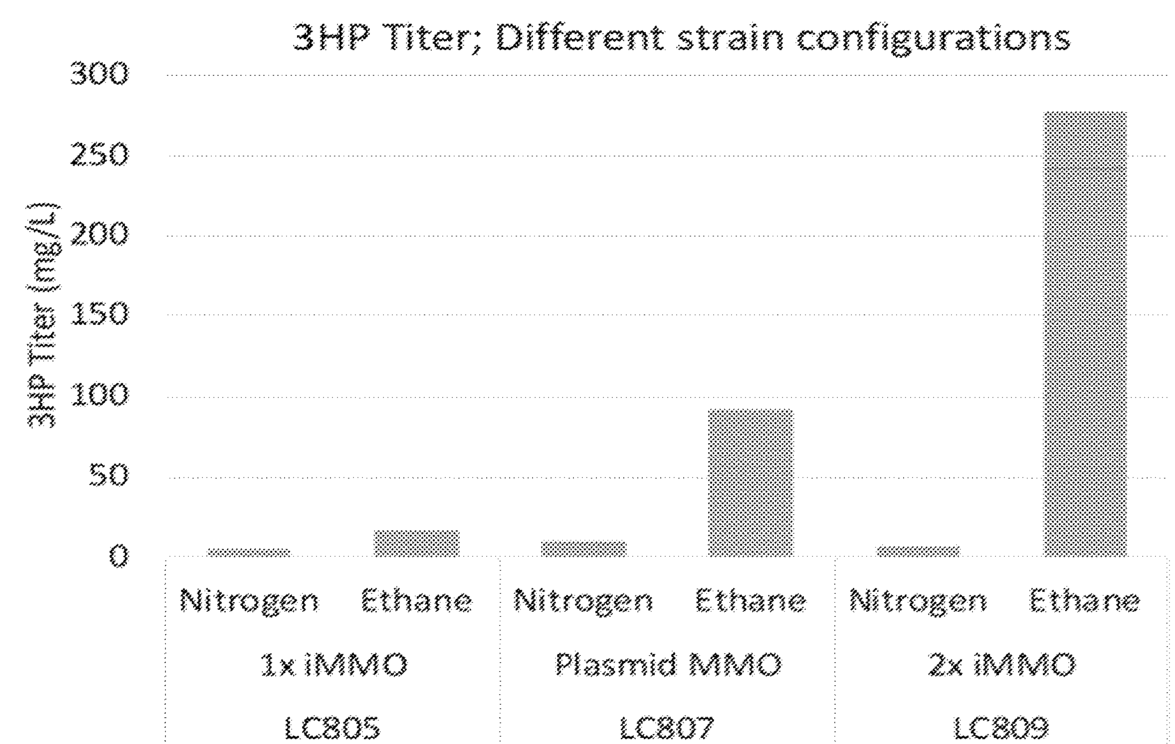

FIG. 1A

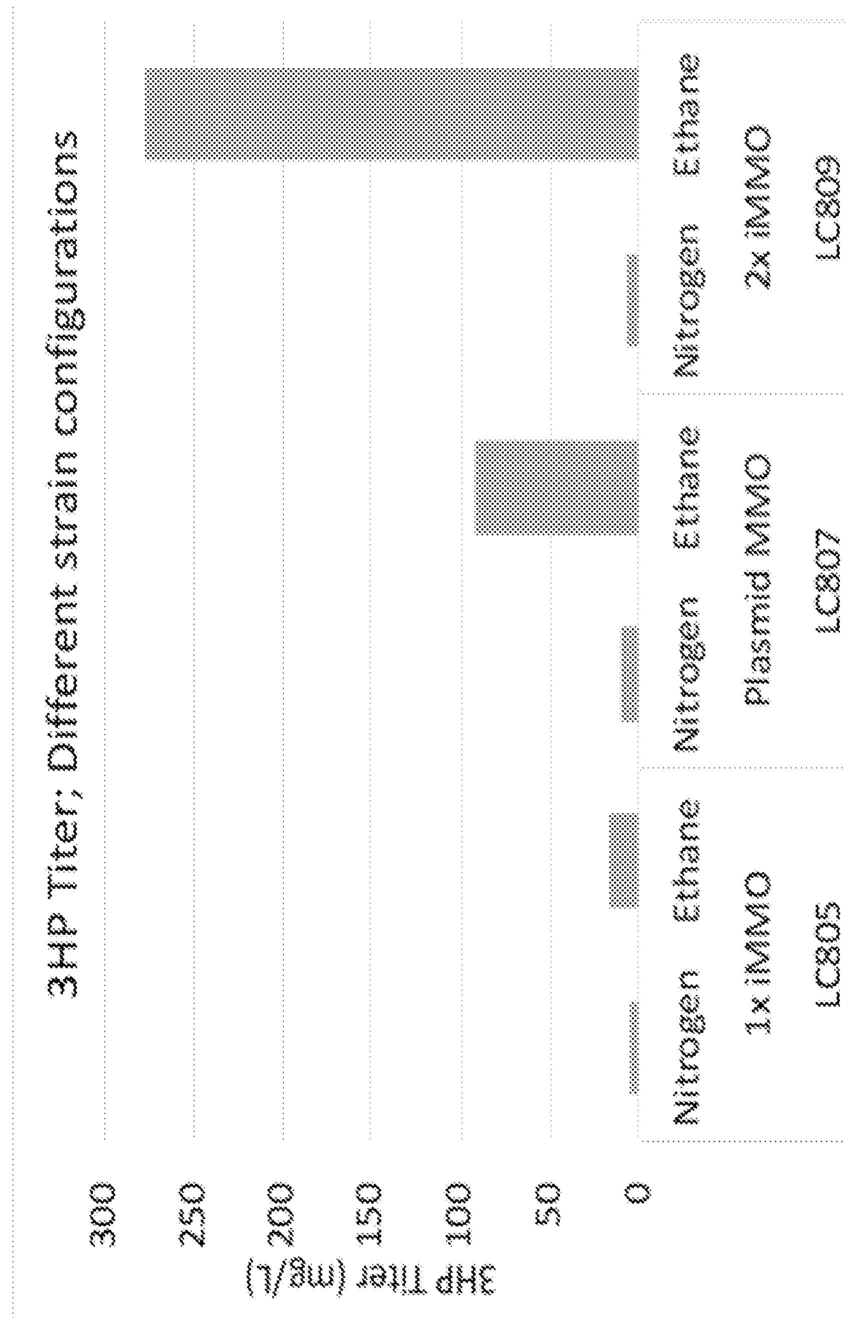
3HP Titer; Different strain configurations
3HP Titer (mg/L)
300
250
200
150
100
50
0
Nitrogen
Ethane
1x iMMO
LC805
Nitrogen
Ethane
Plasmid MMO
LC807
Nitrogen
Ethane
2x iMMO
LC809

FIG. 1B

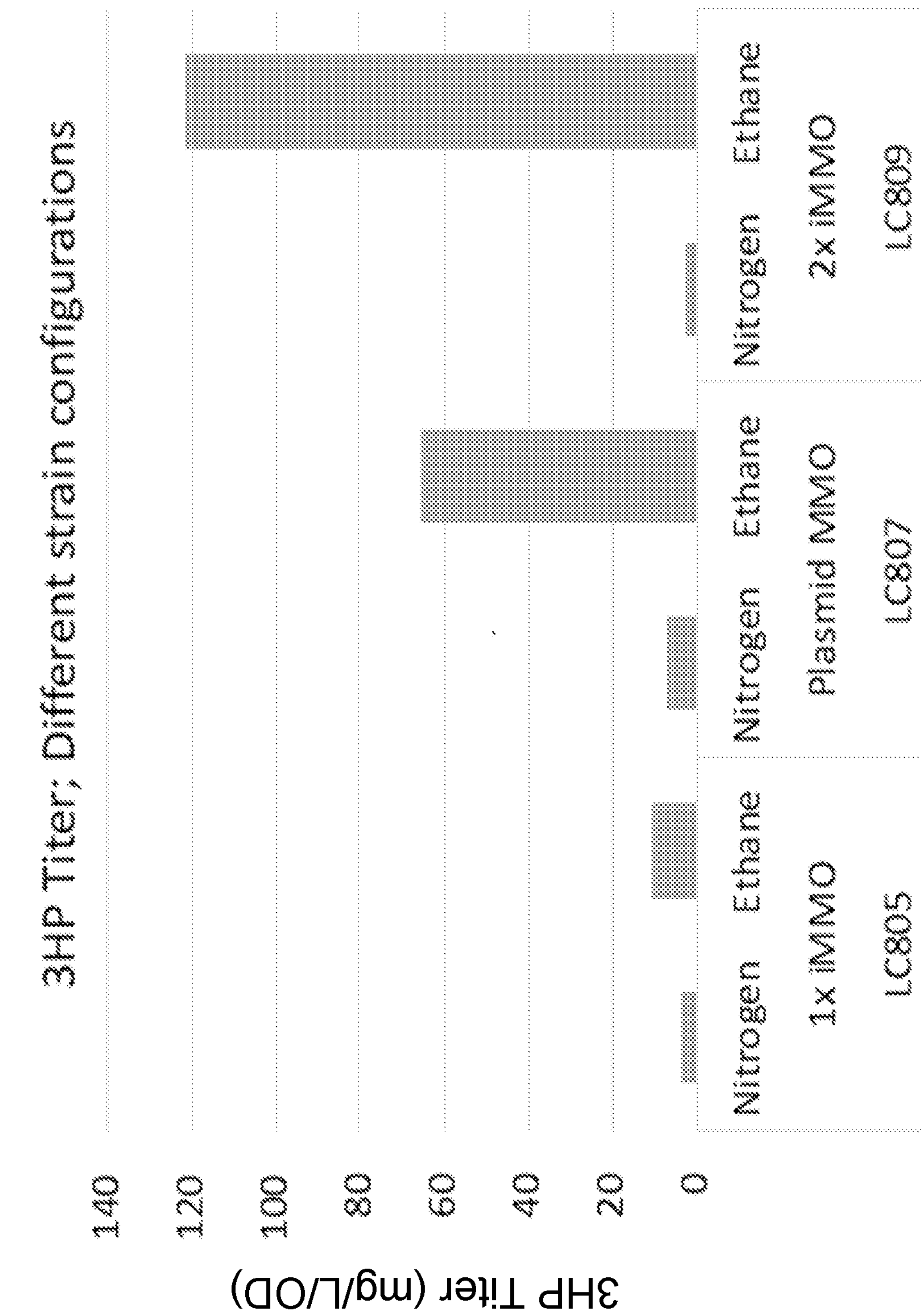
3HP Titer; Different strain configurations
3HP Titer (mg/L/OD)
0
20
40
60
80
100
120
140
Nitrogen
Ethane
1x iMMO
LC805
Nitrogen
Ethane
Plasmid MMO
LC807
Nitrogen
Ethane
2x iMMO
LC809

FIG. 1C

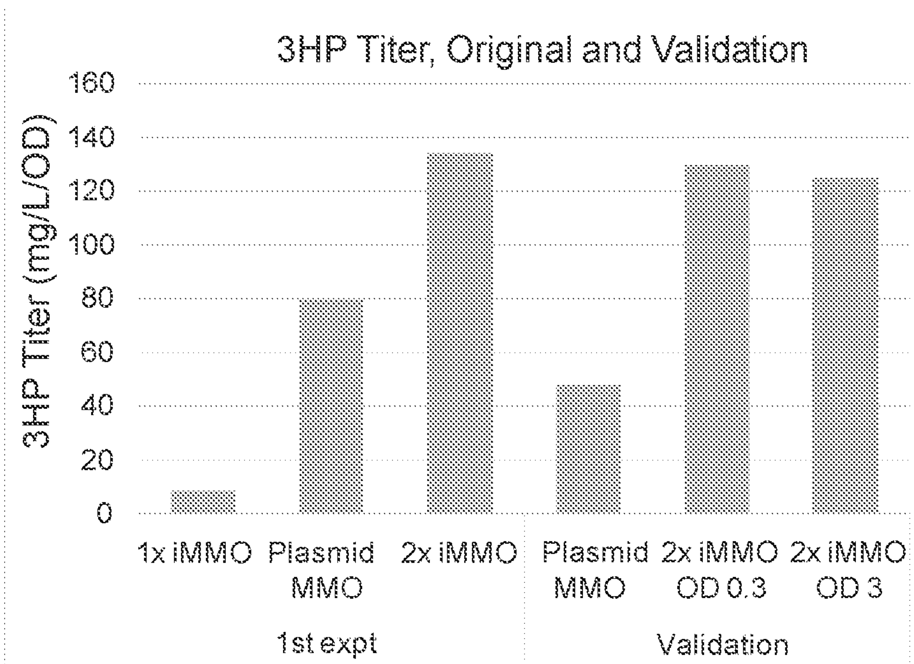
3HP Titer, Original and Validation
3HP Titer (mg/L/OD)
160
140
120
100
80
60
40
20
0
1x iMMO
Plasmid MMO
2x iMMO
Plasmid MMO
2x iMMO OD 0.3
2x iMMO OD 3
1st expt
Validation

FIG. 2

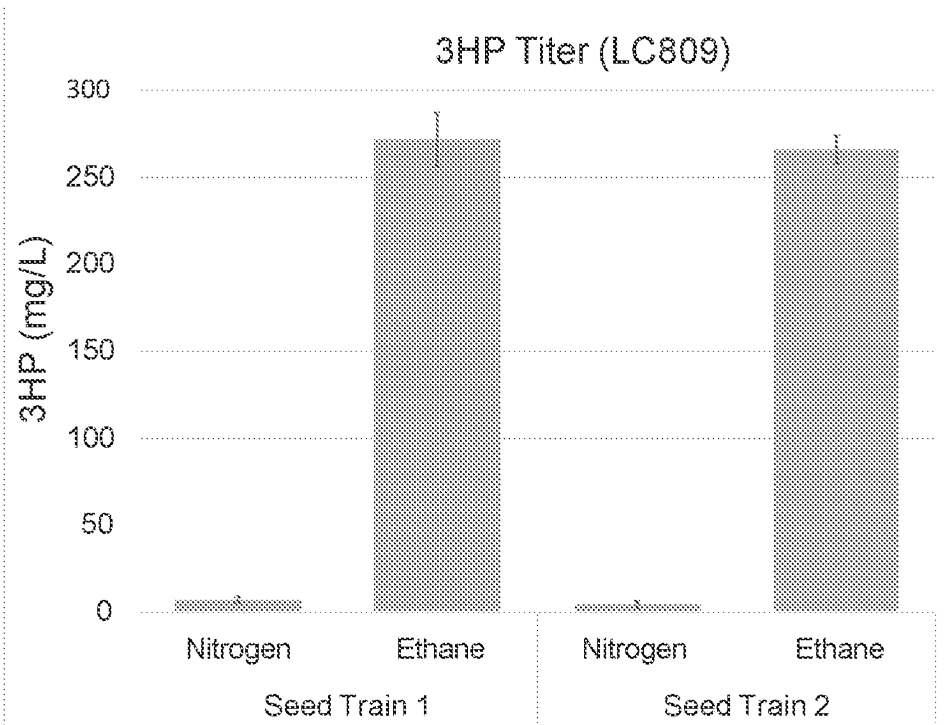
3HP Titer (LC809)
3HP (mg/L)
300
250
200
150
100
50
0
Nitrogen
Ethane
Nitrogen
Ethane
Seed Train 1
Seed Train 2

FIG. 3

	Plasmid Name	Purpose	Origin	AB resistance	Promoter	Gene(s)	2nd promoter	2nd gene(s)
This application IMI012a	pYZ40	protein folding chaperones	CloDF13	Spec	J23100 library hit	groES, groEL from E. Coli; groES, groEL-2 from M. capsulatus		
	pNH296	ethylotrophy (ethanol-->acetyl-CoA)	pMB1	Carg	Ptrc	adhA, adhH from C. glutamicum		
	pTRIM20	3HP production (malonyl-CoA --> 3HP) PLUS adhA (ethanol --> acetaldehyde)	p15a	Kan	J23119	mcrC***, mcrN from Chloroflexus aurantiacus	J23100	adhA (C. glutamicum)
	pBZ15	MMO (m)ethane--> (m)ethanol	p15a	Kan	pBAD	mmoXYBZDC from M. capsulatus		
	pBZ13	protein folding chaperones	CloDF13	Spec	J23100 library hit	groES, groEL from E. Coli; groES, groEL-2 from M. capsulatus		
	pLC99	ethylotrophy (ethanol-->acetyl-CoA)	pMB1	Carb	Ptrc	mdh[F147L] from B. stearothermophilus; acdH from Clostridium kluyveri		
	pLC130	methylotrophy (methanol --> formaldehyde --> RuMP cycle)	CloDF13	Spec	pBAD	HPS, PHI, mdh2 from B. methanolicus MGA3		
	pLC158	methylotrophy (methanol --> formaldehyde --> RuMP cycle)	CloDF13	Spec	pBAD	HPS, PHI from B. methanolicus MGA3, adhA from C. glutamicum		
	pNH241	ACC (acetyl-CoA --> malonyl-CoA)	p15a	Kan	pBAD	accD, accA, accB, accC from E. coli		
	pNH243	3HP production (malonyl-CoA-->3HP)	pMB1	Carb	Ptrc	mcrC***, mcrN from Chloroflexus aurantiacus		
	pNH265	MMO + chaperones	CloDF13*	Spec	pBAD	mmoXYBZDC from M. capsulatus	J23100 library hit	groES, groEL from E. coli; groES, groEL-2 from M. capsulatus

ETHANE OR ETHANOL INTO 3-HYDROXYPROPIONATE USING AN ENGINEERED MICROORGANISM

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a 35 U.S.C. 371 National Stage of International Application Number PCT/US2022/014288, Filed Jan. 28, 2022 which claims priority to and the benefit of U.S. provisional patent application No. 63/143,553, filed on Jan. 29, 2021, the entirety of which is hereby incorporated by reference.

GOVERNMENT INTEREST

[0002] This invention was made in part with government support under Grant No. DE-SC0018831 awarded by the Department of Energy. The government has certain rights in the invention.

SEQUENCE LISTING

[0003] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jan. 26, 2022, is named 1107367_000040 SL.txt and is 148,177 bytes in size.

[0004] Provided are synthetic organisms and methods for the conversion of ethane and related substrates into 3-hydroxypropionate and related products.

BACKGROUND

[0005] Microorganisms can be engineered to manipulate the use of raw materials or substrates required and/or the range of products formed. An engineered microorganism can often provide products at a low cost and with a low environmental impact.

[0006] Sugars, such as simple sugars, disaccharides, starches, carbohydrates, cellulosic sugars, and sugar alcohols, are often used as a raw material for engineered microorganisms. But, sugar can be expensive. Microorganisms engineered to use raw materials or substrates other than sugar would be useful.

[0007] For example, two-carbon compounds, such as ethane, could be used as a raw material for engineered microorganisms. Ethane is plentiful and is significantly less expensive than sugar. And, ethane is expected to remain inexpensive for decades to come. Microorganisms engineered to use ethane rather than sugar would be extremely useful.

[0008] As a product, it would be extremely advantageous to be able to make 3-hydroxypropionate (and 3-hydroxypropionic acid) with engineered microorganisms. 3-hydroxypropionate can be easily converted to a range of bulk chemicals, such as acrylic acid, ethyl acrylate, butyl acrylate, 1,3-propanediol (1,3-PD), 3-hydroxypropionaldehyde (3-HPA), and/or malonic acid. Such bulk chemicals can be used to make high-performance plastics, water-soluble paints, coatings, fibers, adhesives, chemicals for industrial water treatment, and super-absorbent polymers for diapers.

[0009] It would be useful to have synthetic organisms capable of using an inexpensive substrate, such as ethane, to make a valuable product, such as 3-hydroxypropionate.

BRIEF DESCRIPTION

[0010] The invention is drawn to synthetic microorganisms comprising exogenous polynucleotides that improve the production of a product from a substrate. The substrate may be ethane and the product may be 3-hydroxypropionate or a related product. The exogenous polynucleotide may encode an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway. Surprisingly, integration of the exogenous polynucleotide substantially improves production of a product in synthetic microorganisms.

[0011] A first aspect provides a synthetic microorganism comprising or consisting of at least one exogenous polynucleotide that encodes one or more polypeptides comprising or consisting of an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway. The at least one exogenous polynucleotide may be inserted or integrated into the genome of the microorganism.

[0012] The ethane monooxygenase comprises or consists of an ethane monooxygenase having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 1-16. The ethylotrophy pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 22-23. The malonyl-CoA-product pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 29-30.

[0013] The synthetic microorganism generates a product comprising or consisting of 3-hydroxypropionic acid. The synthetic microorganism may generate a product comprising or consisting of a polymer of 3-hydroxypropionic acid. The synthetic microorganism may generate one or more products comprising or consisting of fatty acids or fatty acid derivatives. The synthetic microorganism may generate a product comprising or consisting of polymer(s) of 3-hydroxy-fatty acids. Finally, the synthetic microorganism may generate a product comprising or consisting of malonic acid.

[0014] The synthetic microorganism may further comprise one or more exogenous polynucleotides which overexpress acetyl-CoA carboxylase. The synthetic microorganism may be derived from any microorganism. In some embodiments, the synthetic microorganism is derived from *Escherichia coli*.

[0015] A second aspect provides a method of producing a product, the method comprising or consisting of culturing a synthetic microorganism comprising or consisting of at least one exogenous polynucleotide that encodes one or more polypeptides comprising or consisting of an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway for a suitable period of time and under conditions sufficient to lead to improved production of the product. The at least one exogenous polynucleotide may be inserted or integrated into the genome of the microorganism.

[0016] The ethane monooxygenase comprises or consists of an ethane monooxygenase having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 1-16. The ethylotrophy pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 22-23. The malonyl-CoA-product pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 29-30.

[0017] The synthetic microorganism generates a product comprising or consisting of 3-hydroxypropionic acid. The synthetic microorganism may generate a product comprising or consisting of a polymer of 3-hydroxypropionic acid. The synthetic microorganism may generate one or more products comprising or consisting of fatty acids or fatty acid derivatives. The synthetic microorganism may generate a product comprising or consisting of polymer(s) of 3-hydroxy-fatty acids. Finally, the synthetic microorganism may generate a product comprising or consisting of malonic acid.

[0018] The synthetic microorganism may further comprise one or more exogenous polynucleotides which overexpress acetyl-CoA carboxylase. The synthetic microorganism may be derived from any derived from any microorganism. In some embodiments, the synthetic microorganism is derived from *Escherichia coli.*

[0019] The synthetic microorganism is cultured for a suitable period of time and under conditions sufficient to lead to improved production of the product.

[0020] A third aspect provides a nucleic acid suitable for integration into a synthetic microorganism, wherein the nucleic acid comprises or consists of at least one polynucleotide that encodes one or more polypeptides comprising or consisting of an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway. The at least one polynucleotide may be inserted or integrated into the genome of a microorganism.

[0021] The ethane monooxygenase comprises or consists of an ethane monooxygenase having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 1-16. The ethylotrophy pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 22-23. The malonyl-CoA-product pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 29-30.

[0022] Any suitable microorganism may be used with the nucleic acid. In some embodiments, the synthetic microorganism is derived from *Escherichia coli.*

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] "FIG. **1**A-C depicts results for experiments in which a culture was split into two vials after which the headspace was injected with either ethane or nitrogen. FIG. **1**A depicts the total amount of 3HP produced for cultures of LC805, LC807, and LC809 with starting OD of ~1.6, 1.4, and 2.3, respectively. FIG. **1**B depicts the same data normalized by starting OD. FIG. **1**C depicts a repeat experiment with LC 805 and LC809 at two starting ODs."

[0024] FIG. **2** depicts results for an experiment with LC805 and LC809 in which cultures were split into two vials after which the headspace was injected with either ethane or nitrogen. Additionally, duplicate cultures, one with 0.5% ethanol in the seed train and one without ethanol were used. The total amount of 3HP produced by cultures was determined.

[0025] FIG. **3** shows a plasmid comparison of the plasmids from the invention.

DETAILED DESCRIPTION

A. Definitions

[0026] The invention is drawn to synthetic microorganisms comprising exogenous polynucleotides that improve the production of a product from a substrate. The substrate may ethane and the product may be 3-hydroxypropionate or a related product. The exogenous polynucleotide may encode an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway. Surprisingly, integration of the exogenous polynucleotide substantially improves production of a product in synthetic microorganisms.

[0027] As used herein, "amino acid" shall mean those organic compounds containing amine ($—NH_2$) and carboxyl (—COOH) functional groups, along with a side chain (R group) specific to each amino acid. The key elements of an amino acid are carbon (C), hydrogen (H), oxygen (O), and nitrogen (N), although other elements are found in the side chains of certain amino acids.

[0028] As used herein, the terms "chaperone," "protein folding chaperone" and "folding chaperone" are intended to mean one or more proteins that improve the folding of polypeptide (amino acid) chains into 3-dimensional structures. Protein folding chaperones help their substrates, namely other proteins, to become properly folded and often more highly soluble. Since most proteins must be folded in a particular shape to be functional, the expression of protein folding chaperones can assist in the proper assembly of certain enzymes in a cell and thereby can result in an increase in the enzymatic activity of the substrate proteins.

[0029] As used herein, "conservative amino acid substitution(s)" or "conservative substitution(s)" refers to a substitution in which an amino acid residue is substituted by another amino acid residue having a side chain (R group) with similar chemical properties (e.g., charge or hydrophobicity). In general, a conservative amino acid substitution should not substantially change the functional properties of a protein. The following six groups each contain amino acids that are often, depending upon context, considered conservative substitutions for one another: 1) Serine (S), Threonine (T); 2) Aspartic Acid (D), Glutamic Acid (E); 3) Asparagine (N), Glutamine (Q); 4) Arginine (R), Lysine (K); 5) Isoleucine (I), Leucine (L), Alanine (A), Valine (V), and 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W).

[0030] As used herein, the term "culturing" is intended to mean the growth or maintenance of a microorganism under laboratory or industrial conditions. The culturing of microorganisms is a standard practice in the field of microbiology.

Microorganisms can be cultured using liquid or solid media as a source of nutrients for the microorganisms. In addition, some microorganisms can be cultured in defined media, in which the liquid or solid media are generated by preparation using purified chemical components. The composition of the culture media can be adjusted to suit the microorganism or the industrial purpose for the culture.

[0031] As used herein, the term "dehydrogenase" is intended to mean an enzyme belonging to the group of oxidoreductases that oxidizes a substrate by a reduction reaction that removes one or more hydrogen atoms from a substrate to an electron acceptor.

[0032] As used herein, the term "enzyme" or "enzymatically" shall refer to biological catalysts. Enzymes accelerate, or catalyze, chemical reactions. Like all catalysts, enzymes increase the rate of reaction by lowering the activation energy.

[0033] As used herein, "exogenous" is intended to mean something, such as a gene or polynucleotide, that originates outside of the organism of concern or study. An exogenous polynucleotide, for example, may be introduced into an organism by introduction into the organism of an encoding nucleic acid, such as, for example, by integration into a host chromosome or by introduction of a plasmid. When used in reference to a biosynthetic activity, the term refers to an activity that is introduced into a reference organism, such as a microorganism or synthetic culture as set forth in the invention. As an example, exogenous expression of an encoding nucleic acid can utilize either or both a heterologous or homologous encoding nucleic acid. A nucleic acid need not include all of its relevant or even complete coding regions on a single polymer and the invention provided herein contemplates having complete or partial coding sequences on different polymers.

[0034] As used herein, the term "exogenous polynucleotides" is intended to mean polynucleotides that are not derived from naturally occurring polynucleotides in a given organism. Exogenous polynucleotides may be derived from polynucleotides present in a different organism. The exogenous polynucleotides can be introduced into the organism by introduction of an encoding nucleic acid into the host genetic material such as by integration into a host chromosome or as non-chromosomal (episomal) genetic material such as a plasmid. Therefore, the term as it is used in reference to expression of an encoding nucleic acid refers to introduction of the encoding nucleic acid in an expressible form into the microbial organism. When used in reference to a biosynthetic activity, the term refers to an activity that is introduced into the host reference organism. The source can be, for example, a homologous or heterologous encoding nucleic acid that expresses the referenced activity following introduction into the host microbial organism.

[0035] The term "heterologous" refers to a molecule or activity derived from a source other than the referenced species. As set forth in the invention a nucleic acid need not include all of its relevant or even complete coding regions on a single polymer and the invention provided herein contemplates having complete or partial coding regions on different polymers.

[0036] As used herein, "integrated," as used in reference to a polynucleotide, refers to an exogenous polynucleotide that has been inserted in a chromosome of an organism, especially a polynucleotide comprising a coding sequence that can be expressed by the organism. In this respect, phrases such as "integrated into the genome of the synthetic microorganism" "integrated into the host genome," "integrated into a host chromosome," and the like have equivalent meanings.

[0037] As used herein, a "gene" is a sequence of DNA or RNA which codes for a molecule that has a function. The DNA is first copied into RNA. The RNA can be directly functional or be the intermediate template for a protein that performs a function.

[0038] As used herein, "modification," "genetic alteration," "genetically altered," "genetic engineering," "genetically engineered," "genetic modification," "genetically modified," "genetic regulation," or "genetically regulated" shall be used interchangeably and refer to direct or indirect manipulation of an organism's genome or genes to produce, for example, a desired effect, such as a desired phenotype. Genetic alteration includes a set of technologies that can be used to change genetic makeup, which ultimately could lead to the suppression or enhancement of phenotype or expression of a gene, as used herein. Genetic alteration shall also include the ability to reduce or prevent expression of a gene or genes. Genetic alteration techniques shall include, for example, but are not be limited to, molecular cloning, gene knockouts, gene targeting, mutation, homologous recombination, gene deletion, gene knockdown, gene silencing, gene addition, genome editing, gene attenuation, or any technique that may be used to suppress or alter the expression of a gene and a phenotype as known to one skilled in the art. Gene modification may be performed by any method known to one skilled in the art such as, for example, without limitation, CRISPR.

[0039] As used herein, "gene deletion" or "deletion" refers to a mutation or genetic modification in which a sequence of DNA is lost, deleted, or modified. A gene may be deleted to alter an organism's genome or to produce a desired effect or desired phenotype. Gene deletion may be used, for example, without limitation, as a method to suppress, alter, or enhance a particular phenotype.

[0040] As used herein, the term "gene knockdown" refers to a technique by which expression of one or more genes are reduced. Reduction can occur by any method known to one skilled in the art such as genetic modification, CRISPR interference, or by treatment with a reagent such as a short DNA or RNA oligonucleotide that has a sequence complimentary to either a gene or an mRNA transcript.

[0041] As used herein, the term "gene knockout" refers to a procedure whereby a gene is made inoperative.

[0042] As used herein, "gene silencing," "silencing," or "silenced" refers to the regulation of a gene, in particular, without limitation, the down regulation of a gene. Specifically, the term refers to the ability to reduce or prevent the expression of a certain gene. Gene silencing can occur at any cellular process, such as, for example, without limitation, during transcription or translation. Any methods of gene silencing well known in the art may be used such as, for example, without limitation, RNA interference and the use of antisense oligonucleotides.

[0043] As used herein, the term "homology" or "homologous" refer to the degree of biological shared ancestry in the evolutionary history of life. Homology or homologous may also refer to sequence homology, the biological homology between protein or polynucleotide sequences with respect to shared ancestry as determined by the closeness of nucleotide or protein sequences. Homology among proteins or polynucleotides is typically inferred from their sequence similarity. Alignments of multiple sequences are used to indicate which regions of each sequence are homologous. The term "percent homology" often refers to "sequence similarity." The percentage of identical residues (percent identity) or the percentage of residues conserved with similar physiochemical properties (percent similarity), e.g. leucine and isoleucine, is usually used to quantify homology. Partial homology can occur where a segment of the compared sequences has a shared origin.

[0044] As used herein, the term "identity" refers to a quantitative measurement of the similarity between two sequences (DNA, amino acid, or otherwise).

[0045] As used herein, the term "improved production of a product from a substrate" is intended to mean a situation in which a synthetic microorganism or synthetic culture has been modified in some way, such as, for example, without limitation, through genetic modification or integration, so that, under a set of conditions and relative to the original strain, the modified strain produces a product from the substrate or produces a product from the substrate faster than the rate or amount, respectively, from an unmodified microorganism or synthetic culture. A direct comparison of two strains can be made by growing the microorganisms or synthetic cultures under identical conditions and measuring the amount of product produced by each.

[0046] As used herein, "ethane monooxygenase" means the class of enzymes and enzyme complexes capable of oxidizing a carbon-hydrogen bond of the ethane molecule to result in a molecule of ethanol. In some embodiments, the ethane monooxygenase comprises or consists of, for example, without limitation, a methane monooxygenase such as, for example, sMMO and/or pMMO, a butane monooxygenase, a propane monooxygenase, an ammonium monooxygenase, p450 (such as, for example, P450 BM3 and/or P450cam), and/or AlkB.

[0047] As used herein, "monooxygenase," "methane monooxygenase enzyme," "MMO" "MO," or "soluble diiron monooxygenase" are intended to mean the class of enzymes and enzyme complexes capable of oxidizing a carbon-hydrogen bond of the methane molecule or other short chain alkane to result methanol or another molecule. Naturally occurring methane-consuming microorganisms have evolved at least two classes of monooxygenase enzymes: soluble monooxygenases ("sMMO") or and particulate monooxygenase ("pMMO"). Any enzyme or enzyme complex of these categories, any mutated enzyme or complex, or any researcher-designed enzyme or enzyme complex that converts methane into methanol would be considered a methane monooxygenase enzyme. Many of these enzymes may also oxidize a wide range of substrates, such as ethane into ethanol, and thus act as an ethane monooxygenase.

[0048] As used herein, "ethylotrophy pathway" is intended to mean a metabolic pathway that utilizes reduced carbon substrates containing a single carbon-carbon bond, such as, for example, without limitation, ethane, ethanol, acetaldehyde, acetate, ethylene, and monoethylene glycol.

[0049] As used herein, "malonyl-CoA-product pathway" is intended to mean a pathway that uses malonyl-CoA as a reactant or intermediate molecule to produce a product of interest, such as, for example, without limitation, malonic acid, 3-hydroxypropionic acid, fatty acids, 3-hydroxyfatty acids, fatty alcohols, fatty esters, adipic acid, 2-butanone, 3-pentanone, or salts or derivatives thereof (see, for example, https://agilebiofoundry.org/wp-content/uploads/2020/09/ABF-Metabolic-Map-12-for-survey.pdf for products derived from malonyl-CoA as Beachhead Molecule 11, which is incorporated by reference herein).

[0050] As used herein, the terms "microbe", "microbial," "microbial organism" or "microorganism" are intended to mean any organism that exists as a microscopic cell that is included within the domains of archaea, bacteria, or eukarya. The term is intended to encompass prokaryotic or eukaryotic cells or organisms having a microscopic size and includes bacteria, archaea, and eubacteria of all species as well as eukaryotic microorganisms such as yeast and fungi. The term also includes cell cultures of any species that can be cultured for the production of a product.

[0051] As used herein, "naturally occurring" shall refer to microorganisms or cultures normally found in nature.

[0052] As used herein, an "operon" shall refer to a functioning unit of genomic DNA containing a cluster of genes under the control of a single promoter. The genes are transcribed together into an mRNA strand and either translated together in the cytoplasm or undergo trans-splicing to create monocistronic mRNAs that are translated separately, i.e. several strands of mRNA that each encode a single gene product. The result of this is that the genes contained in the operon are either expressed together or not at all. Several genes may be co-transcribed to define an operon.

[0053] The terms "polynucleotide," "oligonucleotide," "nucleotide sequence," and "nucleic acid sequence" are intended to mean one or more polymers of nucleic acids and include, but are not limited to, coding regions, which are transcribed and translated into a polypeptide such as a chaperone, appropriate regulatory or control sequences, e.g., translational start and stop codons, promoter sequences, ribosome binding sites, polyadenylation signals, transcription factor binding sites, termination sequences, and regulatory domains and enhancers, among others. A polynucleotide, as used herein, need not include all of its relevant or even complete coding regions on a single polymer and the invention provided herein contemplates having complete or partial coding region on different polymers.

[0054] As used herein, a "peptide" refers to short chains of amino acid monomers linked by peptide (amide) bonds. Covalent chemical bonds are formed when the carboxyl group of one amino acid reacts with the amino group of another.

[0055] As used herein, a "polypeptide" or "protein" is a long, continuous, and unbranched peptide chain. Peptides are normally distinguished from polypeptides and proteins on the basis of size, and as an arbitrary benchmark can be understood to contain approximately 50 or fewer amino acids. Proteins consist of one or more polypeptides arranged in a biologically functional way, often bound to ligands such as coenzymes and cofactors, or to another protein or other macromolecule, such as, for example, DNA or RNA.

[0056] Amino acids that have been incorporated into peptides are termed "residues" due to the release of either a hydrogen ion from the amine end or a hydroxyl ion from the carboxyl end, or both, as a water molecule is released during formation of each amide bond. All peptides except cyclic peptides have an N-terminal and C-terminal residue at the ends of the peptide.

[0057] As used herein, "product" shall refer to 3-hydroxypropionate and 3-hydroxypropionic acid and related molecules and derivatives. Related molecules include, for example, without limitation, acrylic acid, ethyl acrylate, butyl acrylate, other acrylic acid esters, 1,3-propanediol (1,3-PD), 3-hydroxypropionaldehyde (3-HPA), and malonic acid. Related products also include polymerized forms of 3-hydroxypropionate, polymerized forms of acrylic acid, and polymerized forms of acrylic acid derivatives. Related products further include substances that derived from malonyl-CoA.

[0058] As used herein, "promoter" shall refer to a region of DNA that initiates transcription of a particular gene. Promoters are located near the transcription start sites of genes, on the same strand and upstream on the DNA (towards the 5' region of the sense strand). Promoters can be about 30-1000 base pairs long.

[0059] As used herein, the term "substrate" shall refer to a chemical species being used in a chemical reaction. In some embodiments, the substrate is ethane or a related molecule.

[0060] As used herein, "sufficient period of time" shall refer to a time period required to grow microorganisms or a synthetic culture to produce a product, such as, for example, a product of interest. In that sense, a sufficient period of time can be the amount of time that enables the microorganisms, or enables the synthetic culture of interest, to produce the product. For example, without limitation, an industrial scale culture may require as little as 5 minutes to begin production of detectable amounts of a product. Some synthetic cultures may be active for weeks.

[0061] As used herein, the term "suitable conditions" is intended to mean any set of culturing parameters that provide the microorganism with an environment that enables the culture to consume the available nutrients. In so doing, the microbiological culture may grow and/or produce products, chemicals, or by-products. Culturing parameters may include, but not be limited to, such features as the temperature of the culture media, the dissolved oxygen concentration, the dissolved carbon dioxide concentration, the rate of stirring of the liquid media, the pressure in the vessel, etc.

[0062] As used herein, the term "synthetic" is intended to mean a culture or microorganism, for example, without limitation, that has been manipulated into a form not naturally occurring or normally found in nature. For example, a synthetic culture or microorganism shall include, without limitation, a culture or microorganism that has been manipulated to express a polypeptide that is not naturally expressed, or transformed to include a synthetic polynucleotide of interest that is not normally included.

[0063] As used herein, the term "synthetic culture" is intended to mean at least one microorganism, or group of microorganisms, that has been manipulated into a form not normally found in nature.

B. Synthetic Cultures

[0064] A first aspect provides a synthetic microorganism comprising or consisting of at least one exogenous polynucleotide that encodes one or more polypeptides comprising or consisting of an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway. The at least one exogenous polynucleotide may be inserted or integrated into the genome of the microorganism.

[0065] The ethane monooxygenase comprises or consists of an ethane monooxygenase having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 1-16. The ethylotrophy pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 22-23. The malonyl-CoA-product pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 29-30.

[0066] The synthetic microorganism generates a product comprising or consisting of 3-hydroxypropionic acid. The synthetic microorganism may generate a product comprising or consisting of a polymer of 3-hydroxypropionic acid. The synthetic microorganism may generate one or more products comprising or consisting of fatty acids or fatty acid derivatives. The synthetic microorganism may generate a product comprising or consisting of polymer(s) of 3-hydroxy-fatty acids. Finally, the synthetic microorganism may generate a product comprising or consisting of malonic acid.

[0067] The synthetic microorganism may further comprise one or more exogenous polynucleotides which overexpress acetyl-CoA carboxylase. The synthetic microorganism may be derived from any derived from any microorganism. In some embodiments, the synthetic microorganism is derived from *Escherichia coli.*

[0068] Seed cultures may initially be grown in a rich medium, such as LB. After expansion of the population, the culture is diluted with fresh rich medium containing inducers to turn on expression of ethylotrophy enzymes encoded by exogenous polynucleotides under the control of inducible promoters. This induction medium may also be supplemented with ethanol to induce expression of endogenous enzymes useful for ethylotrophy. The bacterial cells are then separated from the rich medium and resuspended in a minimal medium, for example BEM6, and grown under an ethane containing atmosphere.

[0069] In some embodiments, minimal medium is supplemented with ethanol instead of fermenting in the presence of ethane. In such embodiments, the microorganism may not have been modified with an ethane monooxygenase or, if the ethane monooxygenase is under the control of an inducible promoter, the inducer may have been omitted from the induction culture.

C. Microorganisms

[0070] Synthetic microorganisms may be derived from any archaea, bacteria, or eukarya known to one skilled in the art. In some embodiments, the synthetic microorganism is derived from at least one of *Escherichia coli, Bacillus subtilis, Bacillus methanolicus, Pseudomonas putida, Saccharomyces cerevisiae, Pichia pastoris, Pichia methanolica, Salmonella enterica, Corynebacterium glutamicum, Klebsiella oxytoca, Anaerobiospirillum succiniciproducens, Actinobacillus succinogenes, Mannheimia succiniciproducens, Rhizobium etli, Gluconobacter oxydans, Zymomonas mobilis, Lactococcus lactis, Lactobacillus plantarum, Streptomyces coelicolor, Clostridium acetobutylicum, Pseudomonas fluorescens, Schizosaccharomyces pombe, Kluyveromyces lactis, Kluyveromyces marxianus, Aspergillus terreus, Aspergillus niger, Yarrowia lipolytica, Hansenula polymor-*

pha, Issatchenkia orientalis, Candida sonorensis, Candida methanosorbosa, and *Candida utilis*. In some embodiments, the microorganism is derived from *Escherichia coli.*

[0071] In some embodiments, the synthetic organism comprises or consists of one or more microorganisms. In some embodiments, for example, without limitation, conversion of ethane into ethanol is catalyzed in one microorganism and conversion of ethanol into 3HP is catalyzed in a second, genetically distinct microorganism. In some embodiments, conversion of ethane into ethanol and conversion of ethanol into 3HP are both catalyzed in a single microorganism. In some embodiments, a single synthetic microorganism comprises or consists of an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway. In some embodiments, the single microorganism is *Escherichia coli.*

D. Genomic Integration of Exogenous Polynucleotides

[0072] In some embodiments, the at least one exogenous polynucleotide is inserted or integrated into the genome of the microorganism. One skilled in the art would have expected that the addition of three exogenous plasmids would show greater expression than two integrated copies. However, counterintuitively, greater production of an end product can be achieved in some instances with a small number of copies of a polypeptide-encoding polynucleotide integrated into the host genome than with the same polynucleotide in a plasmid with higher copy number. For example, as can be seen from the examples, two copies of ethane monooxygenase operons integrated into an *E. Coli* host genome led to production of more 3HP than plasmid-borne ethane monooxygenase operons as set forth in the examples.

E. Ethane Monooxygenase

[0073] In some embodiments, the at least one exogenous polynucleotide encodes one or more polypeptides comprising or consisting of an ethane monooxygenase. An ethane monooxygenase, as contemplated in the current invention, is an enzyme that can oxidize a two-carbon molecule such as, for example, without limitation, ethane. In some embodiments, the ethane monooxygenase comprises or consists of a methane monooxygenase such as, for example, sMMO and/or pMMO. In some embodiments, an ethane monooxygenase comprises or consists of a butane monooxygenase. In some embodiments, an ethane monooxygenase comprises or consists of a propane monooxygenase. In some embodiments, an ethane monooxygenase comprises or consists of an ammonium monooxygenase. In some embodiments, an ethane monooxygenase comprises or consists of a P450. In some embodiments, an ethane monooxygenase comprises or consists of AlkB.

[0074] In some embodiments, the ethane monooxygenase comprises or consists of one or more methane monooxygenase(s). The methane monooxygenases are enzymes complexes capable of oxidizing a carbon-hydrogen bond of the methane molecule and they can also often oxidize ethane to ethanol. Naturally occurring methane-consuming microorganisms have evolved at least two classes of methane monooxygenase enzymes: soluble monooxygenases (sMMO) and particulate monooxygenases (pMMO). Any enzyme or enzyme complex of these categories, any mutated enzyme or complex, or any researcher-designed enzyme or enzyme complex that converts methane into methanol would be considered a methane monooxygenase enzyme. Many of these enzymes are known to also oxidize a wide range of substrates, such as methane to methanol or ethane into ethanol, and thus, are relevant as embodiments of the invention.

[0075] In some embodiments, an ethane monooxygenase comprises or consists of an sMMO. The sMMO from *Methylococcus capsulatus* (Bath) is well-studied. The *Methylococcus capsulatus* (Bath) can hydroxylate a large number of substrates (See, Petroleum Biotechnology by Vazquez-Duhalt and Quintero-Romero in 2004, which is incorporated by reference in its entirety herein). The sMMO from *Methylococcus capsulatus* (Bath) is able to hydroxylate dozens of substrates into an even larger number of products, when assayed in vitro. In some embodiments, the ethane monooxygenase comprises or consists of the monooxygenase from *Methylococcus capsulatus* (Bath).

[0076] In some embodiments, the ethane monooxygenase comprises or consists of a methane monooxygenase from one or more *Methylomonas methanica, Methylocaldum* sp. 175, *Methyloferula stellata, Methylocystis* LW5, *Solimonas aquatica* (DSM 25927), *Methylovulum miyakonense, Rhodococcus ruber* IGEM 231, and/or *Conexibacter woesei.*

[0077] In some embodiments, the ethane monooxygenase comprises or consists of a monooxygenase as set forth in the following table:

Organism	Gene names	Accession number
Pseudomonas mendocina KR1	tmoABCDEF	AY552601.1
Methylocella silvestris BL2	Msil1651-1647	NC_011666.1
Mycobacterium NBB4	smoXYC1B1Z, groL (Mycch_5901 - Mycch_5897, Mycch_5390)	CP003054.1
Thauera butanivorans	bmoXYBZDC	AAM19732.1, AAM19731.1, AAM19730.1, AAM19729.1, AAM19728.1, AAM19727.1, ABU68845.2
Mycobacterium smegmatis mc2-155	mimABCD	CP000480.1
Gordonia TY-5	prmABCDG	AB112920.1
Pseudonocardia autotrophica	WP_037052656.1 to WP_037052662.1	NZ_ JNYD01000036.1
Amycolatopsis methanolica 239	AMETH_2368-2375	CP009110.1
Mycobacterium HXN-1500	CYP153A6 (ahpGHI)	AJ783967.1
Bacillus megaterium	P450-BM3	WP_034650526.1
Pseudomonas putida	P450cam	WP_032492633.1
Methylocella silvestris BL2	mmoXYBZDC (Msil1262-Msil1267)	NC_011666.1
Methylococcus capsulatus (Bath)	mmoXYBZDC_G	AF525283.1, M90050.3
Methylosinus trichosporium OB3b	mmoXYBZDC, groEL	X55394.3, EF685207.1
Methylococcus capsulatus (Bath)	pmoCAB	L40804.2
Methylosinus trichosporium OB3b	pmoCAB	U31650.2
Pseudomonas putida (OCT plasmid)	alkBFGHJKLNST	NG_035191.1
Rhodococcus corallinus B-276	amoABCD	D37875.1

[0078] The particulate methane monooxygenase (pMMO) may also be used to oxidize ethane to ethanol. This protein complex is composed of three subunits and resides in the inner membrane of the native organism. In some embodiments, the ethane monooxygenase comprises or consists of a pMMO.

[0079] In some embodiments, the ethane monooxygenase comprises or consists of a butane monooxygenase. The butane monooxygenase of Thauera butanivorans is most active on butane, and maintains some activity against shorter alkanes. In some embodiments, the ethane monooxygenase comprises or consists of the butane monooxygenase of Thauera butanivorans (See, Sluis et al, Molecular analysis of the soluble butane monooxygenase from *Pseudomonas butanovora*, Microbiology, 148, 3617-3629 (2002), which is incorporated by reference in its entirety herein). The monooxygenase from *Pseudomonas putida* has also shown bacterial oxidation for propane and butane (See, Johnson, E. et al, Propane and n-Butane oxidation by *Pseudomonas putida* GPo1, Applied and environmental microbiology, 950-952 (2006), which is incorporated by reference in its entirety herein).

[0080] In some embodiments, the ethane monooxygenase comprises or consists of a propane monooxygenase. The propane monooxygenase of Pseudonocardia or *Mycobacterium* may be useful with shorter alkanes (See, for example, Furuya, et al., The mycobacterial iron monooxygenases require a specific chaperon-like protein for function expression in a heterologous host, FEBS Journal, 280, 817-826 (2013), which is incorporated by reference in its entirety herein). In some embodiments, the ethane monooxygenase comprises or consists of the propane monooxygenase from *Mycobacterium* NBB3 or from *Mycobacterium* NBB4 or the propane monooxygenase from Pseudonocardia TY-7 (See, for example, T. Kotani et al., Gene structure and regulation of alkane monooxygenase in propane utilizing *Mycobacterium* sp. TY-6 and *Pseudonocardia* sp. TY-7, Journal of bioscience and bioengineering, Vol. 102, No. 3, 184-192 (2006); and Martin, K. et al, SmoXYB1C1Z of *Mycobacterium* sp. Strain NBB4: a soluble methane monooxygenase (sMMO)-like enzyme, active on C2 and C4 alkanes and alkenes, Applied and Environmental Microbiology, p. 5801-5806, Vol. 80, No. 18 (2014), each of which is incorporated by reference in its entirety herein).

[0081] In some embodiments, the ethane monooxygenase comprises or consists of the propane monooxygenase from *Gordonia* (See, for example, T. Kotani et al, Propane monooxygenase and NAD+-dependent secondary alcohol dehydrogenase in propane metabolism by *Gordonia* sp. Strain TY-5, which is incorporated by reference herein in its entirety). In some embodiments, the ethane monooxygenase comprises or consists of a monooxygenase from any Gram-positive organism.

[0082] Ammonium monooxygenase (AMO) can also be useful according to the invention. Ammonia monooxygenase can oxidize ammonium and possibly ethane (See, Tavormina et al., A novel family of functional operons encoding methane/ammonia monooxygenase-related proteins in gammaproteobacterial methanotrophs, Environmental Microbiology reports (2011) 3(1), 91-100, which is incorporated by reference in its entirety herein). In some embodiments, the ethane monooxygenase comprises or consists of an ammonia monooxygenase.

[0083] In some embodiments, the ethane monooxygenase comprises or consists of P450. P450 may be engineered, according to the invention, to oxidize ethane (See F. Xu et al., *The Heme Monooxygenase Cytochrome P*450, 4029-4032, 2005; and P. Meinhold et al., *Direct Conversion of Ethane to Ethanol by Engineered Cytochrome,* 0017 1765-1768, 2005, each of which is incorporated by reference herein). In some embodiments, the ethane monooxygenase comprises or consists of P450 BM-3 (See, Arnold, F. et al, A Panel of Cytochrome P450 BM3 variants to produce drug metabolites and diversify lead compounds, Chemistry. 2009 Nov. 2; 15(43): 11723-11729, which is incorporated by reference in its entity herein). Cytochrome P450 BM3 is a Prokaryote Cytochrome P450 enzyme originally from *Bacillus megaterium*, catalyzes the hydroxylation of several long-chain fatty acids at the ω-1 through ω-3 positions. This bacterial enzyme belongs to CYP family CYP102, with the CYP Symbol CYP102A1. This CYP family constitutes a natural fusion between the CYP domain and an NADPH-dependent cytochrome P450 reductase. In some embodiments, the ethane monooxygenase comprises or consists of P450cam (See, for example, Sevrioukova et al., Structural Biology of Redox Partner Interactions in P450cam Monooxygenase: A Fresh Look at an Old System, Arch Biochem Biophys. 2011 Mar. 1; 507 (1): 66-74, which is incorporated by reference herein in its entirety).

[0084] In some embodiments, the ethane monooxygenase comprises or consists of a pMMO. The particulate methane monooxygenase (pMMO) may also oxidize ethane to ethanol. This protein complex is composed of three subunits and resides in the inner membrane of the native organism. To successfully express the pMMO in *E. coli*, correct N-terminal leader sequences must be properly fused to each of the three subunits. In some embodiments, the ethane monooxygenase comprises or consists of the pMMO from *Methylococcus capsulatus* (See, Elliot, S. et al, Regio- and Stereo selectivity of particulate methane monooxygenase from *Methylococcus capsulatus* (Bath), J. Am. Chem. Soc. 119, 9949-9955 (1997), which is incorporated by reference in its entirety herein).

[0085] AlkB may also be useful as an ethane monooxygenase of the invention. In some embodiments, the ethane monooxygenase comprises or consists of alkB (See, for example, Nie, Y, Diverse alkane hydroxylase genes in microorganisms and environments, *Scientific Reports* volume 4, Article number: 4968 (2014); and Koch, D. et al., In Vivo evolution of butane oxidation by terminal alkane hydroxylases AlkB and CYP153A6, applied and environmental biology, 337-344, (2009), each of which is incorporated by reference herein in its entirety). The integral-membrane alkane monooxygenase (AlkB)-related alkane hydroxylases are so-far the most commonly found alkane hydroxylases distributed in both Gram-negative and Gram-positive bacteria. In some embodiments, the ethane monooxygenase comprises or consists of alkB.

[0086] In some embodiments, the ethane monooxygenase comprises or consists of an ethane monooxygenase having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID NOs: 1-16.

F. Ethylotrophy Pathway

[0087] In some embodiments, the at least exogenous polynucleotide encodes one or more polypeptides comprising or consisting of an ethylotrophy pathway. An ethylotrophy pathway utilizes reduced carbon substrates containing a single carbon-carbon bond, such as, for example, without limitation, ethane, ethanol, acetaldehyde, acetate, ethylene, and monoethylene glycol. One skilled in the art would understand that any ethylotrophy pathway may be used.

[0088] In some embodiments, the ethylotrophy pathway comprises or consists one or more alcohol dehydrogenases. In some embodiments, the one or more polypeptides comprise an alcohol dehydrogenase, for example adhA (EC 1.1.1.1). adhA catalyzes the reaction: ethanol+NAD$^+$ ⟵⟶acetaldehyde+NADH. In some embodiments, adhA comprises or consists of an adhA from *Corynebacterium glutamicum*. In some embodiments, the ethylotrophy pathway comprises or consists of one or more aldehyde dehydrogenases, for example adhH (EC 1.2.1.10). adhH catalyzes the reaction: acetaldehyde+CoA+NAD$^+$←Θacetyl-CoA+NADH. In some embodiments, adhH comprises or consists of an adhH from *Corynebacterium glutamicum*.

[0089] In some embodiments, the ethylotrophy pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID NOs: 22-23.

G. Malonyl-CoA-Product Pathway

[0090] In some embodiments, the at least exogenous polynucleotide encodes one or more polypeptides comprising or consisting of a malonyl-CoA-product pathway. A malonyl-CoA-product pathway uses malonyl-CoA as a reactant or intermediate molecule to produce a product of interest, such as, for example, without limitation, malonic acid, 3-hydroxypropionic acid, fatty acids, 3-hydroxyfatty acids, fatty alcohols, fatty esters, adipic acid, 2-butanone, 3-pentanone, or salts or derivatives thereof (see, for example, https://agilebiofoundry.org/wp-content/uploads/2020/09/ABF-Metabolic-Map-12-for-survey.pdf for products derived from malonyl-CoA as Beachhead Molecule 11, which is incorporated by reference in its entirety herein). In some embodiments, the one or more polypeptides comprising or consisting of a malonyl-CoA-product pathway comprise or consist of one or more of mcrC*** and/or mcrN from *Chloroflexus aurantiacus* (See, Liu et al., "Functional balance between enzymes in malonyl-CoA pathway for 3-hydroxypropionate biosynthesis", Metabolic Engineering, Vol. 34, p. 104-110, (2016); and Lama et al., Production of a 3-hydroxypropionate acid from acetate using metabolically-engineered and glucose-grown *Escherichia coli*, Bioresource Technology, Vol. 320, Part A, 124362 (2021), each of which is incorporated by reference in herein in their entirety).

[0091] In some embodiments, the malonyl-CoA-product pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID NOs: 29-30.

H. Chaperones

[0092] Protein folding chaperones are proteins that improve the folding of polypeptide (amino acid) chains into 3-dimensional structures. Protein folding chaperones help their substrates, namely other proteins, to become properly folded and often more highly soluble. Since most proteins must be folded in a particular shape to be functional, the expression of protein folding chaperones can assist in the proper assembly of certain enzymes in a cell and thereby can result in an increase in the enzymatic activity of the substrate proteins.

[0093] In some embodiments, the at least one exogenous polynucleotide comprises or consists of one or more modifications. In some embodiments, the one or more modifications comprises or consists of polynucleotides encoding, and capable of expressing, one or more chaperone proteins. In some embodiment, the one or more chaperones comprises or consists of groEL and/or groES. In some embodiments, the groEL and/or groES are *E. coli* groEL and/or groES, *M. capsulatus* groEL and/or groES, or both. In some embodiments, the one or more chaperones comprise one or more polypeptides, each of the one or more polypeptides having an amino acid sequence, the amino acid sequence being more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of one of SEQ ID NOs: 36-39, respectively.

I. Method of Producing a Product

[0094] A second aspect provides a method of producing a product, the method comprising or consisting of culturing a synthetic microorganism comprising or consisting of at least one exogenous polynucleotide that encodes one or more polypeptides comprising or consisting of an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway for a suitable period of time and under conditions sufficient to lead to improved production of the product. The at least one exogenous polynucleotide may be inserted or integrated into the genome of the microorganism.

[0095] The ethane monooxygenase comprises or consists of an ethane monooxygenase having an amino acid sequence that is more about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 1-16. The ethylotrophy pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 22-23. The malonyl-CoA-product pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 29-30.

J. Product(s)

[0096] Synthetic organisms may make valuable products from substrates as set forth herein. The synthetic microorganism generates a product comprising or consisting of 3-hydroxypropionic acid. The synthetic microorganism may generate a product comprising or consisting of a polymer of 3-hydroxypropionic acid. The synthetic microorganism may generate one or more products comprising or consisting of fatty acids or fatty acid derivatives. The synthetic microorganism may generate a product comprising or consisting of polymer(s) of 3-hydroxy-fatty acids. Finally, the synthetic microorganism may generate a product comprising or consisting of malonic acid.

K. Expression of Nucleic Acid(s)

[0097] Expression of one or more exogenous nucleic acids in a microorganism or synthetic culture can be accomplished by introducing into a microorganism or culture a nucleic acid comprising a nucleotide sequence encoding the one or more polypeptides under the control of regulatory elements that permit expression in the microorganism or culture.

[0098] Nucleic acids encoding the one or more polypeptides can be introduced into a microorganism or culture by any method known to one of skill in the art without limitation (See, for example, Hinnen et al. (1978) Proc. Natl. Acad. Sci. USA 75:1292-3; Cregg et al. (1985) Mol. Cell. Biol. 5:3376-3385; Goeddel et al. eds, 1990, Methods in Enzymology, vol. 185, Academic Press, Inc., CA; Krieger, 1990, Gene Transfer and Expression—A Laboratory Manual, Stockton Press, NY; Sambrook et al., 1989, Molecular Cloning—A Laboratory Manual, Cold Spring Harbor Laboratory, NY; and Ausubel et al., eds., Current Edition, Current Protocols in Molecular Biology, Greene Publishing Associates and Wiley Interscience, NY, each of which is incorporated by reference in its entirety herein). Exemplary techniques include, but are not limited to, spheroplasting, electroporation, PEG 1000 mediated transformation, and lithium acetate- or lithium chloride-mediated transformation. In some embodiments, the nucleic acid is an extrachromosomal plasmid. In some embodiments, the nucleic acid is a chromosomal integration vector that can integrate the nucleotide sequence into the chromosome of the microorganism or culture.

L. Modification

[0099] Expression of genes and genomes may be modified. In some embodiments, expression of the one of more exogenous or endogenous nucleic acids is modified. For example, the copy number of an enzyme or one or more polypeptides in a microorganism or culture may be altered by modifying the transcription of the gene that encodes the enzyme or one or more polypeptides. This can be achieved, for example, by modifying the copy number of the nucleotide sequence encoding the enzyme or one or more polypeptides (e.g., by using a higher or lower copy number expression vector comprising the nucleotide sequence, or by introducing additional copies of the nucleotide sequence into the genome of the microorganism or culture, or by introducing additional nucleotide sequences into the genome of the microorganism or culture that express the same or similar polypeptide, or by genetically modifying or deleting or disrupting the nucleotide sequence in the genome of the microorganism or culture), by changing the order of coding sequences on a polycistronic mRNA of an operon, or by breaking up an operon into individual genes, each with its own control elements. The strength of the promoter, enhancer, or operator to which the nucleotide sequence is operably linked may also be manipulated or increased or decreased or different promoters, enhancers, or operators may be introduced.

[0100] Counterintuitively, greater production of an end product can be achieved in some instances with a small number of copies of a polypeptide-encoding polynucleotide integrated into the host genome than with the same polynucleotide in a plasmid with higher copy number. For example, as can be seen from the examples, two copies of ethane monooxygenase operons integrated into an *E. Coli* host genome led to production of more 3HP than plasmid-borne methane monooxygenase operons as set forth in the examples.

[0101] Alternatively, or in addition, the copy number of the one or more polypeptides may be altered by modifying the level of translation of an mRNA that encodes the one or more polypeptides. This can be achieved, for example, by modifying the stability of the mRNA, modifying the sequence of the ribosome binding site, modifying the distance or sequence between the ribosome binding site and the start codon of the enzyme coding sequence, modifying the entire intercistronic region located upstream of or adjacent to the 5' side of the start codon of the enzyme coding region, stabilizing the 3'-end of the mRNA transcript using hairpins and specialized sequences, modifying the codon usage of an enzyme, altering expression of rare codon tRNAs used in the biosynthesis of the enzyme, and/or increasing the stability of an enzyme, as, for example, via mutation of its coding sequence.

[0102] Expression of exogenous or endogenous nucleic acids may be modified or regulated by targeting particular genes. For example, without limitation, in some embodiments of the methods described herein, a microorganism, culture, or synthetic microorganism is contacted with one or more nucleases capable of cleaving, i.e., causing a break at a designated region within a selected site. In some embodiments, the break is a single-stranded break, that is, one but not both strands of the target site is cleaved. In some embodiments, the break is a double-stranded break. In some embodiments, a break-inducing agent is used. A break-inducing agent is any agent that recognizes and/or binds to a specific polynucleotide recognition sequence to produce a break at or near a recognition sequence. Examples of break-inducing agents include, but are not limited to, endonucleases, site-specific recombinases, transposases, topoisomerases, and zinc finger nucleases, and include modified derivatives, variants, and fragments thereof.

[0103] In some embodiments, a recognition sequence within a selected target site can be endogenous or exogenous to a microorganism, culture, or synthetic microorganism's genome. When the recognition site is an endogenous or exogenous sequence, it may be a recognition sequence recognized by a naturally occurring, or native break-inducing agent. Alternatively, an endogenous or exogenous recognition site could be recognized and/or bound by a modified or engineered break-inducing agent designed or selected to specifically recognize the endogenous or exogenous recognition sequence to produce a break. In some embodiments, the modified break-inducing agent is derived from a native, naturally occurring break-inducing agent. In other embodiments, the modified break-inducing agent is artificially created or synthesized. Methods for selecting such modified or engineered break-inducing agents are known in the art.

[0104] In some embodiments, the one or more nucleases is a CRISPR/Cas-derived RNA-guided endonuclease. CRISPR may be used to recognize, genetically modify, and/or silence genetic elements at the RNA or DNA level or to express heterologous or homologous genes. CRISPR may also be used to regulate endogenous or exogenous nucleic acids. Any CRISPR/Cas system known in the art finds use as a nuclease in the methods and compositions provided herein. CRISPR systems that find use in the methods and compositions provided herein also include those described in International Publication Numbers WO 2013/142578 A1, WO 2013/098244 A1 and Nucleic Acids Res (2017) 45 (1): 496-508, the contents of which are hereby incorporated in their entireties.

[0105] In some embodiments, the one or more nucleases is a TAL-effector DNA binding domain-nuclease fusion protein (TALEN). TAL effectors of plant pathogenic bacteria in the genus *Xanthomonas* play important roles in disease, or trigger defense, by binding host DNA and activating effector-specific host genes. (See, e.g., Gu et al. (2005) Nature 435:1122-5; Yang et al., (2006) Proc. Natl. Acad. Sci. USA 103:10503-8; Kay et al., (2007) Science 318:648-51; Sugio et al., (2007) Proc. Natl. Acad. Sci. USA 104:10720-5; Romer et al., (2007) Science 318:645-8; Boch et al., (2009) Science 326 (5959): 1509-12; and Moscou and Bogdanove, (2009) 326 (5959): 1501, each of which is incorporated by reference in their entirety). A TAL effector comprises a DNA binding domain that interacts with DNA in a sequence-specific manner through one or more tandem repeat domains. The repeated sequence typically comprises 34 amino acids, and the repeats are typically 91-100% homologous with each other. Polymorphism of the repeats is usually located at positions 12 and 13, and there appears to be a one-to-one correspondence between the identity of repeat variable-diresidues at positions 12 and 13 with the identity of the contiguous nucleotides in the TAL-effector's target sequence.

[0106] The TAL-effector DNA binding domain may be engineered to bind to a desired sequence and fused to a nuclease domain, e.g., from a type II restriction endonuclease, typically a nonspecific cleavage domain from a type II restriction endonuclease such as FokI (See, e.g., Kim et al. (1996) Proc. Natl. Acad. Sci. USA 93:1156-1160, which is incorporated by reference in its entirety herein). Other useful endonucleases may include, for example, HhaI, HindIII, Nod, BbvCI, EcoRI, BglI, and AlwI. Thus, in some embodiments, the TALEN comprises a TAL effector domain comprising a plurality of TAL effector repeat sequences that, in combination, bind to a specific nucleotide sequence in a target DNA sequence, such that the TALEN cleaves target DNA within or adjacent to the specific nucleotide sequence. TALENS useful for the methods provided herein include those described in WO10/079430 and U.S. Patent Application Publication No. 2011/0145940, which is incorporated by reference herein in its entirety.

[0107] In some embodiments, the one or more of the nucleases is a zinc-finger nuclease (ZFN). ZFNs are engineered break-inducing agents comprised of a zinc finger DNA binding domain and a break-inducing agent domain. Engineered ZFNs consist of two zinc finger arrays (ZFAs), each of which is fused to a single subunit of a non-specific endonuclease, such as the nuclease domain from the FokI enzyme, which becomes active upon dimerization.

[0108] Useful zinc-finger nucleases include those that are known and those that are engineered to have specificity for one or more sites. Zinc finger domains are amenable for designing polypeptides which specifically bind a selected polynucleotide recognition sequence. Thus, they are amenable to modifying or regulating expression by targeting particular genes.

[0109] In some embodiments, the activity of one or more genes native to the microorganism, culture, or synthetic microorganism is modified. The activity of one or more genes native to the microorganism or synthetic culture can be modified in a number of other ways, including, but not limited to, gene silencing or any other form of genetic modification, expressing a modified form of the polypeptides or one or more polypeptides that exhibits increased or decreased solubility in the microorganism, culture, or synthetic microorganism, expressing an altered form of the polypeptides or one or more polypeptides that lacks a domain through which the activity of a polypeptide is inhibited, expressing a modified form of the polypeptides that has a higher or lower k_{cat} or a lower or higher K_m for a substrate, or expressing an altered form of the one or more polypeptides or protein product of one or more genes native to the microorganism, culture, or synthetic microorganism that is more or less affected by feed-back or feed-forward regulation by another molecule in the pathway.

M. Nucleic Acid(s)

[0110] A third aspect provides a nucleic acid suitable for integration into a microorganism, wherein the nucleic acid comprises or consists of at least one polynucleotide that encodes one or more polypeptides comprising or consisting of an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway. The at least one polynucleotide may be inserted or integrated into the genome of a microorganism.

[0111] The ethane monooxygenase comprises or consists of an ethane monooxygenase having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 1-16. The ethylotrophy pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 22-23. The malonyl-CoA-product pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID Nos: 29-30.

[0112] Any suitable microorganism may be used with the nucleic acid. In some embodiments, the synthetic microorganism is derived from *Escherichia coli*.

[0113] In some embodiments, the one or more polypeptides or one or more genes native to the microorganism, culture, or synthetic microorganism are modified. It will be recognized by one skilled in the art that absolute identity to the one or more polypeptides or one or more genes native to the microorganism, culture, or synthetic microorganism is not strictly necessary. For example, changes in a particular gene or polynucleotide comprising a sequence encoding a polypeptide or the one or more polypeptides can be performed and screened for activity. Such modified or mutated polynucleotides and polypeptides can be screened for expression or function using methods known in the art.

[0114] Those of skill in the art will recognize that, due to the degenerate nature of the genetic code, a variety of polynucleotides differing in their nucleotide sequences can be used to encode one or more genes native to the microorganism, culture, or synthetic microorganism or one or more polypeptides of the disclosure. Due to the inherent degeneracy of the genetic code, other polynucleotides, which encode substantially the same or functionally equivalent polypeptides, can also be used. The disclosure includes polynucleotides of any sequence that encode the amino acid sequences of the polypeptides and proteins of the one or more polypeptides utilized in the methods of the disclosure.

[0115] In similar fashion, a polypeptide can typically tolerate one or more amino acid substitutions, deletions, and insertions in its amino acid sequence without loss or significant loss of a desired activity. The disclosure includes such one or more polypeptides with different amino acid sequences than the specific proteins described herein so long as the modified or variant polypeptides have an activity that is identical or similar to the referenced polypeptide. Accordingly, the amino acid sequences encoded by the polynucleotide sequences shown herein merely illustrate embodiments of the disclosure.

[0116] The disclosure also includes one or more polypeptides with different amino acid sequences than the specific proteins described herein if the modified or variant polypeptides have an activity that is desirable yet different from referenced polypeptide. In some embodiments, an enzyme may be altered by modifying the gene that encodes the enzyme so that the expressed protein is more or less active than the wild type version. As an example, any of the ethane monooxygenases, an ethylotrophy pathway polypeptides, and/or a malonyl-CoA-product pathway polypeptides may be more or less active according to substitutions.

[0117] As will be understood by those of skill in the art, it can be advantageous to modify a coding sequence to enhance expression in a particular host, such as, without limitation, *Escherichia coli*. The genetic code is redundant with 64 possible codons, but most organisms typically use a subset of these codons. Codons can be substituted, without any resultant change to the amino acid sequence of the corresponding protein, to increase or decrease the translation rate of the sequence, in a process sometimes called "codon optimization".

[0118] Optimized coding sequences can be prepared, for example, to increase the rate of translation or to produce recombinant RNA transcripts having desirable properties, such as a longer half-life, as compared with transcripts produced from a non-optimized sequence. Translation stop codons can also be modified to reflect host preference.

[0119] In addition, homologs of enzymes or the one or more polypeptides or the proteins encoded by the one or more genes native to the microorganism, culture, or synthetic microorganism useful for the compositions and methods provided herein are encompassed by the disclosure. To determine the percent identity of two amino acid sequences, or of two nucleic acid sequences, the sequences are aligned for optimal comparison purposes (e.g., gaps can be introduced in one or both of a first and a second amino acid or nucleic acid sequence for optimal alignment and non-homologous sequences can be disregarded for comparison purposes). The amino acid residues or nucleotides at corresponding amino acid positions or nucleotide positions are then compared. When a position in the first sequence is occupied by the same amino acid residue or nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps, and the length of each gap, which needs to be introduced for optimal alignment of the two sequences.

[0120] It is recognized that residue positions that are not identical often differ by conservative amino acid substitutions. In cases where two or more amino acid sequences differ from each other by conservative substitutions, the percent sequence identity or degree of homology may practically be adjusted upwards to correct for the conservative nature of the substitution. Means for making this adjustment are well known to those of skill in the art (See, e.g., Pearson W. R., 1994, Methods in Mol Biol 25:365-89, which is incorporated in its entirety herein).

[0121] Sequence homology and sequence identity for polypeptides is typically measured using sequence analysis software. A typical algorithm used to compare a molecular sequence to a database containing a large number of sequences from different organisms is the computer program BLAST. When searching a database containing sequences from a large number of different organisms, it is typical to compare amino acid sequences.

[0122] Furthermore, any of the one or more genes native to the microorganism, culture, or synthetic microorganism or genes encoding the enzymes or one or more polypeptides or genes native to the microorganism, culture, or synthetic microorganism (or any others mentioned herein (or any of the regulatory elements that control or modulate expression thereof)) may be optimized by genetic/protein engineering techniques, such as directed evolution or rational mutagenesis, which are known to those of ordinary skill in the art. Such action allows those of ordinary skill in the art to optimize the enzymes for expression and activity in yeast, bacteria, or any other suitable cell or organism.

[0123] For example, amino acid sequence variants of the one or more polypeptides can be prepared by mutations in the DNA. Methods for mutagenesis and nucleotide sequence alterations include, for example, Kunkel, (1985) Proc Natl Acad Sci USA 82:488-92; Kunkel, et al., (1987) Meth Enzymol 154:367-82; U.S. Pat. No. 4,873,192; Walker and Gaastra, eds. (1983) Techniques in Molecular Biology (MacMillan Publishing Company, New York) and the references cited therein. Guidance regarding amino acid substitutions not likely to affect biological activity of the protein is found, for example, in the model of Dayhoff, et al., (1978) Atlas of Protein Sequence and Structure (Natl Biomed Res Found, Washington, D.C.), each of which is incorporated by reference in their entirety.

[0124] Techniques known to those skilled in the art may be suitable to identify additional homologous (or otherwise analogous) genes and homologous (or otherwise analogous) enzymes. Generally, analogous genes and/or analogous enzymes can be identified by functional analysis and will have functional similarities. As an example, to identify homologous or analogous biosynthetic pathway genes, proteins, or enzymes, techniques may include, but are not limited to, cloning a gene by PCR using primers based on a published sequence of a gene/enzyme of interest or by degenerate PCR using degenerate primers designed to amplify a conserved region among a gene of interest.

[0125] Further, one skilled in the art can use techniques to identify homologous or analogous genes, proteins, or enzymes with functional homology or similarity. Techniques include examining a cell or cell culture for the catalytic activity of an enzyme through in vitro enzyme assays for the activity (e.g. as described herein or in Kiritani, K., Branched-Chain Amino Acids Methods Enzymology, 1970), then isolating the enzyme with the activity through purification, determining the protein sequence of the enzyme through techniques such as Edman degradation, design of PCR primers to the likely nucleic acid sequence, amplification of the DNA sequence through PCR, and cloning of the nucleic acid sequence. To identify homologous or similar genes and/or homologous or similar proteins, analogous genes and/or analogous proteins, techniques also include comparison of data concerning a candidate gene or enzyme with databases such as BRENDA, KEGG, or MetaCYC. The candidate gene or proteins may be identified within the above-mentioned databases in accordance with the teachings herein.

[0126] In some embodiments, the microorganism, culture, or synthetic microorganism expressing the one or more polypeptides has one or more genes native to the microorganism, culture, or synthetic microorganism that have been genetically modified, deleted, or whose expression has been reduced or eliminated. In some embodiments, the araBAD genes have been deleted.

[0127] Reduction or elimination of expression may occur through any method known to one skilled in the art and all ways of genetically modifying, deleting, and/or of reducing or eliminating expression of genes native to the microorganism, culture, or synthetic microorganism are provided herein. In particular, one skilled in the art will understand that any form of genetic alteration or genetic engineering or genetic modification, such as those set forth above related to expression, may be used as an alternative to deletion. In some embodiments, other forms of genetic modification that may be used as an alternative to deletion include, for example, without limitation, gene knockouts, mutation, gene targeting, homologous recombination, gene knockdown, gene silencing, gene addition, molecular cloning, gene attenuation, genome editing, CRISPR interference or any technique that may be used to suppress or alter or enhance a particular phenotype.

[0128] In some embodiments, the one or more genes native to the microorganism, culture, or synthetic microorganism can be altered in other ways, including, but not limited to, expressing a modified form of a polypeptide where the modified form of the polypeptide exhibits increased or decreased solubility in the microorganism or synthetic culture, expressing an altered form of a polypeptide that lacks a domain through which activity is inhibited, or expressing an altered form of a polypeptide that is more or less affected by feed-back or feed-forward regulation by another molecule in a pathway expressed in the microorganism, culture, or synthetic microorganism. In some embodiments, the strength of the promoter, enhancer, or operator to which the nucleotide sequence for the one or more genes native to the microorganism, culture, or synthetic microorganism is operably linked may also be manipulated, decreased, or increased or different promoters, enhancers, or operators may be introduced.

S. Sequences

TABLE S

SEQ ID NO	Molecule	Region and/or Designation	Sequence
1.	mmoX	*M. capsulatus* (Bath)	MALSTATKAATDALAANRAPTSGNAQEVHRWLQSFNW DFKNNRTKYATKYKMANETKEQFKLIAKEYARMEAVK DERQFGSLQDALTRLNAGVRVHPKWNETMKVVSNFLE VGEYNAIAATGMLWDSAQAAEQKNGYLAQVLDEIRHT HQCAYVNYYFAKNGQDPAGHNDARRTRTIGPLWKGMK RVFSDGFISGDAVECSLNLQLVGEACFTNPLIVAVTE WAAANGDEITPTVFLSIETDELRHMANGYQTVVSIAN DPASAKYLNTDLNNAFWTQQKYFTPVLGMLFEYGSKF KVEPWVKTWNRWVYEDWGGIWIGRLGKYGVESPRSLK DAKQDAYWAHHDLYLLAYALWPGGFFRLALPDQEEME WFEANYPGWYDHYGKIYEEWRARGCEDPSSGFIPLMW FIENNHPIYIDRVSQVPFCPSLAKGASTLRVHEYNGQ MHTFSDQWGERMWLAEPERYECQNIFEQYEGRELSEV IAELHGLRSDGKTLIAQPHVRGDKLWTLDDIKRLNCV FKNPVKAFN*
2.	mmoY	*M. capsulatus* (Bath)	MSMLGERRRGLTDPEMAAVILKALPEAPLDGNNKMGY FVTPRWKRLTEYEALTVYAQPNADWIAGGLDWGDWTQ KFHGGRPSWGNETTELRTVDWFKHRDPLRRWHAPYVK DKAEEWRYTDRFLQGYSADGQIRAMNPTWRDEFINRY WGAFLFNEYGLFNAHSQGAREALSDVTRVSLAFWGFD KIDIAQMIQLERGFLAKIVPGFDESTAVPKAEWTNGE VYKSARLAVEGLWQEVFDWNESAFSVHAVYDALFGOF VRREFFQRLAPRFGDNLTPFFINQAQTYFQIAKQGVQ DLYYNCLGDDPEFSDYNRTVMRNWTGKWLEPTIAALR DFMGLFAKLPAGTTDKEEITASLYRVVDDWIEDYASR IDFKADRDQIVKAVLAGLK*
3.	mmoB	*M. capsulatus* (Bath)	MSVNSNAYDAGIMGLKGKDFADQFFADENQVVHESDT VVLVLKKSDEINTFIEEILLTDYKKNVNPTVNVEDRA GYWWIKANGKIEVDCDEISELLGRQFNVYDFLVDVSS TIGRAYTLGNKFTITSELMGLDRKLEDSHA*
4.	mmoZ	*M. capsulatus* (Bath)	MAKLGIHSNDTRDAWVNKIAQLNTLEKAAEMLKQFRM DHTTPFRNSYELDNDYLWIEAKLEEKVAVLKAEAFNE VDFRHKTAFGEDAKSVLDGTVAKMNAAKDKWEAEKIH IGFRQAYKPPIMPVNYFLDGERQLGTRLMELRNLNYY DTPLEELRKQRGVRVVHLQSPH*

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
5.	mmoD	*M. capsulatus* (Bath)	MVESAFQPFSGDADEWFEEPRPQAGFFPSADWHLLKR DETYAAYAKDLDFMWRWVIVREERIVQEGCSISLESS IRAVTHVLNYFGMTEQRAPAEDRTGGVQH*
6.	mmoC	*M. capsulatus* (Bath)	MQRVHTITAVTEDGESLRFECRSDEDVITAALRQNIF LMSSCREGGCATCKALCSEGDYDLKGCSVQALPPEEE EEGLVLLCRTYPKTDLEIELPYTHCRISFGEVGSFEA EVVGLNWVSSNTVQFLLQKRPDECGNRGVKFEPGQFM DLTIPGTDVSRSYSPANLPNPEGRLEFLIRVLPEGRF SDYLRNDARVGQVLSVKGPLGVFGLKERGMAPRYFVA GGTGLAPVVSMVRQMQEWTAPNETRIYFGVNTEPELF YIDELKSLERSMRNLTVKACVWHPSGDWEGEQGSPID ALREDLESSDANPDIYLCGPPGMIDAACELVRSRGIP GEQVFFEKFLPSGAA*
7.	pmoC/ ammonia monooxygenase (pMMO)	*M. capsulatus* (Bath)	MAATTIGGAAAAEAPLLDKKWLTFALAIYTVFYLWVR WYEGVYGWSAGLDSFAPEFETYWMNFLYTEIVLEIVT ASILWGYLWKTRDRNLAALTPREELRRNFTHLVWLVA YAWAIYWGASYFTEQDGTWHQTIVRDTDFTPSHIIEF YLSYPIYIITGFAAFIYAKTRLPFFAKGISLPYLVLV VGPFMILPNVGLNEWGHTFWFMEELFVAPLHYGFVIF GWLALAVMGTLTQTFYSFAQGGLGQSLCEAVDEGLIA K
8.	P450 CYP153A6	*Mycobacterium* sp. HXN-1500	MTEMTVAASDATNAAYGMALEDIDVSNPVLFRDNTWH PYFKRLREEDPVHYCKSSMFGPYWSVTKYRDIMAVET NPKVFSSEAKSGGITIMDDNAAASLPMFIAMDPPKHD VQRKTVSPIVAPENLATMESVIRQRTADLLDGLPINE EFDWVHRVSIELTTKMLATLFDFPWDDRAKLTRWSDV TTALPGGGIIDSEEQRMAELMECATYFTELWNQRVNA EPKNDLISMMAHSESTRH MAPEEYLGNIVLLIVGGNDTTRNSMTGGVLALNEFPD EYRKLSANPALISSMVSEIIRWQTPLSHMRRTALEDI EFGGKHIRQGDKVVMWYVSGNRDPEAIDNPDTFIIDR AKPRQHLSFGFGIHRCVGNRLAELQLNILWEEILKRW PDPLQIQVLQEPTRVLSPFVKGYESLPVRINA
9.	P450cam	*Pseudomonas putida*	MTTETIQSNANLAPLPPHVPEHLVFDFDMYNPSNLSA GVQEAWAVLQESNVPDLVWTRONGGHWIATRGQLIRE AYEDYR HFSSECPFIPREAGEAYDFIPTSMDPPEQRQFRALAN QVVGMPVVDKLENRIQELACSLIESLRPQGQCNFTED YAEPFPIRIFMLLAGLPEEDIPHLKYLTDQMTRPDGS MTFAEAKEALYDYLIPIIEQRROKPGTDAISIVANGQ VNGRPITSDEAKRMCGLLLVGGLDTVVNFLSFSMEFL AKSPEHRQELIERPERIPAACEELLRRFSLVADGRIL TSDYEFHGVQLKKGDQILLPQMLSGLDERENACPMHV DFSROKVSHTTFGHGSHLCLGQHLARREIIVTLKEWL TRIPDFSIAPGAQIQHKSGIVSGVQALPLVWDPATTK AV
10.1	P450cam w/ mutations for ethane	*Pseudomonas putida*	MTTETIQSNANLAPLPPHVPEHLVFDFDMYNPSNLSA GVQEAWAVLQESNVPDLVWTRONGGHWIATRGQLIRE AYEDYRHFSSECPWIPREAGEAFDFIPLSMDPPEQRQ FRALANQVVGMPVVDKLENRIQELACSLIESLRPQGQ CNFTEDYAEPFPIRIFMLLAGLPEEDIPHLKYLTDQM MRPDGSMTFAEAKEALYDYLIPIIEQRROKPGTDAIS IVANGQVNGRPITSDEAKRMCGMLLLAGLDTVVNFLS FSMEFLAKSPEHRQELIERPERIPAACEELLRRFSMV ADGRILTSDYEFHGVQLKKGDQILLPQMLSGLDEREN ACPMHVDFSRQKVSHTTFGHGSHLCPGQHLARREIIV TLKEWLTRIPDFSIAPGAQIQHKSGIVSGVQALPLVW DPATTKAV*
11.	alkB	*Pseudomonas putida*	MLEKHRVLDSAPEYVDKKKYLWILSTLWPATPMIGIW LANETGWGIFYGLVLLVWYGALPLLDAMFGEDFNNPP EEVVPKLEKERYYRVLTYLTVPMHYAALIVSAWWVGT QPMSWLEIGALALSLGIVNGLALNTGHELGHKKETFD RWMAKIVLAVVGYGHFFIEHNKGHHRDVATPMDPATS RMGESIYKFSIREIPGAFIRAWGLEEQRLSRRGQSVW SFDNEILQPMIITVILYAVLLALFGPKMLVFLPIQMA FGWWQLTSANYIEHYGLLRQKMEDGRYEHQKPHHSWN

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			SNHIVSNLVLFHLQRHSDHHAHPTRSYQSLRDFPGLP ALPTGYPGAFLMAMIPQWFRSVMDPKVVDWAGGDLNK IQIDDSMRETYLKKFGTSSAGHSSSTSAVA S
12.	Propane monooxygenase	*Gordonia* TY-5 prmA	MSRQSLTKAHAKITELSWEPTFATPATRFGTDYTFEK APKKDPLKQIMRSYFPMEEEKDNRVYGAMDGAIRGNM FRQVQERWLEWOKLFLSIIPFPEISAARAMPMAIDAV PNPEIHNGLAVOMIDEVRHSTIQMNLKKLYMNNYIDP AGFDITEKAFANNYAGTIGRQFGEGFITGDAITAANI YLTVVAETAFTNTLFVAMPDEAAANGDYLLPTVFHSV QSDESRHISNGYSILLMALADERNRPLLERDLRYAWW NNHCVVDAAIGTFIEYGTKDRRKDRESYAEMWRRWIY DDYYRSYLLPLEKYGLTIPHDLVEEAWNRIVDKHYVH EVARFFATGWPVNYWRIDAMTDTDFEWFEEKYPGWYN KFGKWWENYNRLAYPGKNKPIAFEDVDYEYPHRCWTC MVPCLIREDMVTDKVDGQWRTYCSETCAWTDKVAFRP EYEGRPTPNMGRLTGFREWETLHHGKDLADIITDLGY VRDDGKTLIPQPHLDLDPKKMWTLDDVRGIPFGSPNV ALNEMSDDEREAHIAAYMANKNGAVTV
13.	Propane monooxygenase	*Pseudonocardia* sp. TY-7 prmA	MSRQSMSKAHKKITELSWEPTFATPAKRFGTDYTFDN APKKDPLKQILRSYFPMEEEKDSRVFGAMDGAIRGNM FRQVQERWMEWQKLFLSIIPFPEISAARAMPMAIDAV PNPEIHNGLAVQMIDEVRHSTIQMNLKRLYMNYYIDP AGFDMTEKAFANNYAGTIGRQFGEGFITGDAITAANI YLTVVAETAFTNTLFVAMPSEAAANGDYLLPTVFHSV QSDESRHISNGYSILLMALSDEDNROLLERDLRYAWW NNHRVVDAAIGTFIEYGTKDRRKDRESYAEMWRRWIY DDYYRAYLIPLEKYGLVIPHDLIEESWKQIWEKGYVH EVAQFFATGWLANYWRIDSMTDEDFEWFEYKYPGWYD KYGKWWENYNRLSKPNGHNPIVFEDVDYVYPARCWTC MSPCWSVRTLVTAEVDGQHRTYCHEVCRWTDVRGFPS DVPGRETPNMGRLVGKREWETLYHGWNWADVVSDMGF VRDDGKTMTPKPHLDLDPKKMWTLDHMRRCPPLQSPN VLFNEMSDAERAAYVADYNKOGPAGRPAPQS
14.	Propane monooxygenase	*Mycobacterium* NBB4	MTSTLPGKVSGHPSAPHASVSGQEVHSWLMDLGWDSD TIRGKYPTKYKFDPNAREQFKLVARDYGRMEGEKDDR QYGSLLDSLARLKAPTRVEPRWAEVMKLLAGALELGE YNAIAGSAVLADTTRSPELRNGYLMQVEDEVRHTTQT HYLAKYYAGQYYDPAGFTDMRKWRYINPLFPPTMQAF GENFCAGDPVFASLNLQLVAEACFTNPLIVAMTEWSA ANGDEITPTIFLSIQSDEMRHMANGYQTIVSVAHDAD NMQYLQTDLENAFWLQHRFATPIVGAGFEYGAVNKLE PWAKVWDRWVYEDWGGIWLGRLEKFGVKSPANLADAK RQAYWGHHYTYAVAYAVWPLLGFRMDPPNARDMEWFE NNYPGWHSEVGHMWDSWREMGLADPANHTLPGOLVSD AKVPIYFCRVCNFPVIIPTLTGALDDVRILELNGRKH PLCSSWCERMFLKEPERYQGENLWEKFDGWNIADVVH AAGAVRSDGKTLLAQPHLNSERMWTLDDLRACHAVIR DPLKTGGVWLETV
15.	Butane monooxygenase	*Thauera butanivorans*	MSANMAVKQALKANPVPSSVDPQEVHKWLQDFTWDFK GKTAKYPTKYEMDVNTREQFKLTAKEYARMESIKEER QYGTLLDGLDRLDAGNKVHPKWGEVMKLVSNFLETGE YGAIAGSALLWDTAQSPEQRNGYLAQVIDEIRHVNQT AYVNYYYGKHYYDPAGHTNMRQLRAINPLYPGVKRAF GEGFLAGDAVESSINLQLVGEACFTNPLIVSLTEWAA ANGDEITPTVFLSIETDELRHMANGYQTIVSIMNNPE TMKYLQTDLDNAFWTQHKFLTPFVGVALEYGSKYKVE PWAKSWNRWVYEDWAGIWLGRLQQFGVKTPKCLPDAK KDAVWAHHDLALLALALWPLTGIRMELPDSLAMEWFE ANYPGWYNHYGKIYEEWRAAGFEDPKSGFCGALWLME RGHGIFVDHASGLPFCPSLAKSSIKPRFTEYNGKRYA FAEPYGERQWLLEPERYEFQNFFEQFEGWELSDLVKA AGGVRSDGKTLIAQPHLRDTDMWTLDDLKRINLTIPD PMKILNWQPVAQ
16.	AMO	*Methylacidiphilum infernorum*	MAQATTQTVSISIPERSAFSWKNLWIALAIITAFELA VNVYERTFAIAKGLDYFTPQYQTYWMSILYTELILEP TTLIALCSWLWVTRDRAMENLAPAEELKRYWNLGLFV

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			VVYTVLLYWGASYYTEQDGTWHQTVIRDTDFTPSHII EFYQSYPIYIVAGVGSMVYAMTRLPQYAKAFSVPYAV LVGSPLMIFPNVGLNEFGHTRWFMEELFVAPLHWGFV MFGWGALAILGTWLQICPRVVELIKOVYYGKPATAPA VVLNDPEKAADPALCEI
17.			
18.			
19.			
20.			
21			
22	adhA	*C. glutamicum*	MTTAAPQEFTAAVVEKFGHDVTVKDIDLPKPGPHQAL VKVLTSGICHTDLHALEGDWPVKPEPPFVPGHEGVGE VVELGPGEHDVKVGDIVGNAWLWSACGTCEYCITGRE TQCNEAEYGGYTONGSFGQYMLVDTRYAARIPDGVDY LEAAPILCAGVTVYKALKVSETRPGQFMVISGVGGLG HIAVQYAAAMGMRVIAVDIADDKLELARKHGAEFTVN ARNEDSGEAVQKYTNGGAHGVLVTAVHEAAFGOALDM ARRAGTIVFNGLPPGEFPASVFNIVFKGLTIRGSLVG TRQDLAEALDFFARGLIKPTVSECSLDEVNGVLDRMR NGKIDGRVAIRF*
23.	adhH	*C. glutamicum*	MTVYANPGTEGSIVNYEKRYENYIGGKWVPPVEGQYL ENISPVTGEVFCEVARGTAADVELALDAAHAAADAWG KTSVAERALILHRIADRMEEHLEEIAVAETWENGKAV RETLAADIPLAIDHFRYFAGAIRAQEDRSSQIDHNTV AYHFNEPIGVVGQIIPWNFPILMATWKLAPALAAGNA IVMKPAEQTPASILYLINIIGDLIPEGVLNIVNGLGG EAGAALSGSNRIGKIAFTGSTEVGKLINRAASDKIIP VTLELGGKSPSIFFSDVLSQDDAFAEKAVEGFAMFAL NQGEVCTCPSRALVHESIADEFLELGVKRVQNIKLGN PLDTETMMGAQASQEQMDKISSYLKIGPEEGAQTLTG GKVNKVDGMENGYYIEPTVFRGTNDMRIFREEIFGPV LSVATFSDFDEAIRIANDTNYGLGAGVWSRDQNTIYR AGRAIQAGRVWVNQYHNYPAHSAFGGYKESGIGRENH LMMLNHYQQTKNLLVSYDPNPTGLF*
24.			
25.			
26			
27.			
28.			
29.	mcrC***	*C. aurantiacus*	MSATTGARSASVGWAESLIGLHLGKVALITGGSAGIG GQIGRLLALSGARVMLAARDRHKLEQMOAMIQSELAE VGYTDVEDRVHIAPGCDVSSEAQLADLVERTLSAFGT VDYLINNAGIAGVEEMVIDMPVEGWRHTLFANLISNY SLMRKLAPLMKKQGSGYILNVSSYFGGEKDAAIPYPN RADYAVSKAGORAMAEVFARFLGPEIQINAIAPGPVE GDRLRGTGERPGLFARRARLILENKRLNELHAALIAA ARTDERSMHELVELLLPNDVAALEONPAAPTALRELA RRFRSEGDPAASSSSALLNRSIAAKLLARLHNGGYVL PADIFANLPNPPDPFFTRAQIDREARKVRDGIMGMLY LQRMPTEFDVAMATVYYLADRVVSGETFHPSGGLRYE RTPTGGELFGLPSPERLAELVGSTVYLIGEHLTEHLN LLARAYLERYGARQVVMIVETETGAETMRRLLHDHVE AGRLMTIVAGDQIEAAIDQAITRYGRPGPVVCTPFRP LPTVPLVGRKDSDWSTVLSEAEFAELCEHQLTHHFRV ARWIALSDGARLALVTPETTATSTTEQFALANFIKTT LHAFTATIGVESERTAQRILINQVDLTRRARAEEPRD PHERQQELERFIEAVLLVTAPLPPEADTRYAGRIHRG RAITV*

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
30	mcrN	*C. aurantiacus*	MSGTGRLAGKIALITGGAGNIGSELTRRFLAEGATVI ISGRNRAKLTALAERMQAEAGVPAKRIDLEVMDGSDP VAVRAGIEAIVARHGQIDILVNNAGSAGAQRRLAEIP LTEAELGPGAEETLHASIANLLGMGWHLMRIAAPHMP VGSAVINVSTIFSRAEYYGRIPYVTPKAALNALSQLA ARELGARGIRVNTIFPGPIESDRIRTVFQRMDQLKGR PEGDTAHHFLNTMRLCRANDQGALERRFPSVGDVADA AVFLASAESAALSGETIEVTHGMELPACSETSLLART DLRTIDASGRTTLICAGDQIEEVMALTGMLRTCGSEV IIGFRSAAALAQFEQAVNESRRLAGADFTPPIALPLD PRDPATIDAVFDWGAGENTGGIHAAVILPATSHEPAP CVIEVDDERVLNFLADEITGTIVIASRLARYWQSQRL TPGARARGPRVIFLSNGADQNGNVYGRIQSAAIGQLI RVWRHEAELDYQRASAAGDHVLPPVWANQIVRFANRS LEGLEFACAWTAQLLHSQRHINEITLNIPANI*
31.			
32.			
33.			
34.			
35.			
36.	groES	*E. coli*	MNIRPLHDRVIVKRKEVETKSAGGIVLTGSAAAKSTR GEVLAVGNGRILENGEVKPLDVKVGDIVIFNDGYGVK SEKIDNEEVLIMSESDILAIVEA*
37.	groEL	*E. Coli*	MAAKDVKFGNDARVKMLRGVNVLADAVKVTLGPKGRN VVLDKSFGAPTITKDGVSVAREIELEDKFENMGAQMV KEVASKANDAAGDGTTTATVLAQAIITEGLKAVAAGM NPMDLKRGIDKAVTAAVEELKALSVPCSDSKAIAQVG TISANSDETVGKLIAEAMDKVGKEGVITVEDGTGLQD ELDVVEGMQFDRGYLSPYFINKPETGAVELESPFILL ADKKISNIREMLPVLEAVAKAGKPLLIIAEDVEGEAL ATLVVNTMRGIVKVAAVKAPGFGDRRKAMLQDIATLT GGTVISEEIGMELEKATLEDLGQAKRVVINKDTTTII DGVGEEAAIQGRVAQIRQQIEEATSDYDREKLQERVA KLAGGVAVIKVGAATEVEMKEKKARVEDALHATRAAV EEGVVAGGGVALIRVASKLADLRGQNEDQNVGIKVAL RAMEAPLRQIVLNCGEEPSVVANTVKGGDGNYGYNAA TEEYGNMIDMGILDPTKVTRSALQYAASVAGLMITTE CMVTDLPKNDAADLGAAGGMGGMM*
38.	groES	*M. capsulatus* (Bath)	VKIRPLHDRVIIKRLEEERTSAGGIVIPDSAAEKPMR GEILAVGNGKVLDNGEVRALQVKVGDKVLFGKYAGTE VKVDGEDVVVMREDDILAVLES*
39.	groEL-2*	*M. capsulatus* (Bath)	MAKEVVYRGSARQRMMQGIEILARAAIPTLGATGPSV MIQHRADGLPPISTRDGVTVANSIVLKDRVANLGARL LRDVAGTMSREAGDGTTTAIVLARHIAREMFKSLAVG ADPIALKRGIDRAVARVSEDIGARAWRGDKESVILGV AAVATKGEPGVGRLLLEALDAVGVHGAVSIELGQRRE DLLDVVDGYRWEKGYLSPYFVTDRARELAELEDVYLL MTDREVVDFIDLVPLLEAVTEAGGSLLIAADRVHEKA LAGLLLNHVRGVFKAVAVTAPGFGDKRPNRLLDLAAL TGGRAVLEAQGDRLDRVTLADLGRVRRAVVSADDTAL LGIPGTEASRARLEGLRLEAEQYRALKPGQGSATGRL HELEEIEARIVGLSGKSAVYRVGGVTDVEMKERMVRI EGAYRSVVSALEEGVLPGGGVGFLGSMPVLAELEARD ADEARGIGIVRSALTEPLRIIGENSGLSGEAVVAKVM DHANPGWGYDQESGSFCDLHARGIWDAAKVLRLALEK AASVAGTFLTTEAVVLEIPDTDAFAGFSAEWAAATRE DPRV*
40.			
41.			

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
42.			
43.			
44.	pNH284		TTGTTTCATCAAGCCTTACGGTCACCGTAACCAGCAA ATCAATATCACTGTGTGGCTTCAGGCCGCCATCCACT GCGGAGCCGTACAAATGTACGGCCAGCAACGTCGGTT CGAGATGGCGCTCGATGACGCCAACTACCTCTGATAG TTGAGTCGATACTTCGGCGATCACCGCTTCCCTCATA CTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGG GTTATTGTCTCATGAGCGGATACATATTTGAATGTAT TTAGAAAAATAAACAAATAGCTAGCTCACTCGGTCGC TACGCTCCGGGCGTGAGACTGCGGCGGGCGCTGCGGA CACATACAAAGTTACCCACAGATTCCGTGGATAAGCA GGGGACTAACATGTGAGGCAAAACAGCAGGGCCGCGC CGGTGGCGTTTTTCCATAGGCTCCGCCCTCCTGCCAG AGTTCACATAAACAGACGCTTTTCCGGTGCATCTGTG GGAGCCGTGAGGCTCAACCATGAATCTGACAGTACGG GCGAAACCCGACAGGACTTAAAGATCCCCACCGTTTC CGGCtGGTCGCTCCCTCTTGCGCTCTCCTGTTCCGAC CCTGCCGTTTACCGGATACCTGTTCCGCCTTTCTCCC TTACGGGAAGTGTGGCGCTTTCTCATAGCTCACACAC TGGTATCTCGGCTCGGTGTAGGTCGTTCGCTCCAAGC TGGGCTGTAAGCAAGAACTCCCCGTTCAGCCCGACTG CTGCGCCTTATCCGGTAACTGTTCACTTGAGTCCAAC CCGGAAAAGCACGGTAAAACGCCACTGGCAGCAGCCA TTGGTAACTGGGAGTTCGCAGAGGATTTGTTTAGCTA AACACGCGGTTGCTCTTGAAGTGTGCGCCAAAGTCCG GCTACACTGGAAGGACAGATTTGGTTGCTGTGCTCTG CGAAAGCCAGTTACCACGGTTAAGCAGTTCCCCAACT GACTTAACCTTCGATCAAACCACCTCCCaatGTGGTT TTTTCGTTTACAGGGCAAAAGATTACGCGCAGAAAAA AAGGATCTCAAGAAGATCCTTTGATCTTTTCTACTGA ACCGCTCTAGATTTCAGTGCAATTTATCTCTTCAAAT GTAGCACCTGAAGTCAGCCCCATACGATATAAGTTGT AATTCTCATGTTAGTCATGCCCCGCGCCCACCGGAAG GAGCTGACTGGGTTGAAGGCTCTCAAGGGCATCGGTC GAGATCCCGGTGCCTAATGAGTGAGCTAACTTTTGAC gGCTAGCTCAGTCCTAGGGAtaATGCTAGCaccagcc TCGAgggaaaccacgtaagctccggcgtTTAAAcacc cataacagatacggactttctcaaaggagagttatca gTGAAAATCCGCCCGTTACATGACCGTGTCATCATCA AACGCTTGGAAGAAGAGCGTACCTCGGCGGGCGGGAT TGTCATTCCAGATAGCGCaGCTGAAAAACCGATGCGT GGTGAAATCCTGGCAGTGGGCAATGGAAAAGTGCTTG ATAATGGAGAGGTACGTGCTTTACAGGTGAAAGTGGG TGATAAAGTGCTCTTTGGGAAATACGCGGGTACGGAG GTTAAAGTAGATGGGGAAGATGTTGTTGTCATGCGTG AAGATGACATTCTGGCTGTGTTAGAATCTTAATCCGC GCACGACACTGAACATACGAATTTAAGGAATAAAGAT AATGGCGAAAGAAGTTGTGTATCGTGGTAGTGCGCGC CAGCGTATGATGCAGGGTATTGAAATTCTCGCTCGCG CCGCTATTCCAACGCTGGGGGCAACCGGCCCGAGCGT CATGATTCAACATCGCGCCGATGGTCTGCCACCCATT TCTACACGCGATGGCGTTACCGTAGCGAATTCTATTG TTTTAAAAGACCGTGTCGCGAACCTGGGTGCCCGCCT GCTGCGCGACGTAGCCGGTACAATGAGCCGTGAAGCC GGCGACGGCACGACGACTGCGATCGTATTGGCCCGCC ACATCGCCCGTGAGATGTTTAAATCGCTGGCCGTGGG TGCAGATCCGATCGCGCTGAAACGTGGTATCGATCGC GCCGTTGCTCGTGTGTCCGAAGATATTGGGGCGCGTG CGTGGCGTGGCGATAAAGAAAGCGTGATCCTGGGTGT CGCTGCTGTGGCGACGAAAGGCGAACCGGGCGTTGGC CGTCTGCTGCTGGAGGCTCTCGATGCAGTGGGTGTTC ACGGTGCCGTTTCTATCGAACTGGGCCAACGTCGTGA AGATCTGCTGGACGTCGTCGATGGCTATCGCTGGGAA AAAGGTTATTTATCTCCCTACTTTGTCACGGACCGTG CCCGCGAACTCGCGGAACTGGAGGATGTCTACCTGCT CATGACCGACCGCGAAGTGGTTGACTTCATCGACCTT GTACCTCTGCTGGAGGCCGTGACGGAAGCAGGAGGCT CCCTGCTGATTGCCGCGGATCGTGTGCACGAAAAGGC CTTAGCGGGGCTGCTTCTGAATCACGTGCGCGGTGTC TTCAAGGCCGTGGCCGTAACCGCTCCGGGTTTTGGCG

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			ACAAACGCCCGAACCGTTTACTTGACCTGGCCGCGTT AACCGGCGGTCGTGCCGTGCTCGAAGCTCAAGGCGAC CGTCTGGACCGTGTTACCCTCGCGGATCTGGGCCGTG TGCGCCGTGCCGTGGTGTCGGCAGATGATACCGCGCT GCTTGGCATCCCGGGCACCGAAGCTAGCCGTGCACGC CTCGAAGGTCTGCGTTTAGAAGCAGAgCAGTACCGTG CGCTGAAACCAGGGCAGGGTTCTGCCACCGGGCGCCT GCACGAACTTGAAGAAATTGAAGCGCGCATTGTGGGT CTgTCCGGAAAGAGCGCCGTTTATCGCGTCGGAGGTG TGACCGATGTGGAAATGAAAGAGCGCATGGTTCGCAT CGAAggaGCTTACCGTTCGGTGGTAAGTGCGCTGGAG GAAGGCGTGCTCCCTGGCGGTGGTGTCGGCTTTCTGG GTAGTATGCCGGTGCTTGCGGAATTGGAGGCCCGCGA CGCAGATGAAGCTCGCGGGATTGGGATTGTACGCAGC GCCTTAACGGAGCCTCTTCGTATTATCGGCGAAAATA GTGGCTTGAGCGGTGAAGCCGTTGTTGCCAAAGTCAT GGATCATGCCAACCCGGGATGGGGTTACGACCAGGAG TCTGGCTCTTTTTGCGACCTGCATGCGCGTGGGATCT GGGATGCTGCTAAAGTGTTACGTCTCGCGTTGGAGAA GGCAGCCTCTGTTGCTGGGACCTTTCTGACAACCGAA GCTGTTGTTCTCGAAATTCCGGATACAGATGCGTTCG CAGGGTTCAGTGCAGAATGGGCTGCCGCCACGCGCGA AGATCCGCGCGTATGAgtttaaacgcggccgcAATTT GAACGcacccataacagatacggactttctcaaagga gagttatcaATGAATATTCGTCCATTGCATGATCGCG TGATCGTCAAGCGTAAAGAAGTTGAAACTAAATCTGC TGGCGGCATCGTTCTGACCGGCTCTGCAGCGGCTAAA TCCACCCGCGGCGAAGTGCTGGCTGTCGGCAATGGCC GTATCCTTGAAAATGGCGAAGTGAAGCCGCTGGATGT GAAAGTTGGCGACATCGTTATTTTCAACGATGGCTAC GGTGTGAAATCTGAGAAGATCGACAATGAAGAAGTGT TGATCATGTCCGAAAGCGACATTCTGGCAATTGTTGA AGCGTAATCCGCGCACGACACTGAACATACGAATTTA AGGAATAAAGATAatgGCAGCTAAAGACGTAAAATTC GGTAACGACGCTCGTGTGAAAATGCTGCGCGGCGTAA ACGTACTGGCAGATGCAGTGAAAGTTACCCTCGGTCC AAAAGGCCGTAACGTAGTTCTGGATAAATCTTTCGGT GCACCGACCATCACCAAAGATGGTGTTTCCGTTGCTC GTGAAATCGAACTGGAAGACAAGTTCGAAAATATGGG TGCGCAGATGGTGAAAGAAGTTGCCTCTAAAGCAAAC GACGCTGCAGGCGACGGTACCACCACTGCAACCGTAC TGGCTCAGGCTATCATCACTGAAGGTCTGAAAGCTGT TGCTGCGGGCATGAACCCGATGGACCTGAAACGTGGT ATCGACAAAGCGGTTACCGCTGCAGTTGAAGAACTGA AAGCGCTGTCCGTACCATGCTCTGACTCTAAAGCGAT TGCTCAGGTTGGTACCATCTCCGCTAACTCCGACGAA ACCGTAGGTAAACTGATCGCTGAAGCGATGGACAAAG TCGGTAAAGAAGGCGTTATCACCGTTGAAGACGGTAC CGGTCTGCAGGACGAACTGGACGTGGTTGAAGGTATG CAGTTCGACCGTGGCTACCTGTCTCCTTACTTCATCA ACAAGCCGGAAACTGGCGCAGTAGAACTGGAAAGCCC GTTCATCCTGCTGGCTGACAAGAAAATCTCCAACATC CGCGAAATGCTGCCGGTTCTGGAAGCTGTTGCCAAAG CAGGCAAACCGCTGCTGATCATCGCTGAAGATGTAGA AGGCGAAGCGCTGGCAACTCTGGTTGTTAACACCATG CGTGGCATCGTGAAAGTCGCTGCGGTTAAAGCACCGG GCTTCGGCGATCGTCGTAAAGCTATGCTGCAGGATAT CGCAACCCTGACTGGCGGTACCGTGATCTCTGAAGAG ATCGGTATGGAGCTGGAAAAAGCAACCCTGGAAGACC TGGGTCAGGCTAAACGTGTTGTGATCAACAAAGACAC CACCACTATCATCGATGGCGTGGGTGAAGAAGCTGCA ATCCAGGGCCGTGTTGCTCAGATCCGTCAGCAGATTG AAGAAGCAACTTCTGACTACGACCGTGAAAAACTGCA GGAACGCGTAGCGAAACTGGCAGGCGGCGTTGCAGTT ATCAAAGTGGGTGCTGCTACCGAAGTTGAAATGAAAG AGAAAAAAGCACGCGTTGAAGATGCCCTGCACGCGAC CCGTGCTGCGGTAGAAGAAGGCGTGGTTGCTGGTGGT GGTGTTGCGCTGATCCGCGTAGCGTCTAAACTGGCTG ACCTGCGTGGTCAGAACGAAGACCAGAACGTGGGTAT CAAAGTTGCACTGCGTGCAATGGAAGCTCCGCTGCGT CAGATCGTATTGAACTGCGGCGAAGAACCGTCTGTTG TTGCTAACACCGTTAAAGGCGGCGACGGCAACTACGG TTACAACGCAGCAACCGAAGAATACGGCAACATGATC

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			GACATGGGTATCCTGGATCCAACCAAAGTAACTCGTT CTGCTCTGCAGTACGCAGCTTCTGTGGCTGGCCTGAT GATCACCACCGAATGCATGGTTACCGACCTGCCGAAA AACGATGCAGCTGACTTAGGCGCTGCTGGCGGTATGG GCGGCATGATGtaagtttaaacgcggccgcAATTTGA ACGCCAGCACATGGACTCtcgaGTCTACTAGCGCAGC TTAATTAACCTAGGCTGCTGCCACCGCTGAGCAATAA CTAGCATAACCCCTTGGGGCCTCTAAACGGGTCTTGA GGGGTTTTTTGCTGAAACCTCAGGCATTTGAGAAGCA CACGGTCACACTGCTTCCGGTAGTCAATAAACCGGTA AACCAGCAATAGACATAAGCGGTGCATAATGTGCCTG TCAAATGGACGAAGCAGGGATTCTGCAAACCCTATGC TACTCCGTCAAGCCGTCAATTGTCTGATTCGTTACCA ATTATGACAACTTGACGGCTACATCATTCACTTTTTC TTCACAACCGGCACGGAACTCGCTCGGGCTGGCCCCG GTGCATTTTTTAAATACCCGCGAGAAATAGAGTTGAT CGTCAAAACCAACATTGCGACCGACGGTGGCGATAGG CATCCGGGTGGTGCTCAAAAGCAGCTTCGCCTGGCTG ATACGTTGGTCCTCGCGCCAGCTTAAGACGCTAATCC CTAACTGCTGGCGGAAAAGATGTGACAGACGCGACGG CGACAAGCAAACATGCTGTGCGACGCTGGCGATATCA AAATTGCTGTCTGCCAGGTGATCGCTGATGTACTGAC AAGCCTCGCGTACCCGATTATCCATCGGTGGATGGAG CGACTCGTTAATCGCTTCCATGCGCCGCAGTAACAAT TGCTCAAGCAGATTTATCGCCAGCAGCTCCGAATAGC GCCCTTCCCCTTGCCCGGCGTTAATGATTTGCCCAAA CAGGTCGCTGAAATGCGGCTGGTGCGCTTCATCCGGG CGAAAGAACCCCGTATTGGCAAATATTGACGGCCAGT TAAGCCATTCATGCCAGTAGGCGCGCGGACGAAAGTA AACCCACTGGTGATACCATTCGCGAGCCTCCGGATGA CGACCGTAGTGATGAATCTCTCCTGGCGGGAACAGCA AAATATCACCCGGTCGGCAAACAAATTCTCGTCCCTG ATTTTTCACCACCCCCTGACCGCGAATGGTGAGATTG AGAATATAACCTTTCATTCCCAGCGGTCGGTCGATAA AAAAATCGAGATAACCGTTGGCCTCAATCGGCGTTAA ACCCGCCACCAGATGGGCATTAAACGAGTATCCCGGC AGCAGGGGATCATTTTGCGCTTCAGCCATACTTTTCA TACTCCCGCCATTCAGAGAAGAAACCAATTGTCCATA TTGCATCAGACATTGCCGTCACTGCGTCTTTTACTGG CTCTTCTCGCTAACCAAACCGGTAACCCCGCTTATTA AAAGCATTCTGTAACAAAGCGGGACCAAAGCCATGAC AAAAACGCGTAACAAAAGTGTCTATAATCACGGCAGA AAAGTCCACATTGATTATTTGCACGGCGTCACACTTT GCTATGCCATAGCATTTTTATCCATAAGATTAGCGGA TCCTACCTGACGCTTTTTATCGCAACTCTggACaaTg TCTCCATACCCGTTTTTTTGGGcgacctcgtcggagg ttgtatgtccggtgttccgtgacgtcatcgggcattc atcattcatagaatgtgttacggaggaaacaagtaat ggcacttagcaccgcaaccaaggccgcgacggacgcg ctggctgccaatcgggcacccaccagcgGgaatgcac aggaagtgcaccgttggctccagagcttcaactggga tttcaagaacaaccggaccaagtacgccaccaagtac aagatggcgaacgagaccaaggaacagttcaagctga tcgccaaggaatatgcgcgcatggaggcagtcaagga cgaaaggcagttcggtagcctgcaggatgcgctgacc cgcctcaacgccggtgttcgcgttcatccgaagtgga acgagaccatgaaagtggtttcgaacttcctggaagt gggcgaatacaacgccatcgccgctaccgggatgctg tgggattccgcccaggcggcggaacagaagaacggct atctggcccaggtgttggatgaaatccgccacaccca ccagtgtgcctacgtcaactactacttcgcgaagaac ggccaggacccggccggtcacaacgatgctcgccgca cccgtaccatcggtccgctgtggaagggcatgaagcg cgtgttttccgacggcttcatttccggcgacgccgtg gaatgctccctcaacctgcagctggtgggtgaggcct gcttcaccaatccgctgatcgtcgcagtgaccgaatg ggctgccgccaacggcgatgaaatcaccccgacggtg ttcctgtcgatcgagaccgacgaactgcgccacatgg ccaacggttaccagaccgtcgtttccatcgccaacga tccggcttccgccaagtatctcaacacggacctgaac aacgccttctggacccagcagaagtacttcacgccgg tgttgggcatgctgttcgagtatggctccaagttcaa ggtcgagccgtgggtcaagacgtggaaccgctgggtg

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
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			gcaagtacggggtggagtcgccgcgcagcctcaagga
			cgccaagcaggacgcttactgggctcaccacgacctg
			tatctgctggcttatgcgctgtggccgGGcggcttct
			tccgtctggcgctgccggatcaggaagaaatggagtg
			gttcgaggccaactaccccggctggtacgaccactac
			ggcaagatctacgaggaatggcgcgcccgcggttgcg
			aggatccgtcctcgggcttcatcccgctgatgtggtt
			catcgaaaacaaccatcccatctacatcgatcgcgtg
			tcgcaagtgccgttctgcccgagcttggccaagggcg
			ccagcaccctgcgcgtgcacgagtacaacggccagat
			gcacaccttcagcgaccagtggggcgagcgcatgtgg
			ctggccgagccggagcgctacgagtgccagaacatct
			tcgaacagtacgaaggacgcgaactgtcggaagtgat
			cgccgaactgcacgggctgcgcagtgatggcaagacc
			ctgatcgcccagccgcatgtccgtggcgacaagctgt
			ggacgttggacgatatcaaacgcctgaactgcgtctt
			caagaacccggtgaaggcattcaattgaaacgggtgt
			cgggctccgtcacagggcggggcccgacgcacgatcg
			ttcgatcaacctcaaaccaaaaaggaacatcgatatg
			agcatgttaggagaaagacgccgcggtctgaccgatc
			cggaaatggcggccgtcattttgaaggcgcttcctga
			agctccgctggacggcaacaacaagatgggttatttc
			gtcacccccccgctggaaacgcttgacggaatatgaag
			ccctgaccgtttatgcgcagcccaacgccgactggat
			cgccggcggcctggactggggcgactggacccagaaa
			ttccacggcggccgcccttcctggggcaacgagacca
			cggagctgcgcaccgtcgactggttcaagcaccgtga
			cccgctccgccgttggcatgcgccgtacgtcaaggac
			aaggccgaggaatggcgctacaccgaccgcttcctgc
			agggttactccgccgacggtcagatccgggcgatgaa
			cccgacctggcgggacgagttcatcaaccggtattgg
			ggcgccttcctgttcaacgaatacggattgttcaacg
			ctcattcgcagggcgcccgggaggcgctgtcggacgt
			aacccgcgtcagcctggctttctggggcttcgacaag
			atcgacatcgcccagatgatccaactcgaacggggtt
			tcctcgccaagatcgtacccggtttcgacgagtccac
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			tacaagagcgcccgtctggccgtggaagggctgtggc
			aggaggtgttcgactggaacgagagcgctttctcggt
			gcacgccgtctatgacgcgctgttcggtcagttcgtc
			cgccgcgagttctttcagcggctggctccccgcttcg
			gcgacaatctgacgccattcttcatcaaccaggccca
			gacatacttccagatcgccaagcagggcgtacaggat
			ctgtattacaactgtctgggtgacgatccggagttca
			gcgattacaaccgtaccgtgatgcgcaactggaccgg
			caagtggctggagcccacgatcgccgctctgcgcgac
			ttcatggggctgtttgcgaagctgccggcgggcacca
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			aacgtcaatccgacggtaaacgtggaagaccgcgcgg
			gttactggtggatcaaggccaacggcaagatcgaggt
			cgattgcgacgagatttccgagctgttggggcggcag
			ttcaacgtctacgacttcctcgtcgacgtttcctcca
			ccatcggccgggcctataccctgggcaacaagttcac
			cattaccagtgagctgatgggcctggaccgcaagctc
			gaagactCtcacgcttaaggagaatgacatggcgaaa
			ctgggtatacacagcaacgacacccgcgacgcctggg
			tgaacaagatcgcgcagctcaacaccctggaaaaagc
			ggccgagatgctgaagcagttccggatggaccacacc
			acgccgttccgcaacagctacgaactggacaacgact
			acctctggatcgaggccaagctcgaagagaaggtcgc
			cgtcctcaaggcaGAAgccttcaacgaggtggacttc
			cgtcataagaccgctttcggcgaggatgccaagtccg
			ttctggacggcaccgtcgcgaagatgaacgcggccaa
			ggacaagtgggaggcggagaagatccatatcggtttc

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			cgccaggcctacaagccgccgatcatgccggtgaact
			atttcctggacggcgagcgTcagttggggacccggct
			gatggaactgcgcaacctcaactactacgacacgccg
			ctggaagaactgcgcaaacagcgcggtgtgcgggtgg
			tgcatctgcagtcgccgcactgaagggaggaagtctc
			gccctggacgcgacggcatcgccgtgaagtccagggg
			gcagggatgccgttccgggTcggcaggctggcccgga
			atctctggttttcagggggcgtgccggtccacggctc
			ccccctccatctttcgtaaggaaatcaccatggtcga
			atcggcatttcagccattttcgggcgacgcagacgaa
			tggttcgaggaaccacggccccaggccggtttcttcc
			cttccgcggactggcatctgctcaaacgggacgagac
			ctacgcagcctatgccaaggatctcgatttcatgtgg
			cggtgggtcatcgtccgggaagaaaggatcgtccagg
			agggttgctcgatcagcctggagtcgtcgatccgcgc
			cgtgacgcacgtactgaattattttggtatgaccgaa
			caacgcgccccggcagaggaccggaccggcggagttc
			aacattgaacaggtaagtttatgcagcgagttcacac
			tatcacggcggtgacggaggatggcgaatcgctccgc
			ttcgaatgccgttcggacgaggacgtcatcaccgccg
			ccctgcgccagaacatctttctgatgtcgtcctgccg
			ggagggcggctgtgcgacctgcaaggccttgtgcagc
			gaaggggactacgacctcaagggctgcagcgttcagg
			cgctgccgccggaagaggaggaggaagggttggtgtt
			gttgtgccggacctacccgaagaccgacctggaaatc
			gaactgccctatacccattgccgcatcagttttggtg
			aggtcggcagtttcgaggcggaggtcgtcggcctcaa
			ctgggtttcgagcaacaccgtccagtttcttttgcag
			aagcggcccgacgagtgcggcaaccgtggcgtgaaat
			tcgaacccggtcagttcatggacctgaccatccccgg
			caccgatgtctcccgctcctactcgccggcgaacctt
			cctaatcccgaaggccgcctggagttcctgatccgcg
			tgttaccggagggacggttttcggactacctgcgcaa
			tgacgcgcgtgtcggacaggtcctctcggtcaaaggg
			ccactgggcgtgttcggtctcaaggagcggggcatgg
			cgccgcgctatttcgtggccggcggcaccgggttggc
			gccggtggtctcgatggtgcggcagatgcaggagtgg
			accgcgccgaacgagacccgcatctatttcggtgtga
			acaccgagccggaattgttctacatcgacgagctcaa
			atccctggaacgatcgatgcgcaatctcaccgtgaag
			gcctgtgtctggcacccgagcggggactgggaaggcg
			agcagggctcgcccatcgatgcgttgcgggaagacct
			ggagtcctccgacgccaacccggacatttatttgtgc
			ggtccgccgggcatgatcgatgccgcctgcgagctgg
			tacgcagccgcggtatccccggcgaacaggtcttctt
			cgaaaaattcctgccgtccggggcggcctgaaccggg
			gaagtaccgtgaccaccgagcagttcccgccccaatt
			cctgcgtgaaatgatcgagcagctggacgccagcatc
			caggagctcgcacgcaaggaaaagggacttgcggcat
			ccctgggcacgggccgggtcgccgagctcaaggaata
			ctgggaccacgttGTTACAACCAATTAACCAATTCTG
			ACTATTTAACGACCCTGCCCTGAACCGACGACCGGGT
			CATCGTGGCCGGATCTTGCGGCCCCTCGGCTTGAACG
			AATTGTTAGACATTATTTGCCGACTACCTTGGTGATC
			TCGCCTTTCACGTAGTGGACAAATTCTTCCAACTGAT
			CTGCGCGCGAGGCCAAGCGATCTTCTTCTTGTCCAAG
			ATAAGCCTGTCTAGCTTCAAGTATGACGGGCTGATAC
			TGGGCCGGCAGGCGCTCCATTGCCCAGTCGGCAGCGA
			CATCCTTCGGCGCGATTTTGCCGGTTACTGCGCTGTA
			CCAAATGCGGGACAACGTAAGCACTACATTTCGCTCA
			TCGCCAGCCCAGTCGGGCGGCGAGTTCCATAGCGTTA
			AGGTTTCATTTAGCGCCTCAAATAGATCCTGTTCAGG
			AACCGGATCAAAGAGTTCCTCCGCCGCTGGACCTACC
			AAGGCAACGCTATGTTCTCTTGCTTTTGTCAGCAAGA
			TAGCCAGATCAATGTCGATCGTGGCTGGCTCGAAGAT
			ACCTGCAAGAATGTCATTGCGCTGCCATTCTCCAAAT
			TGCAGTTCGCGCTTAGCTGGATAACGCCACGGAATGA
			TGTCGTCGTGCACAACAATGGTGACTTCTACAGCGCG
			GAGAATCTCGCTCTCTCCAGGGGAAGCCGAAGTTTCC
			AAAAGGTCGTTGATCAAAGCTCGCCGCG
45.	pYZ40		CGGCGGGCGCTGCGGACACATACAAAGTTACCCACAG
			ATTCCGTGGATAAGCAGGGGACTAACATGTGAGGCAA

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			AACAGCAGGGCCGCGCCGGTGGCGTTTTTCCATAGGC
			TCCGCCCTCCTGCCAGAGTTCACATAAACAGACGCTT
			TTCCGGTGCATCTGTGGGAGCCGTGAGGCTCAACCAT
			GAATCTGACAGTACGGGCGAAACCCGACAGGACTTAA
			AGATCCCCACCGTTTCCGGCTGGTCGCTCCCTCTTGC
			GCTCTCCTGTTCCGACCCTGCCGTTTACCGGATACCT
			GTTCCGCCTTTCTCCCTTACGGGAAGTGTGGCGCTTT
			CTCATAGCTCACACACTGGTATCTCGGCTCGGTGTAG
			GTCGTTCGCTCCAAGCTGGGCTGTAAGCAAGAACTCC
			CCGTTCAGCCCGACTGCTGCGCCTTATCCGGTAACTG
			TTCACTTGAGTCCAACCCGGAAAAGCACGGTAAAACG
			CCACTGGCAGCAGCCATTGGTAACTGGGAGTTCGCAG
			AGGATTTGTTTAGCTAAACACGCGGTTGCTCTTGAAG
			TGTGCGCCAAAGTCCGGCTACACTGGAAGGACAGATT
			TGGTTGCTGTGCTCTGCGAAAGCCAGTTACCACGGTT
			AAGCAGTTCCCCAACTGACTTAACCTTCGATCAAACC
			ACCTCCCAATGTGGTTTTTTCGTTTACAGGGCAAAAG
			ATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTT
			TGATCTTTTCTACTGAACCGCTCTAGATTTCAGTGCA
			ATTTATCTCTTCAAATGTAGCACCTGAAGTCAGCCCC
			ATACGATATAAGTTGTAATTCTCATGTTAGTCATGCC
			CCGCGCCCACCGGAAGGAGCTGACTGGGTTGAAGGCT
			CTCAAGGGCATCGGTCGAGATCCCGGTGCCTAATGAG
			TGAGCTAACTTTTGACGGCTAGCTCAGTCCTAGGGAT
			AATGCTAGCACCAGCCTCGAGGGAAACCACGTAAGCT
			CCGGCGTTTAAACACCCATAACAGATACGGACTTTCT
			CAAAGGAGAGTTATCAGTGAAAATCCGCCCGTTACAT
			GACCGTGTCATCATCAAACGCTTGGAAGAAGAGCGTA
			CCTCGGCGGGCGGGATTGTCATTCCAGATAGCGCAGC
			TGAAAAACCGATGCGTGGTGAAATCCTGGCAGTGGGC
			AATGGAAAAGTGCTTGATAATGGAGAGGTACGTGCTT
			TACAGGTGAAAGTGGGTGATAAAGTGCTCTTTGGGAA
			ATACGCGGGTACGGAGGTTAAAGTAGATGGGGAAGAT
			GTTGTTGTCATGCGTGAAGATGACATTCTGGCTGTGT
			TAGAATCTTAATCCGCGCACGACACTGAACATACGAA
			TTTAAGGAATAAAGATAATGGCGAAAGAAGTTGTGTA
			TCGTGGTAGTGCGCGCCAGCGTATGATGCAGGGTATT
			GAAATTCTCGCTCGCGCCGCTATTCCAACGCTGGGGG
			CAACCGGCCCGAGCGTCATGATTCAACATCGCGCCGA
			TGGTCTGCCACCCATTTCTACACGCGATGGCGTTACC
			GTAGCGAATTCTATTGTTTTAAAAGACCGTGTCGCGA
			ACCTGGGTGCCCGCCTGCTGCGCGACGTAGCCGGTAC
			AATGAGCCGTGAAGCCGGCGACGGCACGACGACTGCG
			ATCGTATTGGCCCGCCACATCGCCCGTGAGATGTTTA
			AATCGCTGGCCGTGGGTGCAGATCCGATCGCGCTGAA
			ACGTGGTATCGATCGCGCCGTTGCTCGTGTGTCCGAA
			GATATTGGGGCGCGTGCGTGGCGTGGCGATAAAGAAA
			GCGTGATCCTGGGTGTCGCTGCTGTGGCGACGAAAGG
			CGAACCGGGCGTTGGCCGTCTGCTGCTGGAGGCTCTC
			GATGCAGTGGGTGTTCACGGTGCCGTTTCTATCGAAC
			TGGGCCAACGTCGTGAAGATCTGCTGGACGTCGTCGA
			TGGCTATCGCTGGGAAAAAGGTTATTTATCTCCCTAC
			TTTGTCACGGACCGTGCCCGCGAACTCGCGGAACTGG
			AGGATGTCTACCTGCTCATGACCGACCGCGAAGTGGT
			TGACTTCATCGACCTTGTACCTCTGCTGGAGGCCGTG
			ACGGAAGCAGGAGGCTCCCTGCTGATTGCCGCGGATC
			GTGTGCACGAAAAGGCCTTAGCGGGGCTGCTTCTGAA
			TCACGTGCGCGGTGTCTTCAAGGCCGTGGCCGTAACC
			GCTCCGGGTTTTGGCGACAAACGCCCGAACCGTTTAC
			TTGACCTGGCCGCGTTAACCGGCGGTCGTGCCGTGCT
			CGAAGCTCAAGGCGACCGTCTGGACCGTGTTACCCTC
			GCGGATCTGGGCCGTGTGCGCCGTGCCGTGGTGTCGG
			CAGATGATACCGCGCTGCTTGGCATCCCGGGCACCGA
			AGCTAGCCGTGCACGCCTCGAAGGTCTGCGTTTAGAA
			GCAGAGCAGTACCGTGCGCTGAAACCAGGGCAGGGTT
			CTGCCACCGGGCGCCTGCACGAACTTGAAGAAATTGA
			AGCGCGCATTGTGGGTCTGTCCGGAAAGAGCGCCGTT
			TATCGCGTCGGAGGTGTGACCGATGTGGAAATGAAAG
			AGCGCATGGTTCGCATCGAAGGAGCTTACCGTTCGGT
			GGTAAGTGCGCTGGAGGAAGGCGTGCTCCCTGGCGGT
			GGTGTCGGCTTTCTGGGTAGTATGCCGGTGCTTGCGG
			AATTGGAGGCCCGCGACGCAGATGAAGCTCGCGGGAT
			TGGGATTGTACGCAGCGCCTTAACGGAGCCTCTTCGT

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			ATTATCGGCGAAAATAGTGGCTTGAGCGGTGAAGCCG TTGTTGCCAAAGTCATGGATCATGCCAACCCGGGATG GGGTTACGACCAGGAGTCTGGCTCTTTTTGCGACCTG CATGCGCGTGGGATCTGGGATGCTGCTAAAGTGTTAC GTCTCGCGTTGGAGAAGGCAGCCTCTGTTGCTGGGAC CTTTCTGACAACCGAAGCTGTTGTTCTCGAAATTCCG GATACAGATGCGTTCGCAGGGTTCAGTGCAGAATGGG CTGCCGCCACGCGCGAAGATCCGCGCGTATGAGTTTA AACGCGGCCGCAATTTGAACGCACCCATAACAGATAC GGACTTTCTCAAAGGAGAGTTATCAATGAATATTCGT CCATTGCATGATCGCGTGATCGTCAAGCGTAAAGAAG TTGAAACTAAATCTGCTGGCGGCATCGTTCTGACCGG CTCTGCAGCGGCTAAATCCACCCGCGGCGAAGTGCTG GCTGTCGGCAATGGCCGTATCCTTGAAAATGGCGAAG TGAAGCCGCTGGATGTGAAAGTTGGCGACATCGTTAT TTTCAACGATGGCTACGGTGTGAAATCTGAGAAGATC GACAATGAAGAAGTGTTGATCATGTCCGAAAGCGACA TTCTGGCAATTGTTGAAGCGTAATCCGCGCACGACAC TGAACATACGAATTTAAGGAATAAAGATAATGGCAGC TAAAGACGTAAAATTCGGTAACGACGCTCGTGTGAAA ATGCTGCGCGGCGTAAACGTACTGGCAGATGCAGTGA AAGTTACCCTCGGTCCAAAAGGCCGTAACGTAGTTCT GGATAAATCTTTCGGTGCACCGACCATCACCAAAGAT GGTGTTTCCGTTGCTCGTGAAATCGAACTGGAAGACA AGTTCGAAAATATGGGTGCGCAGATGGTGAAAGAAGT TGCCTCTAAAGCAAACGACGCTGCAGGCGACGGTACC ACCACTGCAACCGTACTGGCTCAGGCTATCATCACTG AAGGTCTGAAAGCTGTTGCTGCGGGCATGAACCCGAT GGACCTGAAACGTGGTATCGACAAAGCGGTTACCGCT GCAGTTGAAGAACTGAAAGCGCTGTCCGTACCATGCT CTGACTCTAAAGCGATTGCTCAGGTTGGTACCATCTC CGCTAACTCCGACGAAACCGTAGGTAAACTGATCGCT GAAGCGATGGACAAAGTCGGTAAAGAAGGCGTTATCA CCGTTGAAGACGGTACCGGTCTGCAGGACGAACTGGA CGTGGTTGAAGGTATGCAGTTCGACCGTGGCTACCTG TCTCCTTACTTCATCAACAAGCCGGAAACTGGCGCAG TAGAACTGGAAAGCCCGTTCATCCTGCTGGCTGACAA GAAAATCTCCAACATCCGCGAAATGCTGCCGGTTCTG GAAGCTGTTGCCAAAGCAGGCAAACCGCTGCTGATCA TCGCTGAAGATGTAGAAGGCGAAGCGCTGGCAACTCT GGTTGTTAACACCATGCGTGGCATCGTGAAAGTCGCT GCGGTTAAAGCACCGGGCTTCGGCGATCGTCGTAAAG CTATGCTGCAGGATATCGCAACCCTGACTGGCGGTAC CGTGATCTCTGAAGAGATCGGTATGGAGCTGGAAAAA GCAACCCTGGAAGACCTGGGTCAGGCTAAACGTGTTG TGATCAACAAAGACACCACCACTATCATCGATGGCGT GGGTGAAGAAGCTGCAATCCAGGGCCGTGTTGCTCAG ATCCGTCAGCAGATTGAAGAAGCAACTTCTGACTACG ACCGTGAAAAACTGCAGGAACGCGTAGCGAAACTGGC AGGCGGCGTTGCAGTTATCAAAGTGGGTGCTGCTACC GAAGTTGAAATGAAAGAGAAAAAAGCACGCGTTGAAG ATGCCCTGCACGCGACCCGTGCTGCGGTAGAAGAAGG CGTGGTTGCTGGTGGTGGTGTTGCGCTGATCCGCGTA GCGTCTAAACTGGCTGACCTGCGTGGTCAGAACGAAG ACCAGAACGTGGGTATCAAAGTTGCACTGCGTGCAAT GGAAGCTCCGCTGCGTCAGATCGTATTGAACTGCGGC GAAGAACCGTCTGTTGTTGCTAACACCGTTAAAGGCG GCGACGGCAACTACGGTTACAACGCAGCAACCGAAGA ATACGGCAACATGATCGACATGGGTATCCTGGATCCA ACCAAAGTAACTCGTTCTGCTCTGCAGTACGCAGCTT CTGTGGCTGGCCTGATGATCACCACCGAATGCATGGT TACCGACCTGCCGAAAAACGATGCAGCTGACTTAGGC GCTGCTGGCGGTATGGGCGGCATGATGTAAGTTTAAA CGCGGCCGCAATTTGAACGCCAGCACATGGACTCTCG AGTCTACTAGCGCAGCTTAATTAACCTAGGCTGCTGC CACCGCTGAGCAATAACTAGCATAACCCCTTGGGGCC TCTAAACGGGTCTTGAGGGGTTTTTTGCTGAAACCTC AGGCATTTGAGAAGCACACGGTCACACTGCTTCCGGT AGTCAATAAACCAGCTCAAGGAATACTGGGACCACGT TGTTACAACCAATTAACCAATTCTGACTATTTAACGA CCCTGCCCTGAACCGACGACCGGGTCATCGTGGCCGG ATCTTGCGGCCCCTCGGCTTGAACGAATTGTTAGACA TTATTTGCCGACTACCTTGGTGATCTCGCCTTTCACG

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			TAGTGGACAAATTCTTCCAACTGATCTGCGCGCGAGG CCAAGCGATCTTCTTCTTGTCCAAGATAAGCCTGTCT AGCTTCAAGTATGACGGGCTGATACTGGGCCGGCAGG CGCTCCATTGCCCAGTCGGCAGCGACATCCTTCGGCG CGATTTTGCCGGTTACTGCGCTGTACCAAATGCGGGA CAACGTAAGCACTACATTTCGCTCATCGCCAGCCCAG TCGGGCGGCGAGTTCCATAGCGTTAAGGTTTCATTTA GCGCCTCAAATAGATCCTGTTCAGGAACCGGATCAAA GAGTTCCTCCGCCGCTGGACCTACCAAGGCAACGCTA TGTTCTCTTGCTTTTGTCAGCAAGATAGCCAGATCAA TGTCGATCGTGGCTGGCTCGAAGATACCTGCAAGAAT GTCATTGCGCTGCCATTCTCCAAATTGCAGTTCGCGC TTAGCTGGATAACGCCACGGAATGATGTCGTCGTGCA CAACAATGGTGACTTCTACAGCGCGGAGAATCTCGCT CTCTCCAGGGGAAGCCGAAGTTTCCAAAAGGTCGTTG ATCAAAGCTCGCCGCGTTGTTTCATCAAGCCTTACGG TCACCGTAACCAGCAAATCAATATCACTGTGTGGCTT CAGGCCGCCATCCACTGCGGAGCCGTACAAATGTACG GCCAGCAACGTCGGTTCGAGATGGCGCTCGATGACGC CAACTACCTCTGATAGTTGAGTCGATACTTCGGCGAT CACCGCTTCCCTCATACTCTTCCTTTTTCAATATTAT TGAAGCATTTATCAGGGTTATTGTCTCATGAGCGGAT ACATATTTGAATGTATTTAGAAAAATAAACAAATAGC TAGCTCACTCGGTCGCTACGCTCCGGGCGTGAGACTG
46.	pNH296		CCGACACCATCGAATGGTGCAAAACCTTTCGCGGTAT GGCATGATAGCGCCCGGAAGAGAGTCAATTCAGGGTG GTGAATGTGAAACCAGTAACGTTATACGATGTCGCAG AGTATGCCGGTGTCTCTTATCAGACCGTTTCCCGCGT GGTGAACCAGGCCAGCCACGTTTCTGCGAAAACGCGG GAAAAAGTGGAAGCGGCGATGGCGGAGCTGAATTACA TTCCCAACCGCGTGGCACAACAACTGGGGGCAAACA GTCGTTGCTGATTGGCGTTGCCACCTCCAGTCTGGCC CTGCACGCGCCGTCGCAAATTGTCGCGGCGATTAAAT CTCGCGCCGATCAACTGGGTGCCAGCGTGGTGGTGTC GATGGTAGAACGAAGCGGCGTCGAAGCCTGTAAAGCG GCGGTGCACAATCTTCTCGCGCAACGCGTCAGTGGGC TGATCATTAACTATCCGCTGGATGACCAGGATGCCAT TGCTGTGGAAGCTGCCTGCACTAATGTTCCGGCGTTA TTTCTTGATGTCTCTGACCAGACACCCATCAACAGTA TTATTTTCTCCCATGAAGACGGTACGCGACTGGGCGT GGAGCATCTGGTCGCATTGGGTCACCAGCAAATCGCG CTGTTAGCGGGCCCATTAAGTTCTGTCTCGGCGCGTC TGCGTCTGGCTGGCTGGCATAAATATCTCACTCGCAA TCAAATTCAGCCGATAGCGGAACGGGAAGGCGACTGG AGTGCCATGTCCGGTTTTCAACAAACCATGCAAATGC TGAATGAGGGCATCGTTCCCACTGCGATGCTGGTTGC CAACGATCAGATGGCGCTGGGCGCAATGCGCGCCATT ACCGAGTCCGGGCTGCGCGTTGGTGCGGATATTTCGG TAGTGGGATACGACGATACCGAAGACAGCTCATGTTA TATCCCGCCGTTAACCACCATCAAACAGGATTTTCGC CTGCTGGGGCAAACCAGCGTGGACCGCTTGCTGCAAC TCTCTCAGGGCCAGGCGGTGAAGGGCAATCAGCTGTT GCCCGTCTCACTGGTGAAAAGAAAAACCACCCTGGCG CCCAATACGCAAACCGCCTCTCCCCGCGCGTTGGCCG ATTCATTAATGCAGCTGGCACGACAGGTTTCCCGACT GGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTAAG TTAGCTCACTCATTAGGCACAATTCTCATGTTTGACA GCTTATCATCGACTGCACGGTGCACCAATGCTTCTGG CGTCAGGCAGCCATCGGAAGCTGTGGTATGGCTGTGC AGGTCGTAAATCACTGCATAATTCGTGTCGCTCAAGG CGCACTCCCGTTCTGGATAATGTTTTTTGCGCCGACA TCATAACGGTTCTGGCAAATATTCTGAAATGAGCTGT TGACAATTAATCATCGGCTCGTATAATGTGTGGAATT GTGAGCGGATAACAATTTCACACAGGAAACAGCCAGT CCGTTTAGGTGTTTTCACGAGCAATTGACCAACAAGG ACGTCAATAGAGTTAAGGAGGGAGGGGATGACCACTG CTGCACCCCAAGAATTTACCGCTGCTGTTGTTGAAAA ATTCGGTCATGACGTGACCGTGAAGGATATTGACCTT CCAAAGCCAGGGCCACACCAGGCATTGGTGAAGGTAC TCACCTCCGGCATCTGCCACACCGACCTCCACGCCTT GGAGGGCGATTGGCCAGTAAAGCCGGAACCACCATTC GTACCAGGACACGAAGGTGTAGGTGAAGTTGTTGAGC

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			TCGGACCAGGTGAACACGATGTGAAGGTCGGCGATAT TGTCGGCAATGCGTGGCTCTGGTCAGCGTGTGGCACC TGCGAATACTGCATCACCGGCAGGGAAACTCAGTGCA ACGAAGCTGAGTATGGTGGCTACACCCAAAATGGATC CTTCGGCCAGTACATGCTGGTGGATACCCGTTACGCC GCTCGCATCCCAGACGGCGTGGACTACCTCGAAGCAG CACCAATTCTGTGTGCAGGCGTGACTGTCTACAAGGC ACTCAAAGTCTCTGAAACCCGCCCGGGCCAATTCATG GTGATCTCCGGTGTCGGCGGACTTGGCCACATCGCAG TCCAATACGCAGCGGCGATGGGCATGCGTGTCATTGC GGTAGATATTGCCGATGACAAGCTGGAACTTGCCCGT AAGCACGGTGCGGAATTTACCGTGAATGCGCGTAATG AAGATTCAGGCGAAGCTGTACAGAAGTACACCAACGG TGGCGCACACGGCGTGCTTGTGACTGCAGTTCACGAG GCAGCATTCGGCCAGGCACTGGATATGGCTCGACGTG CAGGAACAATTGTGTTCAACGGTCTGCCACCGGGAGA GTTCCCAGCATCCGTGTTCAACATCGTATTCAAGGGC CTGACCATCCGTGGATCCCTCGTGGGAACCCGCCAAG ACTTGGCCGAAGCGCTCGATTTCTTTGCACGCGGACT AATCAAGCCAACCGTGAGTGAGTGCTCCCTCGATGAG GTCAATGGTGTGCTTGACCGCATGCGAAACGGCAAGA TCGATGGTCGTGTGGCGATTCGTTTCTAACATGCTAA GGTGCTGGCTGCATGCTAAGTTGATACGCCTGCGACA AATTTTTCTAGGAGCGTTAGTATGACTGTCTACGCAA ATCCAGGAACCGAAGGCTCGATCGTTAACTATGAAAA GCGCTACGAGAACTACATTGGTGGCAAGTGGGTTCCA CCGGTAGAGGGCCAGTACCTTGAGAACATTTCACCTG TCACTGGTGAAGTTTTCTGTGAGGTCGCACGTGGCAC CGCAGCGGACGTGGAGCTTGCACTGGATGCTGCACAT GCAGCCGCTGATGCGTGGGGCAAGACTTCTGTCGCTG AACGTGCTCTGATCCTGCACCGCATTGCGGACCGCAT GGAAGAGCACCTGGAAGAAATCGCAGTTGCAGAAACC TGGGAGAACGGCAAGGCAGTCCGTGAGACTCTTGCTG CAGATATCCCACTGGCAATCGACCACTTCCGCTACTT TGCTGGCGCGATCCGTGCTCAGGAAGATCGTTCCTCA CAGATCGACCACAACACTGTTGCTTACCACTTCAACG AGCCAATCGGTGTTGTTGGTCAGATCATTCCTTGGAA CTTCCCAATCCTCATGGCTACCTGGAAGCTCGCACCG GCACTTGCTGCAGGTAACGCGATCGTCATGAAGCCAG CTGAGCAGACCCCAGCATCCATTTTGTATCTGATTAA CATCATCGGCGATCTCATCCCAGAGGGCGTCCTCAAC ATCGTCAACGGACTCGGCGGTGAAGCAGGCGCTGCAC TGTCCGGCTCTAATCGGATTGGCAAGATTGCTTTCAC CGGTTCCACCGAGGTCGGCAAGCTGATCAACCGCGCT GCATCCGACAAGATCATTCCTGTCACCCTGGAGCTCG GCGGTAAGTCCCCATCCATCTTCTTCTCCGATGTTCT GTCACAGGATGACGCCTTCGCAGAGAAGGCAGTTGAA GGCTTCGCGATGTTCGCCCTCAATCAGGGTGAAGTTT GTACCTGTCCTTCCCGTGCACTTGTTCATGAGTCCAT CGCTGATGAATTCCTCGAGCTTGGCGTGAAGCGAGTT CAGAACATCAAGCTGGGTAACCCACTTGATACTGAAA CCATGATGGGTGCTCAGGCGTCCCAGGAGCAGATGGA CAAGATCTCCTCCTACCTGAAGATCGGCCCAGAAGAA GGCGCTCAAACCCTCACTGGTGGCAAGGTCAACAAGG TTGATGGCATGGAGAACGGTTACTACATTGAGCCAAC CGTTTTCCGCGGCACCAACGACATGAGGATCTTCCGC GAGGAAATCTTCGGACCAGTCCTTTCTGTTGCTACCT TCAGCGACTTCGATGAGGCCATCCGTATTGCAAACGA CACCAACTACGGCCTCGGCGCTGGTGTCTGGAGCCGT GACCAAAACACCATTTATCGTGCAGGTCGCGCAATCC AGGCTGGTCGAGTTTGGGTCAACCAGTACCACAACTA CCCAGCGCACTCCGCTTTCGGTGGATACAAGGAGTCC GGCATCGGCCGTGAGAACCACCTCATGATGCTGAACC ACTACCAGCAGACCAAGAACCTGTTGGTCTCCTACGA TCCAAACCCAACCGGACTGTTCTGAGTTTGTCGGTGA ACGCTCTCCTGAGTAGGACAAATCCGCCGGGAGCGGA TTTGAACGTTGCGAAGCAACGGCCCGGAGGGTGGCGG GCAGGACGCCCGCCATAAACTGCCAGGCATCAAATTA AGCAGAAGGCCATCCTGACGGATGGCCTTTTTGCGTT TCTACAAACTCTTTCGGTCCGTTGTTTATTTTTCTAA ATACATTCAAATATGTATCCGCTCATGAGACAATAAC CCTGATAAATGCTTCAATAATATTGAAAAAGGAAGAG TATGAGTATTCAACATTTCCGTGTCGCCCTTATTCCC

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			TTTTTTGCGGCATTTTGCCTTCCTGTTTTTGCTCACC CAGAAACGCTGGTGAAAGTAAAAGATGCTGAAGATCA GTTGGGTGCACGAGTGGGTTACATCGAACTGGATCTC AACAGCGGTAAGATCCTTGAGAGTTTTCGCCCCGAAG AACGTTTTCCAATGATGAGCACTTTTAAAGTTCTGCT ATGTGGCGCGGTATTATCCCGTGTTGACGCCGGGCAA GAGCAACTCGGTCGCCGCATACACTATTCTCAGAATG ACTTGGTTGAGTACTCACCAGTCACAGAAAAGCATCT TACGGATGGCATGACAGTAAGAGAATTATGCAGTGCT GCCATAACCATGAGTGATAACACTGCGGCCAACTTAC TTCTGACAACGATCGGAGGACCGAAGGAGCTAACCGC TTTTTTGCACAACATGGGGGATCATGTAACTCGCCTT GATCGTTGGGAACCGGAGCTGAATGAAGCCATACCAA ACGACGAGCGTGACACCACGATGCCTGTAGCAATGGC AACAACGTTGCGCAAACTATTAACTGGCGAACTACTT ACTCTAGCTTCCCGGCAACAATTAATAGACTGGATGG AGGCGGATAAAGTTGCAGGACCACTTCTGCGCTCGGC CCTTCCGGCTGGCTGGTTTATTGCTGATAAATCTGGA GCCGGTGAGCGTGGGTCTCGCGGTATCATTGCAGCAC TGGGGCCAGATGGTAAGCCCTCCCGTATCGTAGTTAT CTACACGACGGGGAGTCAGGCAACTATGGATGAACGA AATAGACAGATCGCTGAGATAGGTGCCTCACTGATTA AGCATTGGTAACTGTCAGACCAAGTTTACTCATATAT ACTTTAGATTGATTTCCTTAGGACTGAGCGTCAACCC CGTAGAAAAGATCAAAGGATCTTCTTGAGATCCTTTT TTTCTGCGCGTAATCTGCTGCTTGCAAACAAAAAAAC CACCGCTACCAGCGGTGGTTTGTTTGCCGGATCAAGA GCTACCAACTCTTTTTCCGAAGGTAACTGGCTTCAGC AGAGCGCAGATACCAAATACTGTCCTTCTAGTGTAGC CGTAGTTAGGCCACCACTTCAAGAACTCTGTAGCACC GCCTACATACCTCGCTCTGCTAATCCTGTTACCAGTG GCTGCTGCCAGTGGCGATAAGTCGTGTCTTACCGGGT TGGACTCAAGACGATAGTTACCGGATAAGGCGCAGCG GTCGGGCTGAACGGGGGGTTCGTGCACACAGCCCAGC TTGGAGCGAACGACCTACACCGAACTGAGATACCTAC AGCGTGAGCTATGAGAAAGCGCCACGCTTCCCGAAGG GAGAAAGGCGGACAGGTATCCGGTAAGCGGCAGGGTC GGAACAGGAGAGCGCACGAGGGAGCTTCCAGGGGGAA ACGCCTGGTATCTTTATAGTCCTGTCGGGTTTCGCCA CCTCTGACTTGAGCGTCGATTTTTGTGATGCTCGTCA GGGGGGCGGAGCCTATGGAAAAACGCCAGCAACGCGG CCTTTTTACGGTTCCTGGCCTTTTGCTGGCCTTTTGC TCACATGTTCTTTCCTGCGTTATCCCCTGATTCTGTG GATAACCGTATTACCGCCTTTGAGTGAGCTGATACCG CTCGCCGCAGCCGAACGACCGAGCGCAGCGAGTCAGT GAGCGAGGAAGCGGAAGAGCGCCTGATGCGGTATTTT CTCCTTACGCATCTGTGCGGTATTTCACACCGCATAT AAGGTGCACTGTGACTGGGTCATGGCTGCGCCCCGAC ACCCGCCAACACCCGCTGACGCGCCCTGACGGGCTTG TCTGCTCCCGGCATCCGCTTACAGACAAGCTGTGACC GTCTCCGGGAGCTGCATGTGTCAGAGGTTTTCACCGT CATCACCGAAACGCGCGAGGCAGCTGCGGTAAAGCTC ATCAGCGTGGTCGTGCAGCGATTCACAGATGTCTGCC TGTTCATCCGCGTCCAGCTCGTTGAGTTTCTCCAGAA GCGTTAATGTCTGGCTTCTGATAAAGCGGGCCATGTT AAGGGCGGTTTTTTCCTGTTTGGTCACTGATGCCTCC GTGTAAGGGGGATTTCTGTTCATGGGGGTAATGATAC CGATGAAACGAGAGAGGATGCTCACGATACGGGTTAC TGATGATGAACATGCCCGGTTACTGGAACGTTGTGAG GGTAAACAACTGGCGGTATGGATGCGGCGGGACCAGA GAAAAATCACTCAGGGTCAATGCCAGCGCTTCGTTAA TACAGATGTAGGTGTTCCACAGGGTAGCCAGCAGCAT CCTGCGATGCAGATCCGGAACATAATGGTGCAGGGCG CTGACTTCCGCGTTTCCAGACTTTACGAAACACGGAA ACCGAAGACCATTCATGTTGTTGCTCAGGTCGCAGAC GTTTTGCAGCAGCAGTCGCTTCACGTTCGCTCGCGTA TCGGTGATTCATTCTGCTAACCAGTAAGGCAACCCCG CCAGCCTAGCCGGGTCCTCAACGACAGGAGCACGATC ATGCGCACCCGTGGCCAGGACCCAACGCTGCCCGAAA TT
47.	pTRIM20	site 1 (IS7)	GCCATATTCAACGGGAAACGTCTTGCTCGAGGCCGCG ATTAAATTCCAACATGGATGCTGATTTATATGGGTAT

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			AAATGGGCTCGCGATAATGTCGGGCAATCAGGTGCGA CAATCTATCGATTGTATGGGAAGCCCGATGCGCCAGA GTTGTTTCTGAAACATGGCAAAGGTAGCGTTGCCAAT GATGTTACAGATGAGATGGTCAGACTAAACTGGCTGA CGGAATTTATGCCTCTTCCGACCATCAAGCATTTTAT CCGTACTCCTGATGATGCATGGTTACTCACCACTGCG ATCCCCGGGAAAACAGCATTCCAGGTATTAGAAGAAT ATCCTGATTCAGGTGAAAATATTGTTGATGCGCTGGC AGTGTTCCTGCGCCGGTTGCATTCGATTCCTGTTTGT AATTGTCCTTTTAACAGCGATCGCGTATTTCGTCTCG CTCAGGCGCAATCACGAATGAATAACGGTTTGGTTGA TGCGAGTGATTTTGATGACGAGCGTAATGGCTGGCCT GTTGAACAAGTCTGGAAAGAAATGCATAAGCTTTTGC CATTCTCACCGGATTCAGTCGTCACTCATGGTGATTT CTCACTTGATAACCTTATTTTTGACGAGGGGAAATTA ATAGGTTGTATTGATGTTGGACGAGTCGGAATCGCAG ACCGATACCAGGATCTTGCCATCCTATGGAACTGCCT CGGTGAGTTTTCTCCTTCATTACAGAAACGGCTTTTT CAAAAATATGGTATTGATAATCCTGATATGAATAAAT TGCAGTTTCATTTGATGCTCGATGAGTTTTTCTAATA AAAAATGCCCTCTTGGGTTATCAAGAGGGTCATTATA TTTCGCAAAAAACCCCGCTTCGGCGGGGTTTTTTCGC TTGACGGCTAGCTCAGTCCTAGGTACAGTGCTAGCAA TTAAAGGAGGCCTAATGACCACTGCTGCACCCCAAGA ATTTACTGCTGCTGTTGTTGAAAAATTCGGTCATGAC GTGACCGTGAAGGATATTGACCTTCCAAAGCCAGGGC CACACCAGGCATTGGTGAAGGTACTCACCTCCGGCAT CTGCCACACCGACCTCCACGCCTTGGAGGGCGATTGG CCAGTAAAGCCGGAACCACCATTCGTACCAGGACACG AAGGTGTAGGTGAAGTTGTTGAGCTCGGACCAGGTGA ACACGATGTGAAGGTCGGCGATATTGTCGGCAATGCG TGGCTCTGGTCAGCGTGTGGCACCTGCGAATACTGCA TCACCGGCAGGGAAACTCAGTGCAACGAAGCTGAGTA TGGTGGCTACACCCAAAATGGATCCTTCGGCCAGTAC ATGCTGGTGGATACCCGTTACGCCGCTCGCATCCCAG ACGGCGTGGACTACCTCGAAGCAGCACCAATTCTGTG TGCAGGCGTGACTGTCTACAAGGCACTCAAAGTCTCT GAAACCCGCCCGGGCCAATTCATGGTGATCTCCGGTG TCGGCGGACTTGGCCACATCGCAGTCCAATACGCAGC GGCGATGGGCATGCGTGTCATTGCGGTAGATATTGCC GATGACAAGCTGGAACTTGCCCGTAAGCACGGTGCGG AATTTACCGTGAATGCGCGTAATGAAGATTCAGGCGA AGCTGTACAGAAGTACACCAACGGTGGCGCACACGGC GTGCTTGTGACTGCAGTTCACGAGGCAGCATTCGGCC AGGCACTGGATATGGCTCGACGTGCAGGAACAATTGT GTTCAACGGTCTGCCACCGGGAGAGTTCCCAGCATCC GTGTTCAACATCGTATTCAAGGGCCTGACCATCCGTG GATCCCTCGTGGGAACCCGCCAAGACTTGGCCGAAGC GCTCGATTTCTTTGCACGCGGACTAATCAAGCCAACC GTGAGTGAGTGCTCCCTCGATGAGGTCAATGGTGTGC TTGACCGCATGCGAAACGGCAAGATTGATGGTCGTGT GGCAATTCGCTACTAAGCATGGACGAGCTGTACAAGT AAGACTGCTAAAGCGTCAAAAGGCCGGATTTTCCGGC CTTTTTTATTACTAAACGCAAAAAGGCCATCCGTCAG GATGGCCTTCTGCTTAATTTGATGCCTGGCAGTTTAT GGCGGGCGTCCTGCCCGCCACCCTCCGGGCCGTTGCT TCGCAACGTTCAAATCCGCTCCCGGCGGATTTGTCCT ACTCAGGAGAGCGCTCCTAAGTCATTTAGATATTGGC TGGGATGTTCAGGGTGATTTCGTTAATATGGCGTTGG GAGTGTAAAAGTTGAGCGGTCCATGCGCAGGCGAACT CCAAACCCTCCAGCGAGCGGTTTGCAAAACGTACGAT TTGGTTTGCCCATACGGGCGGCAGAACATGGTCGCCC GCGGCAGACGCGCGTTGATAGTCCAGCTCGGCTTCGT GGCGCCAAACACGAATTAATTGACCGATGGCTGCCGA CTGAATGCGGCCATAAACATTTCCGTTCTGATCTGCC CCATTAGACAGAAAGATCACACGTGGTCCGCGGGCGC GCGCCCCAGGCGTCAGGCGTTGTGACTGCCAGTAGCG AGCAAGACGGCTAGCAATGACAATCGTCCCGGTGATT TCGTCCGCAAGAAAGTTCAGCACACGTTCGTCATCTA CTTCAATTACACACGGTGCGGGCTCATGGGATGTCGC GGGCAGGATAACCGCCGCATGAATACCCCCAGTGTTT TCTCCAGCTCCCCAATCGAATACAGCGTCGATCGTAG CAGGGTCACGCGGATCAAGAGGTAAGGCGATCGGCGG

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			GGTAAAGTCTGCGCCAGCTAAACGACGCGACTCATTT ACTGCCTGTTCAAACTGTGCTAATGCGGCGGCGCTAC GAAAACCAATAATTACCTCCGATCCGCATGTGCGCAG CATACCCGTTAAAGCCATTACCTCCTCAATCTGATCC CCGGCACAGATAAGGGTGGTACGGCCAGAGGCATCGA TAGTGCGCAGATCAGTGCGCGCTAACAGACTCGTTTC CGAACAGGCCGGCAGCTCCATTCCATGGGTCACCTCA ATCGTTTCCCCAGATAATGCTGCAGACTCAGCAGAAG CCAAGAAGACCGCGGCATCGGCGACATCGCCGACGGA AGGGAAGCGGCGTTCTAATGCCCCTTGGTCGTTTGCG CGACACAGGCGCATCGTGTTCAAGAAATGGTGCGCCG TGTCACCCTCAGGACGGCCCTTCAGCTGATCCATACG TTGAAATACAGTGCGAATGCGGTCCGATTCGATGGGA CCCGGAAAGATTGTATTTACGCGAATTCCACGCGCGC CCAGTTCGCGCGCAGCCAGCTGGCTTAACGCATTCAG TGCGGCTTTGGGGGTCACATATGGGATACGTCCGTAG TATTCGGCACGAGAAAAGATGGTGGACACATTGATAA CAGCTGATCCAACGGGCATGTGCGGGGCCGCAATGCG CATCAGATGCCAGCCCATTCCAAGAAGATTGGCGATA CTAGCGTGCAATGTTTCCTCTGCTCCGGGACCCAATT CTGCTTCAGTTAAAGGGATTTCCGCTAAACGACGCTG AGCACCAGCGCTACCGGCGTTGTTGACAAGAATGTCA ATCTGGCCGTGGCGAGCAACAATAGCCTCAATGCCTG CGCGCACGGCAACTGGGTCGCTTCCGTCCATAACTTC CAAGTCAATGCGTTTAGCGGGGACGCCGGCCTCAGCT TGCATACGCTCGGCCAGCGCAGTCAGCTTAGCACGAT TACGTCCAGAGATAATGACTGTCGCACCCTCTGCCAG GAAGCGACGGGTTAATTCGGAGCCAATATTGCCGGCA CCTCCCGTAATCAAGGCAATTTTTCCCGCCAAGCGCC CTGTTCCTGACATTGAGTACCTCCTTTCTTAATTATA CCGTAATCGCGCGACCGCGGTGGATACGTCCGGCATA ACGAGTGTCCGCTTCGGGTGGCAACGGTGCGGTAACT AACAGCACGGCTTCGATGAAGCGCTCCAATTCCTGTT GACGTTCGTGAGGGTCACGCGGCTCCTCTGCGCGAGC ACGACGAGTTAAATCGACTTGATTGATCAGGATGCGC TGCGCTGTGCGTTCAGACTCTACACCGATCGTTGCGG TGAAAGCATGAAGCGTTGTCTTAATGAAGTTGGCCAA GGCGAACTGCTCGGTTGTGCTAGTTGCCGTAGTCTCG GGCGTGACTAAGGCAAGACGTGCACCGTCAGACAGGG CGATCCAACGTGCTACACGGAAGTGATGCGTTAACTG ATGCTCACATAACTCAGCGAATTCTGCCTCGCTCAAG ACGGTAGACCAATCGCTGTCCTTGCGGCCTACAAGAG GAACAGTGGGCAATGGACGAAAAGGGGTACAGACTAC GGGACCTGGACGTCCGTAACGAGTAATCGCTTGGTCA ATCGCGGCCTCGATCTGGTCACCAGCCACAATCGTCA TCAAGCGTCCTGCCTCTACGTGGTCGTGCAGTAAACG ACGCATGGTTTCAGCTCCAGTTTCCGTTTCTACGATC ATAACTACCTGGCGTGCCCCGTAGCGCTCTAAGTAAG CGCGTGCAAGCAAGTTTAAATGCTCAGTTAAATGTTC ACCGATCAGATAGACTGTGGAGCCGACAAGCTCTGCT AAACGTTCCGGCGAAGGCAGGCCAAACAATTCTCCCC CTGTCGGGGTGCGTTCGTAGCGCAAACCTCCAGACGG ATGGAAAGTCTCGCCTGAGACAACACGGTCCGCCAAA TAATAAACCGTCGCCATCGCAACATCAAATTCCGTAG GCATGCGCTGAAGATAAAGCATTCCCATAATTCCATC GCGTACCTTACGAGCCTCGCGGTCGATCTGGGCGCGG GTGAAAAAAGGGTCTGGTGGGTTCGGCAAATTGGCAA AAATGTCTGCTGGCAGAACATATCCACCATTGTGCAA ACGAGCAAGTAATTTCGCCGCAATGGAGCGATTTAAT AAAGCACTGGACGAAGACGCTGCAGGGTCTCCCTCAC TGCGGAAGCGACGCGCCAGCTCACGAAGTGCGGTAGG GGCAGCGGGGTTTTGTTCAAGCGCCGCGACGTCATTC GGCAACAACAATTCAACCAACTCATGCATCGAACGTT CATCGGTACGAGCTGCCGCGATTAACGCAGCATGAAG CTCATTTAAACGCTTGTTCTCCAAGATAAGGCGTGCA CGGCGAGCGAACAATCCAGGACGCTCACCAGTTCCGC GCAGACGATCTCCCTCTACGGGTCCAGGTGCAATAGC ATTGATCTGAATCTCGGGTCCCAAAAAGCGCGCAAAG ACTTCAGCCATAGCGCGCTGGCCGGCTTTACTAACAG CATAGTCGGCACGGTTGGGATACGGAATTGCTGCGTC TTTCTCCCCTCCGAAATACGAGCTTACGTTGAGGATG TACCCCGACCCCTGCTTCTTCATCAGTGGTGCCAGCT TACGCATAAGGCTGTAGTTCGAGATGAGGTTTGCGAA

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			CAGCGTATGACGCCAACCCTCCACGGGCATGTCGATA ACCATCTCCTCTACACCCGCAATGCCGGCGTTATTGA TTAAATAGTCAACGGTACCGAAAGCTGACAATGTGCG TTCAACCAAGTCTGCAAGTTGGGCCTCACTTGAAACG TCGCACCCTGGTGCGATATGAACGCGATCCTCCACAT CTGTGTAGCCAACTTCCGCTAATTCACTCTGAATCAT AGCTTGCATCTGTTCCAATTTGTGGCGATCACGTGCC GCCAGCATCACACGTGCACCAGATAAAGCCAATAAGC GACCAATCTGCCCACCAATGCCGGCAGATCCACCCGT AATCAGGGCGACCTTGCCCAAATGCAACCCAATCAGT GATTCTGCCCATCCAACAGAGGCGCTACGTGCGCCCG TCGTCGCCGACATTAATACCTCCTGTCCTTGTTGGTC AATTGCTCGTGAAAACACCTAAACGGACTGGCTGTTT CCTGTGTGAGCTAGCATTATACCTAGGACTGAGCTAG CTGTCAATTAATAAGATGATCTTCTTGAGATCGTTTT GGTCTGCGCGTAATCTCTTGCTCTGAAAACGAAAAAA CCGCCTTGCAGGGCGGTTTTTCGAAGGTTCTCTGAGC TACCAACTCTTTGAACCGAGGTAACTGGCTTGGAGGA GCGCAGTCACCAAAACTTGTCCTTTCAGTTTAGCCTT AACCGGCGCATGACTTCAAGACTAACTCCTCTAAATC AATTACCAGTGGCTGCTGCCAGTGGTGCTTTTGCATG TCTTTCCGGGTTGGACTCAAGACGATAGTTACCGGAT AAGGCGCAGCGGTCGGACTGAACGGGGGGTTCGTGCA TACAGTCCAGCTTGGAGCGAACTGCCTACCCGGAACT GAGTGTCAGGCGTGGAATGAGACAAACGCGGCCATAA CAGCGGAATGACACCGGTAAACCGAAAGGCAGGAACA GGAGAGCGCACGAGGGAGCCGCCAGGGGGAAACGCCT GGTATCTTTATAGTCCTGTCGGGTTTCGCCACCACTG ATTTGAGCGTCAGATTTCGTGATGCTTGTCAGGGGGG CGGAGCCTATGGAAAAACGGCTTTGCCGCGGCCCTCT CACTTCCCTGTTAAGTATCTTCCTGGCATCTTCCAGG AAATCTCCGCCCCGTTCGTAAGCCATTTCCGCTCGCC GCAGTCGAACGACCGAGCGTAGCGAGTCAGTGAGCGA GGAAGCGGAATATATCCTGTATCACATATTCTGCTGA CGCACCGGTGCAGCCTTTTTTCTCCTGCCACATGAAG CACTTCACTGACACCCTCATCAGTGCCAACATAGTAA GCCAGTATACACTCCGCTAGCGCTGAGGTCTGCCTCG TGAAGAAGGTGTTGCTGACTCATACCAGGCCTGAATC GCCCCATCATCCAGCCAGAAAGTGAGGGAGCCACGGT TGATGAGAGCTTTGTTGTAGGTGGACCAGTTGGTGAT TTTGAACTTTTGCTTTGCCACGGAACGGTCTGCGTTG TCGGGAAGATGCGTGATCTGATCCTTCAACTCAGCAA AAGTTCGATTTATTCAACAAAGCCACGTTGTGTCTCA AAATCTCTGATGTTACATTGCACAAGATAAAAATATA TCATCATGAACAATAAAACTGTCTGCTTACATAAACA GTAATACAAGGGGTGTTATGA
48.	IMMO		GAACCCCCAGGATGCGCATAGCGCTTATATTCGCGGT GACGTGGAGTTGGTGCGGATTCGTGATGCCGAAGGGC GAATTGCGGCAGAAGGGGCGTTGCCTTATCCACCTGG CGTGCTTTGCGTGGTACCCGGGGAAGTCTGGGGTGGG GCGGTTCAACGTTATTTCCTTGCACTGGAAGAAGGGG TGAATTTGTTGCCGGGATTTTCGCCGGAGCTGCAAGG TGTTTATAGCGAAACCGATGCGGATGGCGTGAAACGG TTGTACGGTTATGTGTTGAAGTAAGAATAAAAAAAAC GGGTCACCTTCTGGCGACCCGTTTTTCTTTGCGTAAT TAGTGGCTAACCGTCTGTGTGCCTGTCGGGACACGAA CGTGTTTATATTTGAACATCGCCATGAACGCGAAGGC CAGAACCACGGAGTAACCAGCGAAAATCAACCATACG GTCTGCCAGTCGGTAATGCCGTTTTGGGTGTACATCT CAACAACTTTACCGCTCACGATGCCGCCGAGGATACA GCCGAAGCCGTTAGTCATCATCAGGAACATCCCTTGT GCACTGGCGCGAATTGCCGGGCTAACTTCTTTTTCGA CAAACACCGAACCAGAGATGTTGAAGAAGTCGAATGC GCAACCGTAAACGATCATCGACAGTACCAGCAGTACA GTACCTCACACTGCTTCCGGTAGTCAATAAACCGGTA AACCAGCAATAGACATAAGCGGTGCATAATGTGCCTG TCAAATGGACGAAGCAGGGATTCTGCAAACCCTATGC TACTCCGTCAAGCCGTCAATTGTCTGATTCGTTACCA ATTATGACAACTTGACGGCTACATCATTCACTTTTTC TTCACAACCGGCACGGAACTCGCTCGGGCTGGCCCCG GTGCATTTTTTAAATACCCGCGAGAAATAGAGTTGAT CGTCAAAACCAACATTGCGACCGACGGTGGCGATAGG

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			CATCCGGGTGGTGCTCAAAAGCAGCTTCGCCTGGCTG
			ATACGTTGGTCCTCGCGCCAGCTTAAGACGCTAATCC
			CTAACTGCTGGCGGAAAAGATGTGACAGACGCGACGG
			CGACAAGCAAACATGCTGTGCGACGCTGGCGATATCA
			AAATTGCTGTCTGCCAGGTGATCGCTGATGTACTGAC
			AAGCCTCGCGTACCCGATTATCCATCGGTGGATGGAG
			CGACTCGTTAATCGCTTCCATGCGCCGCAGTAACAAT
			TGCTCAAGCAGATTTATCGCCAGCAGCTCCGAATAGC
			GCCCTTCCCCTTGCCCGGCGTTAATGATTTGCCCAAA
			CAGGTCGCTGAAATGCGGCTGGTGCGCTTCATCCGGG
			CGAAAGAACCCCGTATTGGCAAATATTGACGGCCAGT
			TAAGCCATTCATGCCAGTAGGCGCGCGGACGAAAGTA
			AACCCACTGGTGATACCATTCGCGAGCCTCCGGATGA
			CGACCGTAGTGATGAATCTCTCCTGGCGGGAACAGCA
			AAATATCACCCGGTCGGCAAACAAATTCTCGTCCCTG
			ATTTTTCACCACCCCCTGACCGCGAATGGTGAGATTG
			AGAATATAACCTTTCATTCCCAGCGGTCGGTCGATAA
			AAAAATCGAGATAACCGTTGGCCTCAATCGGCGTTAA
			ACCCGCCACCAGATGGGCATTAAACGAGTATCCCGGC
			AGCAGGGGATCATTTTGCGCTTCAGCCATACTTTTCA
			TACTCCCGCCATTCAGAGAAGAAACCAATTGTCCATA
			TTGCATCAGACATTGCCGTCACTGCGTCTTTTACTGG
			CTCTTCTCGCTAACCAAACCGGTAACCCCGCTTATTA
			AAAGCATTCTGTAACAAAGCGGGACCAAAGCCATGAC
			AAAAACGCGTAACAAAAGTGTCTATAATCACGGCAGA
			AAAGTCCACATTGATTATTTGCACGGCGTCACACTTT
			GCTATGCCATAGCATTTTTATCCATAAGATTAGCGGA
			TCCTACCTGACGCTTTTTATCGCAACTCTGGACAATG
			TCTCCATACCCGTTTTTTTGGGCGACCTCGTCGGAGG
			TTGTATGTCCGGTGTTCCGTGACGTCATCGGGCATTC
			ATCATTCATAGAATGTGTTACGGAGGAAACAAGTAAT
			GGCACTTAGCACCGCAACCAAGGCCGCGACGGACGCG
			CTGGCTGCCAATCGGGCACCCACCAGCGGGAATGCAC
			AGGAAGTGCACCGTTGGCTCCAGAGCTTCAACTGGGA
			TTTCAAGAACAACCGGACCAAGTACGCCACCAAGTAC
			AAGATGGCGAACGAGACCAAGGAACAGTTCAAGCTGA
			TCGCCAAGGAATATGCGCGCATGGAGGCAGTCAAGGA
			CGAAAGGCAGTTCGGTAGCCTGCAGGATGCGCTGACC
			CGCCTCAACGCCGGTGTTCGCGTTCATCCGAAGTGGA
			ACGAGACCATGAAAGTGGTTTCGAACTTCCTGGAAGT
			GGGCGAATACAACGCCATCGCCGCTACCGGGATGCTG
			TGGGATTCCGCCCAGGCGGCGGAACAGAAGAACGGCT
			ATCTGGCCCAGGTGTTGGATGAAATCCGCCACACCCA
			CCAGTGTGCCTACGTCAACTACTACTTCGCGAAGAAC
			GGCCAGGACCCGGCCGGTCACAACGATGCTCGCCGCA
			CCCGTACCATCGGTCCGCTGTGGAAGGGCATGAAGCG
			CGTGTTTTCCGACGGCTTCATTTCCGGCGACGCCGTG
			GAATGCTCCCTCAACCTGCAGCTGGTGGGTGAGGCCT
			GCTTCACCAATCCGCTGATCGTCGCAGTGACCGAATG
			GGCTGCCGCCAACGGCGATGAAATCACCCCGACGGTG
			TTCCTGTCGATCGAGACCGACGAACTGCGCCACATGG
			CCAACGGTTACCAGACCGTCGTTTCCATCGCCAACGA
			TCCGGCTTCCGCCAAGTATCTCAACACGGACCTGAAC
			AACGCCTTCTGGACCCAGCAGAAGTACTTCACGCCGG
			TGTTGGGCATGCTGTTCGAGTATGGCTCCAAGTTCAA
			GGTCGAGCCGTGGGTCAAGACGTGGAACCGCTGGGTG
			TACGAGGACTGGGGCGGCATCTGGATCGGCCGTCTGG
			GCAAGTACGGGGTGGAGTCGCCGCGCAGCCTCAAGGA
			CGCCAAGCAGGACGCTTACTGGGCTCACCACGACCTG
			TATCTGCTGGCTTATGCGCTGTGGCCGGGCGGCTTCT
			TCCGTCTGGCGCTGCCGGATCAGGAAGAAATGGAGTG
			GTTCGAGGCCAACTACCCCGGCTGGTACGACCACTAC
			GGCAAGATCTACGAGGAATGGCGCGCCCGCGGTTGCG
			AGGATCCGTCCTCGGGCTTCATCCCGCTGATGTGGTT
			CATCGAAAACAACCATCCCATCTACATCGATCGCGTG
			TCGCAAGTGCCGTTCTGCCCGAGCTTGGCCAAGGGCG
			CCAGCACCCTGCGCGTGCACGAGTACAACGGCCAGAT
			GCACACCTTCAGCGACCAGTGGGGCGAGCGCATGTGG
			CTGGCCGAGCCGGAGCGCTACGAGTGCCAGAACATCT
			TCGAACAGTACGAAGGACGCGAACTGTCGGAAGTGAT
			CGCCGAACTGCACGGGCTGCGCAGTGATGGCAAGACC
			CTGATCGCCCAGCCGCATGTCCGTGGCGACAAGCTGT
			GGACGTTGGACGATATCAAACGCCTGAACTGCGTCTT

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			CAAGAACCCGGTGAAGGCATTCAATTGAAACGGGTGT CGGGCTCCGTCACAGGGGGGGGCCCGACGCACGATCG TTCGATCAACCTCAAACCAAAAAGGAACATCGATATG AGCATGTTAGGAGAAAGACGCCGCGGTCTGACCGATC CGGAAATGGCGGCCGTCATTTTGAAGGCGCTTCCTGA AGCTCCGCTGGACGGCAACAACAAGATGGGTTATTTC GTCACCCCCCGCTGGAAACGCTTGACGGAATATGAAG CCCTGACCGTTTATGCGCAGCCCAACGCCGACTGGAT CGCCGGCGGCCTGGACTGGGGCGACTGGACCCAGAAA TTCCACGGCGGCCGCCCTTCCTGGGGCAACGAGACCA CGGAGCTGCGCACCGTCGACTGGTTCAAGCACCGTGA CCCGCTCCGCCGTTGGCATGCGCCGTACGTCAAGGAC AAGGCCGAGGAATGGCGCTACACCGACCGCTTCCTGC AGGGTTACTCCGCCGACGGTCAGATCCGGGCGATGAA CCCGACCTGGCGGGACGAGTTCATCAACCGGTATTGG GGCGCCTTCCTGTTCAACGAATACGGATTGTTCAACG CTCATTCGCAGGGCGCCCGGGAGGCGCTGTCGGACGT AACCCGCGTCAGCCTGGCTTTCTGGGGCTTCGACAAG ATCGACATCGCCCAGATGATCCAACTCGAACGGGGTT TCCTCGCCAAGATCGTACCCGGTTTCGACGAGTCCAC AGCGGTGCCGAAGGCCGAATGGACGAACGGGGAGGTC TACAAGAGCGCCCGTCTGGCCGTGGAAGGGCTGTGGC AGGAGGTGTTCGACTGGAACGAGAGCGCTTTCTCGGT GCACGCCGTCTATGACGCGCTGTTCGGTCAGTTCGTC CGCCGCGAGTTCTTTCAGCGGCTGGCTCCCCGCTTCG GCGACAATCTGACGCCATTCTTCATCAACCAGGCCCA GACATACTTCCAGATCGCCAAGCAGGGCGTACAGGAT CTGTATTACAACTGTCTGGGTGACGATCCGGAGTTCA GCGATTACAACCGTACCGTGATGCGCAACTGGACCGG CAAGTGGCTGGAGCCCACGATCGCCGCTCTGCGCGAC TTCATGGGGCTGTTTGCGAAGCTGCCGGCGGGCACCA CTGACAAGGAAGAAATCACCGCGTCCCTGTACCGGGT GGTCGACGACTGGATCGAGGACTACGCCAGCAGGATC GACTTCAAGGCGGACCGCGATCAGATCGTTAAAGCGG TTCTGGCAGGATTGAAATAATAGAGGAACTATTACGA TGAGCGTAAACAGCAACGCATACGACGCCGGCATCAT GGGCCTGAAAGGCAAGGACTTCGCCGATCAGTTCTTT GCCGACGAAAACCAAGTGGTCCATGAAAGCGACACGG TCGTTCTGGTCCTCAAGAAGTCGGACGAGATCAATAC CTTTATCGAGGAGATCCTTCTGACGGACTACAAGAAG AACGTCAATCCGACGGTAAACGTGGAAGACCGCGCGG GTTACTGGTGGATCAAGGCCAACGGCAAGATCGAGGT CGATTGCGACGAGATTTCCGAGCTGTTGGGGCGGCAG TTCAACGTCTACGACTTCCTCGTCGACGTTTCCTCCA CCATCGGCCGGGCCTATACCCTGGGCAACAAGTTCAC CATTACCAGTGAGCTGATGGGCCTGGACCGCAAGCTC GAAGACTCTCACGCTTAAGGAGAATGACATGGCGAAA CTGGGTATACACAGCAACGACACCCGCGACGCCTGGG TGAACAAGATCGCGCAGCTCAACACCCTGGAAAAAGC GGCCGAGATGCTGAAGCAGTTCCGGATGGACCACACC ACGCCGTTCCGCAACAGCTACGAACTGGACAACGACT ACCTCTGGATCGAGGCCAAGCTCGAAGAGAAGGTCGC CGTCCTCAAGGCAGAAGCCTTCAACGAGGTGGACTTC CGTCATAAGACCGCTTTCGGCGAGGATGCCAAGTCCG TTCTGGACGGCACCGTCGCGAAGATGAACGCGGCCAA GGACAAGTGGGAGGCGGAGAAGATCCATATCGGTTTC CGCCAGGCCTACAAGCCGCCGATCATGCCGGTGAACT ATTTCCTGGACGGCGAGCGTCAGTTGGGGACCCGGCT GATGGAACTGCGCAACCTCAACTACTACGACACGCCG CTGGAAGAACTGCGCAAACAGCGCGGTGTGCGGGTGG TGCATCTGCAGTCGCCGCACTGAAGGGAGGAAGTCTC GCCCTGGACGCGACGGCATCGCCGTGAAGTCCAGGGG GCAGGGATGCCGTTCCGGGCCGGCAGGCTGGCCCGGA ATCTCTGGTTTTCAGGGGGCGTGCCGGTCCACGGCTC CCCCCTCCATCTTTCGTAAGGAAATCACCATGGTCGA ATCGGCATTTCAGCCATTTTCGGGCGACGCAGACGAA TGGTTCGAGGAACCACGGCCCCAGGCCGGTTTCTTCC CTTCCGCGGACTGGCATCTGCTCAAACGGGACGAGAC CTACGCAGCCTATGCCAAGGATCTCGATTTCATGTGG CGGTGGGTCATCGTCCGGGAAGAAAGGATCGTCCAGG AGGGTTGCTCGATCAGCCTGGAGTCGTCGATCCGCGC CGTGACGCACGTACTGAATTATTTTGGTATGACCGAA CAACGCGCCCCGGCAGAGGACCGGACCGGCGGAGTTC

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
			AACATTGAACAGGTAAGTTTATGCAGCGAGTTCACAC TATCACGGCGGTGACGGAGGATGGCGAATCGCTCCGC TTCGAATGCCGTTCGGACGAGGACGTCATCACCGCCG CCCTGCGCCAGAACATCTTTCTGATGTCGTCCTGCCG GGAGGGCGGCTGTGCGACCTGCAAGGCCTTGTGCAGC GAAGGGGACTACGACCTCAAGGGCTGCAGCGTTCAGG CGCTGCCGCCGGAAGAGGAGGAGGAAGGGTTGGTGTT GTTGTGCCGGACCTACCCGAAGACCGACCTGGAAATC GAACTGCCCTATACCCATTGCCGCATCAGTTTTGGTG AGGTCGGCAGTTTCGAGGCGGAGGTCGTCGGCCTCAA CTGGGTTTCGAGCAACACCGTCCAGTTTCTTTTGCAG AAGCGGCCCGACGAGTGCGGCAACCGTGGCGTGAAAT TCGAACCCGGTCAGTTCATGGACCTGACCATCCCCGG CACCGATGTCTCCCGCTCCTACTCGCCGGCGAACCTT CCTAATCCCGAAGGCCGCCTGGAGTTCCTGATCCGCG TGTTACCGGAGGGACGGTTTTCGGACTACCTGCGCAA TGACGCGCGTGTCGGACAGGTCCTCTCGGTCAAAGGG CCACTGGGCGTGTTCGGTCTCAAGGAGCGGGGCATGG CGCCGCGCTATTTCGTGGCCGGCGGCACCGGGTTGGC GCCGGTGGTCTCGATGGTGCGGCAGATGCAGGAGTGG ACCGCGCCGAACGAGACCCGCATCTATTTCGGTGTGA ACACCGAGCCGGAATTGTTCTACATCGACGAGCTCAA ATCCCTGGAACGATCGATGCGCAATCTCACCGTGAAG GCCTGTGTCTGGCACCCGAGCGGGGACTGGGAAGGCG AGCAGGGCTCGCCCATCGATGCGTTGCGGGAAGACCT GGAGTCCTCCGACGCCAACCCGGACATTTATTTGTGC GGTCCGCCGGGCATGATCGATGCCGCCTGCGAGCTGG TACGCAGCCGCGGTATCCCCGGCGAACAGGTCTTCTT CGAAAAATTCCTGCCGTCCGGGGCGGCCTGAACCGGG GAAGTACCGTGACCACCGAGCAGTTCCCGCCCCAATT CCTGCGTGAAATGATCGAGCAGCTGGACGCCAGCATC CAGGAGCTCGCACGCAAGGAAAAGGGACTTGCGGCAT CCCTGGGCACGGGCCGGGTCGCCGAGCTCAAGGAATA CTGGGACCACGTAGTCGGGTCGCCGTAAGCAAACAGC GCAAAACGCAGGATCCACGCCACAATACTGATCATCA TTACGTTCTTAATACCGTAGCGGCTTAAGAAGAACGG GATGGTCAGAATGAACAGGGTTTCAGAGATCTGCGAA ATCGACATGATGATTGACGCATGCTGCACAATAAAGC TGCTGGCAAACATCGGATCTTTGTCGAAGCTGTGCAG GAAGGTATTACCGAACATGTTGGTAATCTGCAGTTCC GCGCCCAGCAGCATTGAGAAGATAAAGAAGATTGCCA TACGCTTGTTTTTAAACAGCGCGAATGCATCGAGGCC CAGCAGGGTTGTCCAGCTCTGATTCGCTTGCTGTTTA GCAACCGGAATATGCGGCAGAGTCAGGGTAAACAGAA CCAGAATGGCGGAAAGTGCTGCGCCAATATACAGCTG CATGTGGCTTAATTCGAAGCCAGACAGGCTCACCACC CACATTGCCATGATAAAGCCGATGGTGCCCCAGATAC GGATTGGC
49.	IMMO	site 2 (SS9)	TTTTCTCAAGTTCTGCCAGCTTCTGCTGGGTTTCTTT GTCTTGTGCAGGATGCTGCATCACCGGCATCATGTTG TCGGCAATAGTGGTCGCCGGTTTAACCAGATACTGGT TTGGGTGATCGTATCCGGCAAAGCCTTTGCGTGCGGT GGCGGTTGACGGGGTATCTGCTGCGTTGGCCATGAGT GGAAGAGCGGCGGCGACAGCTACAACAAGAAGTTTGG GTGAAAACGAAAATTCCATGCAAAATGCTCCGGTTTC ATGTCGTCAAAATGTTGACGTAATTAAGCATTGATAA TTGATAATTGAGATCCCTCTCCCTGACAGGATGATTG CATAAATAATAGTGATGAAAATAAATTATTTATTTAT CCAGAAAATGAATTGGAAAATCAGGAGAGCGTTTTCA ATCCTACCCCTAACTGACTATGGCCTTCTGTAGTCAC ACTGCTTCCGGTAGTCAATAAACCGGTAAACCAGCAA TAGACATAAGCGGTGCATAATGTGCCTGTCAAATGGA CGAAGCAGGGATTCTGCAAACCCTATGCTACTCCGTC AAGCCGTCAATTGTCTGATTCGTTACCAATTATGACA ACTTGACGGCTACATCATTCACTTTTTCTTCACAACC GGCACGGAACTCGCTCGGGCTGGCCCCGGTGCATTTT TTAAATACCCGCGAGAAATAGAGTTGATCGTCAAAAC CAACATTGCGACCGACGGTGGCGATAGGCATCCGGGT GGTGCTCAAAAGCAGCTTCGCCTGGCTGATACGTTGG TCCTCGCGCCAGCTTAAGACGCTAATCCCTAACTGCT GGCGGAAAAGATGTGACAGACGCGACGGCGACAAGCA AACATGCTGTGCGACGCTGGCGATATCAAAATTGCTG

TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
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TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
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			TTTATGCGCAGCCCAACGCCGACTGGATCGCCGGCGG
			CCTGGACTGGGGCGACTGGACCCAGAAATTCCACGGC
			GGCCGCCCTTCCTGGGGCAACGAGACCACGGAGCTGC
			GCACCGTCGACTGGTTCAAGCACCGTGACCCGCTCCG
			CCGTTGGCATGCGCCGTACGTCAAGGACAAGGCCGAG
			GAATGGCGCTACACCGACCGCTTCCTGCAGGGTTACT
			CCGCCGACGGTCAGATCCGGGCGATGAACCCGACCTG
			GCGGGACGAGTTCATCAACCGGTATTGGGGCGCCTTC
			CTGTTCAACGAATACGGATTGTTCAACGCTCATTCGC
			AGGGCGCCCGGGAGGCGCTGTCGGACGTAACCCGCGT
			CAGCCTGGCTTTCTGGGGCTTCGACAAGATCGACATC
			GCCCAGATGATCCAACTCGAACGGGGTTTCCTCGCCA
			AGATCGTACCCGGTTTCGACGAGTCCACAGCGGTGCC
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			GCCCGTCTGGCCGTGGAAGGGCTGTGGCAGGAGGTGT
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			TTCTTTCAGCGGCTGGCTCCCCGCTTCGGCGACAATC
			TGACGCCATTCTTCATCAACCAGGCCCAGACATACTT
			CCAGATCGCCAAGCAGGGCGTACAGGATCTGTATTAC
			AACTGTCTGGGTGACGATCCGGAGTTCAGCGATTACA
			ACCGTACCGTGATGCGCAACTGGACCGGCAAGTGGCT
			GGAGCCCACGATCGCCGCTCTGCGCGACTTCATGGGG
			CTGTTTGCGAAGCTGCCGGCGGGCACCACTGACAAGG
			AAGAAATCACCGCGTCCCTGTACCGGGTGGTCGACGA
			CTGGATCGAGGACTACGCCAGCAGGATCGACTTCAAG
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			AGGCAAGGACTTCGCCGATCAGTTCTTTGCCGACGAA
			AACCAAGTGGTCCATGAAAGCGACACGGTCGTTCTGG
			TCCTCAAGAAGTCGGACGAGATCAATACCTTTATCGA
			GGAGATCCTTCTGACGGACTACAAGAAGAACGTCAAT
			CCGACGGTAAACGTGGAAGACCGCGCGGGTTACTGGT
			GGATCAAGGCCAACGGCAAGATCGAGGTCGATTGCGA
			CGAGATTTCCGAGCTGTTGGGGCGGCAGTTCAACGTC
			TACGACTTCCTCGTCGACGTTTCCTCCACCATCGGCC
			GGGCCTATACCCTGGGCAACAAGTTCACCATTACCAG
			TGAGCTGATGGGCCTGGACCGCAAGCTCGAAGACTCT
			CACGCTTAAGGAGAATGACATGGCGAAACTGGGTATA
			CACAGCAACGACACCCGCGACGCCTGGGTGAACAAGA
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			GCTGAAGCAGTTCCGGATGGACCACACCACGCCGTTC
			CGCAACAGCTACGAACTGGACAACGACTACCTCTGGA
			TCGAGGCCAAGCTCGAAGAGAAGGTCGCCGTCCTCAA
			GGCAGAAGCCTTCAACGAGGTGGACTTCCGTCATAAG
			ACCGCTTTCGGCGAGGATGCCAAGTCCGTTCTGGACG
			GCACCGTCGCGAAGATGAACGCGGCCAAGGACAAGTG
			GGAGGCGGAGAAGATCCATATCGGTTTCCGCCAGGCC
			TACAAGCCGCCGATCATGCCGGTGAACTATTTCCTGG
			ACGGCGAGCGTCAGTTGGGGACCCGGCTGATGGAACT
			GCGCAACCTCAACTACTACGACACGCCGCTGGAAGAA
			CTGCGCAAACAGCGCGGTGTGCGGGTGGTGCATCTGC
			AGTCGCCGCACTGAAGGGAGGAAGTCTCGCCCTGGAC
			GCGACGGCATCGCCGTGAAGTCCAGGGGGCAGGGATG
			CCGTTCCGGGCCGGCAGGCTGGCCCGGAATCTCTGGT
			TTTCAGGGGGCGTGCCGGTCCACGGCTCCCCCCTCCA
			TCTTTCGTAAGGAAATCACCATGGTCGAATCGGCATT
			TCAGCCATTTTCGGGCGACGCAGACGAATGGTTCGAG
			GAACCACGGCCCCAGGCCGGTTTCTTCCCTTCCGCGG
			ACTGGCATCTGCTCAAACGGGACGAGACCTACGCAGC
			CTATGCCAAGGATCTCGATTTCATGTGGCGGTGGGTC
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			CGTACTGAATTATTTTGGTATGACCGAACAACGCGCC
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			CAGGTAAGTTTATGCAGCGAGTTCACACTATCACGGC
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TABLE S-continued

SEQ ID NO	Molecule	Region and/or Designation	Sequence
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EXAMPLES

Example 1: Construction of *E. Coli* Strains with Integrated Methane Monooxygenase (MMO)

[0129] The MMO used in the construction of these *E. coli* strains is a modified version of mmoXYBZDC from *M. capsulatus*. The enzyme was improved for the conversion of methane to methanol by directed evolution and contains four amino acid substitutions relative to the wild type sequence. Briefly, a library of single mutations was screened for activity. Beneficial mutations were identified and recombined in random combinations averaging four mutations per plasmid in plasmids encoding both MMO and chaperones, and the plasmids screened again using methane as a substrate. The variant converting methane to methanol at the highest rate, containing amino acid substitutions of mmoX (V23G, T356G), mmoZ (R70E), mmoB (Y139S) was adopted. A mutation (N409G) in the groEL-2 (mmoG) protein chaperone was also incorporated. This plasmid was named pNH284 (SEQ ID NO: 44) and this strain was named NH848.

[0130] Using a CRISPR-based strategy as described by Bassalo, et al. (*ACS Synth. Biol.* 5 (7): 561-568, 2016, which is incorporated by reference herein, including any drawings) the coding sequence for the optimized MMO under the control of the arabinose promoter was inserted into the *E. coli* genome. The strain NH283 was constructed by the deletion of a region of DNA from the *E. coli* genome that contains the genes araBAD using the method of Datsenko and Wanner (See, K. Datsenko and B. Wanner, *One-step inactivation of chromosomal genes in Escherichia coli K-*12 *using PCR Products*, Proceedings of the National Academy of Sciences, Vol 97, Issue 12, p.6640-5, 2000, which is incorporated by reference herein in its entirety). In one strain, YZ55, a single copy of MMO was inserted at the IS7 site (as defined in Bassalo et al.; position 3099988). In a second strain, YZ71, two copies of MMO were inserted, one at IS7 and a second at SS9 (as defined in Bassalo et al.; position 3979535). Proper insertion was confirmed by sequencing.

Example 2: Production of 3-Hydroxypropionate from Ethane Using an Engineered Microorganism

[0131] FIG. **3** shows a plasmid of comparison of the plasmids used in this invention.

[0132] 3-hydroxypropionate (3HP) was produced from an ethane feedstock via the fermentation of three engineered strains of *Escherichia coli*. Each strain harbored three plasmids: 1) pYZ40 (SEQ ID NO: 45), conferring expression of the chaperones groEL and groES from each of *E. coli* and *M. capsulatus;* 2) pNH296 (SEQ ID NO: 46) conferring expression of adhA and adhH from *C. glutamicum* for the conversion of ethanol to acetyl-CoA (ethylotrophy), and 3) pTRIM20 (SEQ ID NO: 47) conferring expression of mcrC*** and mcrN from *Chloroflexus aurantiacus*, to enable production of 3HP from malonyl-CoA.

[0133] pYZ40 (SEQ ID NO: 45) contains a CloDF13 origin of replication, spectinomycin resistance, and the chaperone operon is under the control of the constitutive J23100 promoter. pNH296 (SEQ ID NO: 46) contains a pMB1 origin of replication, carbenicillin resistance, and the alcohol dehydrogenase operon is under the control of the IPTG-inducible pTrc promoter. pTRIM20 (SEQ ID NO: 47) contains a p15a origin of replication, kanamycin resistance, and the malonyl-CoA reductase operon is under the control of the constitutive J23119 promoter. pTRIM20 additionally contains a coding sequence for adhA under the control of the constitutive J23100 promoter.

[0134] One of the three strains for the fermentation, LC805, was derived from YZ71 (with one integrated copy of MMO) by addition of the three plasmids. Another strain, LC809, was derived from YZ55 (with two integrated copies of MMO) by addition of the three plasmids. The third strain, LC807, was derived from a strain with plasmid-borne MMO, NH848, by addition of the two plasmids pNH296 and pTRIM20. NH848 is the base strain NH283 transformed with the plasmid pNH284 (SEQ ID NO: 44).

[0135] The three strains (LC805, LC807, and LC809) were induced with arabinose and IPTG in LB for 4 hours and resuspended in BEM6. (The minimal media called BEM6 contains (in ddH_2O): 50 mM KH_2PO_4, 50 mM Na_2HPO_4*7 H_2O, 1 mM $MgSO_4$, 0.15% LB, 1.5625 mM glutamine, 80 μM $FeSO_4$, 0.1 mM $CaCl_2$), 1 mM IPTG, 0.1% of the 1000× metals solution, and 1 mM L-arabinose.) The 1000× metals solution contains (in ddH_2O): 50 mM $FeCl_3$, 20 mM $CaCl_2$, 10 mM $MnCl_2$, 10 mM $ZnSO_4$, 2 mM $CoCl_2$, 2 mM $NiCl_2$, 2 mM Na_2MoO_4, 2 mM Na_2SeO_3, 2 mM H_3BO_3. This cell culture was split between two serum bottles; each bottle was stoppered and injected with 90 mL of either nitrogen or ethane. The serum bottles were shaken for 96 hours at 37 C. Samples were taken and supernatant was subjected to NMR analysis. The 3HP titer was determined and the titer from the ethane condition was compared to that from the nitrogen condition either as total concentration (FIG. **1**A) or normalized by starting OD_{600} (FIG. **1**B). This procedure was repeated using LC805 and ten-fold different starting ODs of LC809 (FIG. **1**C).

[0136] LC809 (having two integrated copies of MMO) consistently produced 120-130 mg/L/OD of 3HP. LC809 shows a substantially higher production of 3HP than LC805, having a single integrated copy of MMO, as might have been expected. Surprisingly, however, LC809 also shows substantially higher production of 3HP than LC807, which had a plasmid-borne MMO.

[0137] When using ethane/ethanol as a primary or sole carbon source, different biochemical pathways need to be active as opposed to when bacteria has a plentiful supply of, for example, 5-carbon sugars or 6-carbon sugars. Better growth can sometimes be obtained if ethanol is included in the seed train, to induce up-regulation of these pathways. Thus, 3HP production by LC809 was compared using a seed train of LB overnight, followed by 1:30 dilution into LB induction, followed by centrifugation and resuspension in BEM6 (starting OD~2.3) (Seed Train 1), with the same supplemented with 0.5% ethanol (Seed Train 2). Equivalent amounts of 3HP were produced in both conditions (FIG. **2**).

[0138] All publications, patents, and patent applications cited in this specification are hereby incorporated by reference as if each individual publication or patent application were specifically and individually indicated to be incorporated by reference. Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be readily apparent to those of ordinary skill in the art in light of the teachings of this invention that certain changes and modifications may be made thereto without departing from the spirit or scope of the appended claims.

SEQUENCE LISTING

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<213> ORGANISM: Methylococcus capsulatus

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Asn Arg Ala Pro Thr Ser Gly Asn Ala Gln Glu Val His Arg Trp Leu
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Gln Ser Phe Asn Trp Asp Phe Lys Asn Asn Arg Thr Lys Tyr Ala Thr
        35                  40                  45

Lys Tyr Lys Met Ala Asn Glu Thr Lys Glu Gln Phe Lys Leu Ile Ala
    50                  55                  60

Lys Glu Tyr Ala Arg Met Glu Ala Val Lys Asp Glu Arg Gln Phe Gly
65                  70                  75                  80

Ser Leu Gln Asp Ala Leu Thr Arg Leu Asn Ala Gly Val Arg Val His
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                85                  90                  95

Pro Lys Trp Asn Glu Thr Met Lys Val Val Ser Asn Phe Leu Glu Val
            100                 105                 110

Gly Glu Tyr Asn Ala Ile Ala Ala Thr Gly Met Leu Trp Asp Ser Ala
        115                 120                 125

Gln Ala Ala Glu Gln Lys Asn Gly Tyr Leu Ala Gln Val Leu Asp Glu
    130                 135                 140

Ile Arg His Thr His Gln Cys Ala Tyr Val Asn Tyr Tyr Phe Ala Lys
145                 150                 155                 160

Asn Gly Gln Asp Pro Ala Gly His Asn Asp Ala Arg Arg Thr Arg Thr
                165                 170                 175

Ile Gly Pro Leu Trp Lys Gly Met Lys Arg Val Phe Ser Asp Gly Phe
            180                 185                 190

Ile Ser Gly Asp Ala Val Glu Cys Ser Leu Asn Leu Gln Leu Val Gly
        195                 200                 205

Glu Ala Cys Phe Thr Asn Pro Leu Ile Val Ala Val Thr Glu Trp Ala
    210                 215                 220

Ala Ala Asn Gly Asp Glu Ile Thr Pro Thr Val Phe Leu Ser Ile Glu
225                 230                 235                 240

Thr Asp Glu Leu Arg His Met Ala Asn Gly Tyr Gln Thr Val Val Ser
                245                 250                 255

Ile Ala Asn Asp Pro Ala Ser Ala Lys Tyr Leu Asn Thr Asp Leu Asn
            260                 265                 270

Asn Ala Phe Trp Thr Gln Gln Lys Tyr Phe Thr Pro Val Leu Gly Met
        275                 280                 285

Leu Phe Glu Tyr Gly Ser Lys Phe Lys Val Glu Pro Trp Val Lys Thr
    290                 295                 300

Trp Asn Arg Trp Val Tyr Glu Asp Trp Gly Gly Ile Trp Ile Gly Arg
305                 310                 315                 320

Leu Gly Lys Tyr Gly Val Glu Ser Pro Arg Ser Leu Lys Asp Ala Lys
                325                 330                 335

Gln Asp Ala Tyr Trp Ala His His Asp Leu Tyr Leu Leu Ala Tyr Ala
            340                 345                 350

Leu Trp Pro Gly Gly Phe Phe Arg Leu Ala Leu Pro Asp Gln Glu Glu
        355                 360                 365

Met Glu Trp Phe Glu Ala Asn Tyr Pro Gly Trp Tyr Asp His Tyr Gly
    370                 375                 380

Lys Ile Tyr Glu Glu Trp Arg Ala Arg Gly Cys Glu Asp Pro Ser Ser
385                 390                 395                 400

Gly Phe Ile Pro Leu Met Trp Phe Ile Glu Asn Asn His Pro Ile Tyr
                405                 410                 415

Ile Asp Arg Val Ser Gln Val Pro Phe Cys Pro Ser Leu Ala Lys Gly
            420                 425                 430

Ala Ser Thr Leu Arg Val His Glu Tyr Asn Gly Gln Met His Thr Phe
        435                 440                 445

Ser Asp Gln Trp Gly Glu Arg Met Trp Leu Ala Glu Pro Glu Arg Tyr
    450                 455                 460

Glu Cys Gln Asn Ile Phe Glu Gln Tyr Glu Gly Arg Glu Leu Ser Glu
465                 470                 475                 480

Val Ile Ala Glu Leu His Gly Leu Arg Ser Asp Gly Lys Thr Leu Ile
                485                 490                 495
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Ala Gln Pro His Val Arg Gly Asp Lys Leu Trp Thr Leu Asp Asp Ile
            500                 505                 510

Lys Arg Leu Asn Cys Val Phe Lys Asn Pro Val Lys Ala Phe Asn
        515                 520                 525


<210> SEQ ID NO 2
<211> LENGTH: 389
<212> TYPE: PRT
<213> ORGANISM: Methylococcus capsulatus

<400> SEQUENCE: 2

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Ala Ala Val Ile Leu Lys Ala Leu Pro Glu Ala Pro Leu Asp Gly Asn
            20                  25                  30

Asn Lys Met Gly Tyr Phe Val Thr Pro Arg Trp Lys Arg Leu Thr Glu
        35                  40                  45

Tyr Glu Ala Leu Thr Val Tyr Ala Gln Pro Asn Ala Asp Trp Ile Ala
    50                  55                  60

Gly Gly Leu Asp Trp Gly Asp Trp Thr Gln Lys Phe His Gly Gly Arg
65                  70                  75                  80

Pro Ser Trp Gly Asn Glu Thr Thr Glu Leu Arg Thr Val Asp Trp Phe
                85                  90                  95

Lys His Arg Asp Pro Leu Arg Arg Trp His Ala Pro Tyr Val Lys Asp
            100                 105                 110

Lys Ala Glu Glu Trp Arg Tyr Thr Asp Arg Phe Leu Gln Gly Tyr Ser
        115                 120                 125

Ala Asp Gly Gln Ile Arg Ala Met Asn Pro Thr Trp Arg Asp Glu Phe
    130                 135                 140

Ile Asn Arg Tyr Trp Gly Ala Phe Leu Phe Asn Glu Tyr Gly Leu Phe
145                 150                 155                 160

Asn Ala His Ser Gln Gly Ala Arg Glu Ala Leu Ser Asp Val Thr Arg
                165                 170                 175

Val Ser Leu Ala Phe Trp Gly Phe Asp Lys Ile Asp Ile Ala Gln Met
            180                 185                 190

Ile Gln Leu Glu Arg Gly Phe Leu Ala Lys Ile Val Pro Gly Phe Asp
        195                 200                 205

Glu Ser Thr Ala Val Pro Lys Ala Glu Trp Thr Asn Gly Glu Val Tyr
    210                 215                 220

Lys Ser Ala Arg Leu Ala Val Glu Gly Leu Trp Gln Glu Val Phe Asp
225                 230                 235                 240

Trp Asn Glu Ser Ala Phe Ser Val His Ala Val Tyr Asp Ala Leu Phe
                245                 250                 255

Gly Gln Phe Val Arg Arg Glu Phe Phe Gln Arg Leu Ala Pro Arg Phe
            260                 265                 270

Gly Asp Asn Leu Thr Pro Phe Phe Ile Asn Gln Ala Gln Thr Tyr Phe
        275                 280                 285

Gln Ile Ala Lys Gln Gly Val Gln Asp Leu Tyr Tyr Asn Cys Leu Gly
    290                 295                 300

Asp Asp Pro Glu Phe Ser Asp Tyr Asn Arg Thr Val Met Arg Asn Trp
305                 310                 315                 320

Thr Gly Lys Trp Leu Glu Pro Thr Ile Ala Ala Leu Arg Asp Phe Met
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                325                 330                 335

Gly Leu Phe Ala Lys Leu Pro Ala Gly Thr Thr Asp Lys Glu Glu Ile
            340                 345                 350

Thr Ala Ser Leu Tyr Arg Val Val Asp Asp Trp Ile Glu Asp Tyr Ala
        355                 360                 365

Ser Arg Ile Asp Phe Lys Ala Asp Arg Asp Gln Ile Val Lys Ala Val
    370                 375                 380

Leu Ala Gly Leu Lys
385


<210> SEQ ID NO 3
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<212> TYPE: PRT
<213> ORGANISM: Methylococcus capsulatus

<400> SEQUENCE: 3

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Gly Lys Asp Phe Ala Asp Gln Phe Phe Ala Asp Glu Asn Gln Val Val
            20                  25                  30

His Glu Ser Asp Thr Val Val Leu Val Leu Lys Lys Ser Asp Glu Ile
        35                  40                  45

Asn Thr Phe Ile Glu Glu Ile Leu Leu Thr Asp Tyr Lys Lys Asn Val
    50                  55                  60

Asn Pro Thr Val Asn Val Glu Asp Arg Ala Gly Tyr Trp Trp Ile Lys
65                  70                  75                  80

Ala Asn Gly Lys Ile Glu Val Asp Cys Asp Glu Ile Ser Glu Leu Leu
                85                  90                  95

Gly Arg Gln Phe Asn Val Tyr Asp Phe Leu Val Asp Val Ser Ser Thr
            100                 105                 110

Ile Gly Arg Ala Tyr Thr Leu Gly Asn Lys Phe Thr Ile Thr Ser Glu
        115                 120                 125

Leu Met Gly Leu Asp Arg Lys Leu Glu Asp Ser His Ala
    130                 135                 140


<210> SEQ ID NO 4
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<213> ORGANISM: Methylococcus capsulatus

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Asn Lys Ile Ala Gln Leu Asn Thr Leu Glu Lys Ala Ala Glu Met Leu
            20                  25                  30

Lys Gln Phe Arg Met Asp His Thr Thr Pro Phe Arg Asn Ser Tyr Glu
        35                  40                  45

Leu Asp Asn Asp Tyr Leu Trp Ile Glu Ala Lys Leu Glu Glu Lys Val
    50                  55                  60

Ala Val Leu Lys Ala Glu Ala Phe Asn Glu Val Asp Phe Arg His Lys
65                  70                  75                  80

Thr Ala Phe Gly Glu Asp Ala Lys Ser Val Leu Asp Gly Thr Val Ala
                85                  90                  95

Lys Met Asn Ala Ala Lys Asp Lys Trp Glu Ala Glu Lys Ile His Ile
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            100                 105                 110

Gly Phe Arg Gln Ala Tyr Lys Pro Pro Ile Met Pro Val Asn Tyr Phe
        115                 120                 125

Leu Asp Gly Glu Arg Gln Leu Gly Thr Arg Leu Met Glu Leu Arg Asn
    130                 135                 140

Leu Asn Tyr Tyr Asp Thr Pro Leu Glu Glu Leu Arg Lys Gln Arg Gly
145                 150                 155                 160

Val Arg Val Val His Leu Gln Ser Pro His
                165                 170


<210> SEQ ID NO 5
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<212> TYPE: PRT
<213> ORGANISM: Methylococcus capsulatus

<400> SEQUENCE: 5

Met Val Glu Ser Ala Phe Gln Pro Phe Ser Gly Asp Ala Asp Glu Trp
1               5                   10                  15

Phe Glu Glu Pro Arg Pro Gln Ala Gly Phe Phe Pro Ser Ala Asp Trp
            20                  25                  30

His Leu Leu Lys Arg Asp Glu Thr Tyr Ala Ala Tyr Ala Lys Asp Leu
        35                  40                  45

Asp Phe Met Trp Arg Trp Val Ile Val Arg Glu Glu Arg Ile Val Gln
    50                  55                  60

Glu Gly Cys Ser Ile Ser Leu Glu Ser Ser Ile Arg Ala Val Thr His
65                  70                  75                  80

Val Leu Asn Tyr Phe Gly Met Thr Glu Gln Arg Ala Pro Ala Glu Asp
                85                  90                  95

Arg Thr Gly Gly Val Gln His
            100


<210> SEQ ID NO 6
<211> LENGTH: 348
<212> TYPE: PRT
<213> ORGANISM: Methylococcus capsulatus

<400> SEQUENCE: 6

Met Gln Arg Val His Thr Ile Thr Ala Val Thr Glu Asp Gly Glu Ser
1               5                   10                  15

Leu Arg Phe Glu Cys Arg Ser Asp Glu Asp Val Ile Thr Ala Ala Leu
            20                  25                  30

Arg Gln Asn Ile Phe Leu Met Ser Ser Cys Arg Glu Gly Gly Cys Ala
        35                  40                  45

Thr Cys Lys Ala Leu Cys Ser Glu Gly Asp Tyr Asp Leu Lys Gly Cys
    50                  55                  60

Ser Val Gln Ala Leu Pro Pro Glu Glu Glu Glu Glu Gly Leu Val Leu
65                  70                  75                  80

Leu Cys Arg Thr Tyr Pro Lys Thr Asp Leu Glu Ile Glu Leu Pro Tyr
                85                  90                  95

Thr His Cys Arg Ile Ser Phe Gly Glu Val Gly Ser Phe Glu Ala Glu
            100                 105                 110

Val Val Gly Leu Asn Trp Val Ser Ser Asn Thr Val Gln Phe Leu Leu
        115                 120                 125

Gln Lys Arg Pro Asp Glu Cys Gly Asn Arg Gly Val Lys Phe Glu Pro
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    130                 135                 140

Gly Gln Phe Met Asp Leu Thr Ile Pro Gly Thr Asp Val Ser Arg Ser
145                 150                 155                 160

Tyr Ser Pro Ala Asn Leu Pro Asn Pro Glu Gly Arg Leu Glu Phe Leu
                165                 170                 175

Ile Arg Val Leu Pro Glu Gly Arg Phe Ser Asp Tyr Leu Arg Asn Asp
            180                 185                 190

Ala Arg Val Gly Gln Val Leu Ser Val Lys Gly Pro Leu Gly Val Phe
        195                 200                 205

Gly Leu Lys Glu Arg Gly Met Ala Pro Arg Tyr Phe Val Ala Gly Gly
    210                 215                 220

Thr Gly Leu Ala Pro Val Val Ser Met Val Arg Gln Met Gln Glu Trp
225                 230                 235                 240

Thr Ala Pro Asn Glu Thr Arg Ile Tyr Phe Gly Val Asn Thr Glu Pro
                245                 250                 255

Glu Leu Phe Tyr Ile Asp Glu Leu Lys Ser Leu Glu Arg Ser Met Arg
            260                 265                 270

Asn Leu Thr Val Lys Ala Cys Val Trp His Pro Ser Gly Asp Trp Glu
        275                 280                 285

Gly Glu Gln Gly Ser Pro Ile Asp Ala Leu Arg Glu Asp Leu Glu Ser
    290                 295                 300

Ser Asp Ala Asn Pro Asp Ile Tyr Leu Cys Gly Pro Pro Gly Met Ile
305                 310                 315                 320

Asp Ala Ala Cys Glu Leu Val Arg Ser Arg Gly Ile Pro Gly Glu Gln
                325                 330                 335

Val Phe Phe Glu Lys Phe Leu Pro Ser Gly Ala Ala
            340                 345


<210> SEQ ID NO 7
<211> LENGTH: 260
<212> TYPE: PRT
<213> ORGANISM: Methylococcus capsulatus

<400> SEQUENCE: 7

Met Ala Ala Thr Thr Ile Gly Gly Ala Ala Ala Ala Glu Ala Pro Leu
1               5                   10                  15

Leu Asp Lys Lys Trp Leu Thr Phe Ala Leu Ala Ile Tyr Thr Val Phe
            20                  25                  30

Tyr Leu Trp Val Arg Trp Tyr Glu Gly Val Tyr Gly Trp Ser Ala Gly
        35                  40                  45

Leu Asp Ser Phe Ala Pro Glu Phe Glu Thr Tyr Trp Met Asn Phe Leu
    50                  55                  60

Tyr Thr Glu Ile Val Leu Glu Ile Val Thr Ala Ser Ile Leu Trp Gly
65                  70                  75                  80

Tyr Leu Trp Lys Thr Arg Asp Arg Asn Leu Ala Ala Leu Thr Pro Arg
                85                  90                  95

Glu Glu Leu Arg Arg Asn Phe Thr His Leu Val Trp Leu Val Ala Tyr
            100                 105                 110

Ala Trp Ala Ile Tyr Trp Gly Ala Ser Tyr Phe Thr Glu Gln Asp Gly
        115                 120                 125

Thr Trp His Gln Thr Ile Val Arg Asp Thr Asp Phe Thr Pro Ser His
    130                 135                 140
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Ile Ile Glu Phe Tyr Leu Ser Tyr Pro Ile Tyr Ile Ile Thr Gly Phe
145                 150                 155                 160

Ala Ala Phe Ile Tyr Ala Lys Thr Arg Leu Pro Phe Phe Ala Lys Gly
                165                 170                 175

Ile Ser Leu Pro Tyr Leu Val Leu Val Val Gly Pro Phe Met Ile Leu
            180                 185                 190

Pro Asn Val Gly Leu Asn Glu Trp Gly His Thr Phe Trp Phe Met Glu
        195                 200                 205

Glu Leu Phe Val Ala Pro Leu His Tyr Gly Phe Val Ile Phe Gly Trp
    210                 215                 220

Leu Ala Leu Ala Val Met Gly Thr Leu Thr Gln Thr Phe Tyr Ser Phe
225                 230                 235                 240

Ala Gln Gly Gly Leu Gly Gln Ser Leu Cys Glu Ala Val Asp Glu Gly
                245                 250                 255

Leu Ile Ala Lys
            260


<210> SEQ ID NO 8
<211> LENGTH: 420
<212> TYPE: PRT
<213> ORGANISM: Mycobacterium sp.

<400> SEQUENCE: 8

Met Thr Glu Met Thr Val Ala Ala Ser Asp Ala Thr Asn Ala Ala Tyr
1               5                   10                  15

Gly Met Ala Leu Glu Asp Ile Asp Val Ser Asn Pro Val Leu Phe Arg
            20                  25                  30

Asp Asn Thr Trp His Pro Tyr Phe Lys Arg Leu Arg Glu Glu Asp Pro
        35                  40                  45

Val His Tyr Cys Lys Ser Ser Met Phe Gly Pro Tyr Trp Ser Val Thr
    50                  55                  60

Lys Tyr Arg Asp Ile Met Ala Val Glu Thr Asn Pro Lys Val Phe Ser
65                  70                  75                  80

Ser Glu Ala Lys Ser Gly Gly Ile Thr Ile Met Asp Asp Asn Ala Ala
                85                  90                  95

Ala Ser Leu Pro Met Phe Ile Ala Met Asp Pro Pro Lys His Asp Val
            100                 105                 110

Gln Arg Lys Thr Val Ser Pro Ile Val Ala Pro Glu Asn Leu Ala Thr
        115                 120                 125

Met Glu Ser Val Ile Arg Gln Arg Thr Ala Asp Leu Leu Asp Gly Leu
    130                 135                 140

Pro Ile Asn Glu Glu Phe Asp Trp Val His Arg Val Ser Ile Glu Leu
145                 150                 155                 160

Thr Thr Lys Met Leu Ala Thr Leu Phe Asp Phe Pro Trp Asp Asp Arg
                165                 170                 175

Ala Lys Leu Thr Arg Trp Ser Asp Val Thr Thr Ala Leu Pro Gly Gly
            180                 185                 190

Gly Ile Ile Asp Ser Glu Glu Gln Arg Met Ala Glu Leu Met Glu Cys
        195                 200                 205

Ala Thr Tyr Phe Thr Glu Leu Trp Asn Gln Arg Val Asn Ala Glu Pro
    210                 215                 220

Lys Asn Asp Leu Ile Ser Met Met Ala His Ser Glu Ser Thr Arg His
225                 230                 235                 240
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Met Ala Pro Glu Glu Tyr Leu Gly Asn Ile Val Leu Leu Ile Val Gly
                245                 250                 255

Gly Asn Asp Thr Thr Arg Asn Ser Met Thr Gly Gly Val Leu Ala Leu
            260                 265                 270

Asn Glu Phe Pro Asp Glu Tyr Arg Lys Leu Ser Ala Asn Pro Ala Leu
        275                 280                 285

Ile Ser Ser Met Val Ser Glu Ile Ile Arg Trp Gln Thr Pro Leu Ser
    290                 295                 300

His Met Arg Arg Thr Ala Leu Glu Asp Ile Glu Phe Gly Gly Lys His
305                 310                 315                 320

Ile Arg Gln Gly Asp Lys Val Val Met Trp Tyr Val Ser Gly Asn Arg
                325                 330                 335

Asp Pro Glu Ala Ile Asp Asn Pro Asp Thr Phe Ile Ile Asp Arg Ala
            340                 345                 350

Lys Pro Arg Gln His Leu Ser Phe Gly Phe Gly Ile His Arg Cys Val
        355                 360                 365

Gly Asn Arg Leu Ala Glu Leu Gln Leu Asn Ile Leu Trp Glu Glu Ile
    370                 375                 380

Leu Lys Arg Trp Pro Asp Pro Leu Gln Ile Gln Val Leu Gln Glu Pro
385                 390                 395                 400

Thr Arg Val Leu Ser Pro Phe Val Lys Gly Tyr Glu Ser Leu Pro Val
                405                 410                 415

Arg Ile Asn Ala
            420


&lt;210&gt; SEQ ID NO 9
&lt;211&gt; LENGTH: 415
&lt;212&gt; TYPE: PRT
&lt;213&gt; ORGANISM: Pseudomonas putida

&lt;400&gt; SEQUENCE: 9

Met Thr Thr Glu Thr Ile Gln Ser Asn Ala Asn Leu Ala Pro Leu Pro
1               5                   10                  15

Pro His Val Pro Glu His Leu Val Phe Asp Phe Asp Met Tyr Asn Pro
            20                  25                  30

Ser Asn Leu Ser Ala Gly Val Gln Glu Ala Trp Ala Val Leu Gln Glu
        35                  40                  45

Ser Asn Val Pro Asp Leu Val Trp Thr Arg Cys Asn Gly Gly His Trp
    50                  55                  60

Ile Ala Thr Arg Gly Gln Leu Ile Arg Glu Ala Tyr Glu Asp Tyr Arg
65                  70                  75                  80

His Phe Ser Ser Glu Cys Pro Phe Ile Pro Arg Glu Ala Gly Glu Ala
                85                  90                  95

Tyr Asp Phe Ile Pro Thr Ser Met Asp Pro Pro Glu Gln Arg Gln Phe
            100                 105                 110

Arg Ala Leu Ala Asn Gln Val Val Gly Met Pro Val Val Asp Lys Leu
        115                 120                 125

Glu Asn Arg Ile Gln Glu Leu Ala Cys Ser Leu Ile Glu Ser Leu Arg
    130                 135                 140

Pro Gln Gly Gln Cys Asn Phe Thr Glu Asp Tyr Ala Glu Pro Phe Pro
145                 150                 155                 160

Ile Arg Ile Phe Met Leu Leu Ala Gly Leu Pro Glu Glu Asp Ile Pro
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                165                 170                 175

His Leu Lys Tyr Leu Thr Asp Gln Met Thr Arg Pro Asp Gly Ser Met
            180                 185                 190

Thr Phe Ala Glu Ala Lys Glu Ala Leu Tyr Asp Tyr Leu Ile Pro Ile
        195                 200                 205

Ile Glu Gln Arg Arg Gln Lys Pro Gly Thr Asp Ala Ile Ser Ile Val
    210                 215                 220

Ala Asn Gly Gln Val Asn Gly Arg Pro Ile Thr Ser Asp Glu Ala Lys
225                 230                 235                 240

Arg Met Cys Gly Leu Leu Leu Val Gly Gly Leu Asp Thr Val Val Asn
                245                 250                 255

Phe Leu Ser Phe Ser Met Glu Phe Leu Ala Lys Ser Pro Glu His Arg
            260                 265                 270

Gln Glu Leu Ile Glu Arg Pro Glu Arg Ile Pro Ala Ala Cys Glu Glu
        275                 280                 285

Leu Leu Arg Arg Phe Ser Leu Val Ala Asp Gly Arg Ile Leu Thr Ser
    290                 295                 300

Asp Tyr Glu Phe His Gly Val Gln Leu Lys Lys Gly Asp Gln Ile Leu
305                 310                 315                 320

Leu Pro Gln Met Leu Ser Gly Leu Asp Glu Arg Glu Asn Ala Cys Pro
                325                 330                 335

Met His Val Asp Phe Ser Arg Gln Lys Val Ser His Thr Thr Phe Gly
            340                 345                 350

His Gly Ser His Leu Cys Leu Gly Gln His Leu Ala Arg Arg Glu Ile
        355                 360                 365

Ile Val Thr Leu Lys Glu Trp Leu Thr Arg Ile Pro Asp Phe Ser Ile
    370                 375                 380

Ala Pro Gly Ala Gln Ile Gln His Lys Ser Gly Ile Val Ser Gly Val
385                 390                 395                 400

Gln Ala Leu Pro Leu Val Trp Asp Pro Ala Thr Thr Lys Ala Val
                405                 410                 415


<210> SEQ ID NO 10
<211> LENGTH: 415
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 10

Met Thr Thr Glu Thr Ile Gln Ser Asn Ala Asn Leu Ala Pro Leu Pro
1               5                   10                  15

Pro His Val Pro Glu His Leu Val Phe Asp Phe Asp Met Tyr Asn Pro
            20                  25                  30

Ser Asn Leu Ser Ala Gly Val Gln Glu Ala Trp Ala Val Leu Gln Glu
        35                  40                  45

Ser Asn Val Pro Asp Leu Val Trp Thr Arg Cys Asn Gly Gly His Trp
    50                  55                  60

Ile Ala Thr Arg Gly Gln Leu Ile Arg Glu Ala Tyr Glu Asp Tyr Arg
65                  70                  75                  80

His Phe Ser Ser Glu Cys Pro Trp Ile Pro Arg Glu Ala Gly Glu Ala
                85                  90                  95

Phe Asp Phe Ile Pro Leu Ser Met Asp Pro Pro Glu Gln Arg Gln Phe
            100                 105                 110
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Arg Ala Leu Ala Asn Gln Val Val Gly Met Pro Val Val Asp Lys Leu
115 120 125

Glu Asn Arg Ile Gln Glu Leu Ala Cys Ser Leu Ile Glu Ser Leu Arg
130 135 140

Pro Gln Gly Gln Cys Asn Phe Thr Glu Asp Tyr Ala Glu Pro Phe Pro
145 150 155 160

Ile Arg Ile Phe Met Leu Leu Ala Gly Leu Pro Glu Glu Asp Ile Pro
165 170 175

His Leu Lys Tyr Leu Thr Asp Gln Met Met Arg Pro Asp Gly Ser Met
180 185 190

Thr Phe Ala Glu Ala Lys Glu Ala Leu Tyr Asp Tyr Leu Ile Pro Ile
195 200 205

Ile Glu Gln Arg Arg Gln Lys Pro Gly Thr Asp Ala Ile Ser Ile Val
210 215 220

Ala Asn Gly Gln Val Asn Gly Arg Pro Ile Thr Ser Asp Glu Ala Lys
225 230 235 240

Arg Met Cys Gly Met Leu Leu Leu Ala Gly Leu Asp Thr Val Val Asn
245 250 255

Phe Leu Ser Phe Ser Met Glu Phe Leu Ala Lys Ser Pro Glu His Arg
260 265 270

Gln Glu Leu Ile Glu Arg Pro Glu Arg Ile Pro Ala Ala Cys Glu Glu
275 280 285

Leu Leu Arg Arg Phe Ser Met Val Ala Asp Gly Arg Ile Leu Thr Ser
290 295 300

Asp Tyr Glu Phe His Gly Val Gln Leu Lys Lys Gly Asp Gln Ile Leu
305 310 315 320

Leu Pro Gln Met Leu Ser Gly Leu Asp Glu Arg Glu Asn Ala Cys Pro
325 330 335

Met His Val Asp Phe Ser Arg Gln Lys Val Ser His Thr Thr Phe Gly
340 345 350

His Gly Ser His Leu Cys Pro Gly Gln His Leu Ala Arg Arg Glu Ile
355 360 365

Ile Val Thr Leu Lys Glu Trp Leu Thr Arg Ile Pro Asp Phe Ser Ile
370 375 380

Ala Pro Gly Ala Gln Ile Gln His Lys Ser Gly Ile Val Ser Gly Val
385 390 395 400

Gln Ala Leu Pro Leu Val Trp Asp Pro Ala Thr Thr Lys Ala Val
405 410 415

<210> SEQ ID NO 11
<211> LENGTH: 401
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 11

Met Leu Glu Lys His Arg Val Leu Asp Ser Ala Pro Glu Tyr Val Asp
1 5 10 15

Lys Lys Lys Tyr Leu Trp Ile Leu Ser Thr Leu Trp Pro Ala Thr Pro
20 25 30

Met Ile Gly Ile Trp Leu Ala Asn Glu Thr Gly Trp Gly Ile Phe Tyr
35 40 45

Gly Leu Val Leu Leu Val Trp Tyr Gly Ala Leu Pro Leu Leu Asp Ala
50 55 60

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Met Phe Gly Glu Asp Phe Asn Asn Pro Pro Glu Glu Val Val Pro Lys
65                  70                  75                  80

Leu Glu Lys Glu Arg Tyr Tyr Arg Val Leu Thr Tyr Leu Thr Val Pro
                85                  90                  95

Met His Tyr Ala Ala Leu Ile Val Ser Ala Trp Trp Val Gly Thr Gln
            100                 105                 110

Pro Met Ser Trp Leu Glu Ile Gly Ala Leu Ala Leu Ser Leu Gly Ile
        115                 120                 125

Val Asn Gly Leu Ala Leu Asn Thr Gly His Glu Leu Gly His Lys Lys
    130                 135                 140

Glu Thr Phe Asp Arg Trp Met Ala Lys Ile Val Leu Ala Val Val Gly
145                 150                 155                 160

Tyr Gly His Phe Phe Ile Glu His Asn Lys Gly His His Arg Asp Val
                165                 170                 175

Ala Thr Pro Met Asp Pro Ala Thr Ser Arg Met Gly Glu Ser Ile Tyr
            180                 185                 190

Lys Phe Ser Ile Arg Glu Ile Pro Gly Ala Phe Ile Arg Ala Trp Gly
        195                 200                 205

Leu Glu Glu Gln Arg Leu Ser Arg Arg Gly Gln Ser Val Trp Ser Phe
    210                 215                 220

Asp Asn Glu Ile Leu Gln Pro Met Ile Ile Thr Val Ile Leu Tyr Ala
225                 230                 235                 240

Val Leu Leu Ala Leu Phe Gly Pro Lys Met Leu Val Phe Leu Pro Ile
                245                 250                 255

Gln Met Ala Phe Gly Trp Trp Gln Leu Thr Ser Ala Asn Tyr Ile Glu
            260                 265                 270

His Tyr Gly Leu Leu Arg Gln Lys Met Glu Asp Gly Arg Tyr Glu His
        275                 280                 285

Gln Lys Pro His His Ser Trp Asn Ser Asn His Ile Val Ser Asn Leu
    290                 295                 300

Val Leu Phe His Leu Gln Arg His Ser Asp His His Ala His Pro Thr
305                 310                 315                 320

Arg Ser Tyr Gln Ser Leu Arg Asp Phe Pro Gly Leu Pro Ala Leu Pro
                325                 330                 335

Thr Gly Tyr Pro Gly Ala Phe Leu Met Ala Met Ile Pro Gln Trp Phe
            340                 345                 350

Arg Ser Val Met Asp Pro Lys Val Val Asp Trp Ala Gly Gly Asp Leu
        355                 360                 365

Asn Lys Ile Gln Ile Asp Asp Ser Met Arg Glu Thr Tyr Leu Lys Lys
    370                 375                 380

Phe Gly Thr Ser Ser Ala Gly His Ser Ser Ser Thr Ser Ala Val Ala
385                 390                 395                 400

Ser


<210> SEQ ID NO 12
<211> LENGTH: 545
<212> TYPE: PRT
<213> ORGANISM: Gordonia sp.

<400> SEQUENCE: 12

Met Ser Arg Gln Ser Leu Thr Lys Ala His Ala Lys Ile Thr Glu Leu
1               5                   10                  15
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Ser Trp Glu Pro Thr Phe Ala Thr Pro Ala Thr Arg Phe Gly Thr Asp
        20                  25                  30

Tyr Thr Phe Glu Lys Ala Pro Lys Lys Asp Pro Leu Lys Gln Ile Met
    35                  40                  45

Arg Ser Tyr Phe Pro Met Glu Glu Glu Lys Asp Asn Arg Val Tyr Gly
    50                  55                  60

Ala Met Asp Gly Ala Ile Arg Gly Asn Met Phe Arg Gln Val Gln Glu
65                  70                  75                  80

Arg Trp Leu Glu Trp Gln Lys Leu Phe Leu Ser Ile Ile Pro Phe Pro
                85                  90                  95

Glu Ile Ser Ala Ala Arg Ala Met Pro Met Ala Ile Asp Ala Val Pro
            100                 105                 110

Asn Pro Glu Ile His Asn Gly Leu Ala Val Gln Met Ile Asp Glu Val
        115                 120                 125

Arg His Ser Thr Ile Gln Met Asn Leu Lys Lys Leu Tyr Met Asn Asn
    130                 135                 140

Tyr Ile Asp Pro Ala Gly Phe Asp Ile Thr Glu Lys Ala Phe Ala Asn
145                 150                 155                 160

Asn Tyr Ala Gly Thr Ile Gly Arg Gln Phe Gly Glu Gly Phe Ile Thr
                165                 170                 175

Gly Asp Ala Ile Thr Ala Ala Asn Ile Tyr Leu Thr Val Val Ala Glu
            180                 185                 190

Thr Ala Phe Thr Asn Thr Leu Phe Val Ala Met Pro Asp Glu Ala Ala
        195                 200                 205

Ala Asn Gly Asp Tyr Leu Leu Pro Thr Val Phe His Ser Val Gln Ser
    210                 215                 220

Asp Glu Ser Arg His Ile Ser Asn Gly Tyr Ser Ile Leu Leu Met Ala
225                 230                 235                 240

Leu Ala Asp Glu Arg Asn Arg Pro Leu Leu Glu Arg Asp Leu Arg Tyr
                245                 250                 255

Ala Trp Trp Asn Asn His Cys Val Val Asp Ala Ala Ile Gly Thr Phe
            260                 265                 270

Ile Glu Tyr Gly Thr Lys Asp Arg Arg Lys Asp Arg Glu Ser Tyr Ala
        275                 280                 285

Glu Met Trp Arg Arg Trp Ile Tyr Asp Asp Tyr Tyr Arg Ser Tyr Leu
    290                 295                 300

Leu Pro Leu Glu Lys Tyr Gly Leu Thr Ile Pro His Asp Leu Val Glu
305                 310                 315                 320

Glu Ala Trp Asn Arg Ile Val Asp Lys His Tyr Val His Glu Val Ala
                325                 330                 335

Arg Phe Phe Ala Thr Gly Trp Pro Val Asn Tyr Trp Arg Ile Asp Ala
            340                 345                 350

Met Thr Asp Thr Asp Phe Glu Trp Phe Glu Glu Lys Tyr Pro Gly Trp
        355                 360                 365

Tyr Asn Lys Phe Gly Lys Trp Trp Glu Asn Tyr Asn Arg Leu Ala Tyr
    370                 375                 380

Pro Gly Lys Asn Lys Pro Ile Ala Phe Glu Asp Val Asp Tyr Glu Tyr
385                 390                 395                 400

Pro His Arg Cys Trp Thr Cys Met Val Pro Cys Leu Ile Arg Glu Asp
                405                 410                 415
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Met Val Thr Asp Lys Val Asp Gly Gln Trp Arg Thr Tyr Cys Ser Glu
420 425 430

Thr Cys Ala Trp Thr Asp Lys Val Ala Phe Arg Pro Glu Tyr Glu Gly
435 440 445

Arg Pro Thr Pro Asn Met Gly Arg Leu Thr Gly Phe Arg Glu Trp Glu
450 455 460

Thr Leu His His Gly Lys Asp Leu Ala Asp Ile Ile Thr Asp Leu Gly
465 470 475 480

Tyr Val Arg Asp Asp Gly Lys Thr Leu Ile Pro Gln Pro His Leu Asp
485 490 495

Leu Asp Pro Lys Lys Met Trp Thr Leu Asp Asp Val Arg Gly Ile Pro
500 505 510

Phe Gly Ser Pro Asn Val Ala Leu Asn Glu Met Ser Asp Asp Glu Arg
515 520 525

Glu Ala His Ile Ala Ala Tyr Met Ala Asn Lys Asn Gly Ala Val Thr
530 535 540

Val
545

<210> SEQ ID NO 13
<211> LENGTH: 549
<212> TYPE: PRT
<213> ORGANISM: Pseudonocardia sp.

<400> SEQUENCE: 13

Met Ser Arg Gln Ser Met Ser Lys Ala His Lys Lys Ile Thr Glu Leu
1 5 10 15

Ser Trp Glu Pro Thr Phe Ala Thr Pro Ala Lys Arg Phe Gly Thr Asp
20 25 30

Tyr Thr Phe Asp Asn Ala Pro Lys Lys Asp Pro Leu Lys Gln Ile Leu
35 40 45

Arg Ser Tyr Phe Pro Met Glu Glu Glu Lys Asp Ser Arg Val Phe Gly
50 55 60

Ala Met Asp Gly Ala Ile Arg Gly Asn Met Phe Arg Gln Val Gln Glu
65 70 75 80

Arg Trp Met Glu Trp Gln Lys Leu Phe Leu Ser Ile Ile Pro Phe Pro
85 90 95

Glu Ile Ser Ala Ala Arg Ala Met Pro Met Ala Ile Asp Ala Val Pro
100 105 110

Asn Pro Glu Ile His Asn Gly Leu Ala Val Gln Met Ile Asp Glu Val
115 120 125

Arg His Ser Thr Ile Gln Met Asn Leu Lys Arg Leu Tyr Met Asn Tyr
130 135 140

Tyr Ile Asp Pro Ala Gly Phe Asp Met Thr Glu Lys Ala Phe Ala Asn
145 150 155 160

Asn Tyr Ala Gly Thr Ile Gly Arg Gln Phe Gly Glu Gly Phe Ile Thr
165 170 175

Gly Asp Ala Ile Thr Ala Ala Asn Ile Tyr Leu Thr Val Val Ala Glu
180 185 190

Thr Ala Phe Thr Asn Thr Leu Phe Val Ala Met Pro Ser Glu Ala Ala
195 200 205

Ala Asn Gly Asp Tyr Leu Leu Pro Thr Val Phe His Ser Val Gln Ser
210 215 220

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Asp Glu Ser Arg His Ile Ser Asn Gly Tyr Ser Ile Leu Leu Met Ala
225                 230                 235                 240

Leu Ser Asp Glu Asp Asn Arg Gln Leu Leu Glu Arg Asp Leu Arg Tyr
                245                 250                 255

Ala Trp Trp Asn Asn His Arg Val Val Asp Ala Ala Ile Gly Thr Phe
            260                 265                 270

Ile Glu Tyr Gly Thr Lys Asp Arg Arg Lys Asp Arg Glu Ser Tyr Ala
        275                 280                 285

Glu Met Trp Arg Arg Trp Ile Tyr Asp Asp Tyr Tyr Arg Ala Tyr Leu
    290                 295                 300

Ile Pro Leu Glu Lys Tyr Gly Leu Val Ile Pro His Asp Leu Ile Glu
305                 310                 315                 320

Glu Ser Trp Lys Gln Ile Trp Glu Lys Gly Tyr Val His Glu Val Ala
                325                 330                 335

Gln Phe Phe Ala Thr Gly Trp Leu Ala Asn Tyr Trp Arg Ile Asp Ser
            340                 345                 350

Met Thr Asp Glu Asp Phe Glu Trp Phe Glu Tyr Lys Tyr Pro Gly Trp
        355                 360                 365

Tyr Asp Lys Tyr Gly Lys Trp Trp Glu Asn Tyr Asn Arg Leu Ser Lys
    370                 375                 380

Pro Asn Gly His Asn Pro Ile Val Phe Glu Asp Val Asp Tyr Val Tyr
385                 390                 395                 400

Pro Ala Arg Cys Trp Thr Cys Met Ser Pro Cys Trp Ser Val Arg Thr
                405                 410                 415

Leu Val Thr Ala Glu Val Asp Gly Gln His Arg Thr Tyr Cys His Glu
            420                 425                 430

Val Cys Arg Trp Thr Asp Val Arg Gly Phe Pro Ser Asp Val Pro Gly
        435                 440                 445

Arg Glu Thr Pro Asn Met Gly Arg Leu Val Gly Lys Arg Glu Trp Glu
    450                 455                 460

Thr Leu Tyr His Gly Trp Asn Trp Ala Asp Val Val Ser Asp Met Gly
465                 470                 475                 480

Phe Val Arg Asp Asp Gly Lys Thr Met Thr Pro Lys Pro His Leu Asp
                485                 490                 495

Leu Asp Pro Lys Lys Met Trp Thr Leu Asp His Met Arg Arg Cys Pro
            500                 505                 510

Pro Leu Gln Ser Pro Asn Val Leu Phe Asn Glu Met Ser Asp Ala Glu
        515                 520                 525

Arg Ala Ala Tyr Val Ala Asp Tyr Asn Lys Gln Gly Pro Ala Gly Arg
    530                 535                 540

Pro Ala Pro Gln Ser
545


<210> SEQ ID NO 14
<211> LENGTH: 531
<212> TYPE: PRT
<213> ORGANISM: Mycobacterium sp.

<400> SEQUENCE: 14

Met Thr Ser Thr Leu Pro Gly Lys Val Ser Gly His Pro Ser Ala Pro
1               5                   10                  15

His Ala Ser Val Ser Gly Gln Glu Val His Ser Trp Leu Met Asp Leu
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            20                  25                  30

Gly Trp Asp Ser Asp Thr Ile Arg Gly Lys Tyr Pro Thr Lys Tyr Lys
        35                  40                  45

Phe Asp Pro Asn Ala Arg Glu Gln Phe Lys Leu Val Ala Arg Asp Tyr
    50                  55                  60

Gly Arg Met Glu Gly Glu Lys Asp Asp Arg Gln Tyr Gly Ser Leu Leu
65                  70                  75                  80

Asp Ser Leu Ala Arg Leu Lys Ala Pro Thr Arg Val Glu Pro Arg Trp
                85                  90                  95

Ala Glu Val Met Lys Leu Leu Ala Gly Ala Leu Glu Leu Gly Glu Tyr
            100                 105                 110

Asn Ala Ile Ala Gly Ser Ala Val Leu Ala Asp Thr Thr Arg Ser Pro
        115                 120                 125

Glu Leu Arg Asn Gly Tyr Leu Met Gln Val Glu Asp Glu Val Arg His
    130                 135                 140

Thr Thr Gln Thr His Tyr Leu Ala Lys Tyr Tyr Ala Gly Gln Tyr Tyr
145                 150                 155                 160

Asp Pro Ala Gly Phe Thr Asp Met Arg Lys Trp Arg Tyr Ile Asn Pro
                165                 170                 175

Leu Phe Pro Pro Thr Met Gln Ala Phe Gly Glu Asn Phe Cys Ala Gly
            180                 185                 190

Asp Pro Val Phe Ala Ser Leu Asn Leu Gln Leu Val Ala Glu Ala Cys
        195                 200                 205

Phe Thr Asn Pro Leu Ile Val Ala Met Thr Glu Trp Ser Ala Ala Asn
    210                 215                 220

Gly Asp Glu Ile Thr Pro Thr Ile Phe Leu Ser Ile Gln Ser Asp Glu
225                 230                 235                 240

Met Arg His Met Ala Asn Gly Tyr Gln Thr Ile Val Ser Val Ala His
                245                 250                 255

Asp Ala Asp Asn Met Gln Tyr Leu Gln Thr Asp Leu Glu Asn Ala Phe
            260                 265                 270

Trp Leu Gln His Arg Phe Ala Thr Pro Ile Val Gly Ala Gly Phe Glu
        275                 280                 285

Tyr Gly Ala Val Asn Lys Leu Glu Pro Trp Ala Lys Val Trp Asp Arg
    290                 295                 300

Trp Val Tyr Glu Asp Trp Gly Gly Ile Trp Leu Gly Arg Leu Glu Lys
305                 310                 315                 320

Phe Gly Val Lys Ser Pro Ala Asn Leu Ala Asp Ala Lys Arg Gln Ala
                325                 330                 335

Tyr Trp Gly His His Tyr Thr Tyr Ala Val Ala Tyr Ala Val Trp Pro
            340                 345                 350

Leu Leu Gly Phe Arg Met Asp Pro Pro Asn Ala Arg Asp Met Glu Trp
        355                 360                 365

Phe Glu Asn Asn Tyr Pro Gly Trp His Ser Glu Val Gly His Met Trp
    370                 375                 380

Asp Ser Trp Arg Glu Met Gly Leu Ala Asp Pro Ala Asn His Thr Leu
385                 390                 395                 400

Pro Gly Gln Leu Val Ser Asp Ala Lys Val Pro Ile Tyr Phe Cys Arg
                405                 410                 415

Val Cys Asn Phe Pro Val Ile Ile Pro Thr Leu Thr Gly Ala Leu Asp
            420                 425                 430
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```
Asp Val Arg Ile Leu Glu Leu Asn Gly Arg Lys His Pro Leu Cys Ser
        435                 440                 445

Ser Trp Cys Glu Arg Met Phe Leu Lys Glu Pro Glu Arg Tyr Gln Gly
    450                 455                 460

Glu Asn Leu Trp Glu Lys Phe Asp Gly Trp Asn Ile Ala Asp Val Val
465                 470                 475                 480

His Ala Ala Gly Ala Val Arg Ser Asp Gly Lys Thr Leu Leu Ala Gln
                485                 490                 495

Pro His Leu Asn Ser Glu Arg Met Trp Thr Leu Asp Asp Leu Arg Ala
            500                 505                 510

Cys His Ala Val Ile Arg Asp Pro Leu Lys Thr Gly Gly Val Trp Leu
        515                 520                 525

Glu Thr Val
    530


<210> SEQ ID NO 15
<211> LENGTH: 530
<212> TYPE: PRT
<213> ORGANISM: Thauera butanivorans

<400> SEQUENCE: 15

Met Ser Ala Asn Met Ala Val Lys Gln Ala Leu Lys Ala Asn Pro Val
1               5                   10                  15

Pro Ser Ser Val Asp Pro Gln Glu Val His Lys Trp Leu Gln Asp Phe
            20                  25                  30

Thr Trp Asp Phe Lys Gly Lys Thr Ala Lys Tyr Pro Thr Lys Tyr Glu
        35                  40                  45

Met Asp Val Asn Thr Arg Glu Gln Phe Lys Leu Thr Ala Lys Glu Tyr
    50                  55                  60

Ala Arg Met Glu Ser Ile Lys Glu Glu Arg Gln Tyr Gly Thr Leu Leu
65                  70                  75                  80

Asp Gly Leu Asp Arg Leu Asp Ala Gly Asn Lys Val His Pro Lys Trp
                85                  90                  95

Gly Glu Val Met Lys Leu Val Ser Asn Phe Leu Glu Thr Gly Glu Tyr
            100                 105                 110

Gly Ala Ile Ala Gly Ser Ala Leu Leu Trp Asp Thr Ala Gln Ser Pro
        115                 120                 125

Glu Gln Arg Asn Gly Tyr Leu Ala Gln Val Ile Asp Glu Ile Arg His
    130                 135                 140

Val Asn Gln Thr Ala Tyr Val Asn Tyr Tyr Tyr Gly Lys His Tyr Tyr
145                 150                 155                 160

Asp Pro Ala Gly His Thr Asn Met Arg Gln Leu Arg Ala Ile Asn Pro
                165                 170                 175

Leu Tyr Pro Gly Val Lys Arg Ala Phe Gly Glu Gly Phe Leu Ala Gly
            180                 185                 190

Asp Ala Val Glu Ser Ser Ile Asn Leu Gln Leu Val Gly Glu Ala Cys
        195                 200                 205

Phe Thr Asn Pro Leu Ile Val Ser Leu Thr Glu Trp Ala Ala Ala Asn
    210                 215                 220

Gly Asp Glu Ile Thr Pro Thr Val Phe Leu Ser Ile Glu Thr Asp Glu
225                 230                 235                 240

Leu Arg His Met Ala Asn Gly Tyr Gln Thr Ile Val Ser Ile Met Asn
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                245                 250                 255

Asn Pro Glu Thr Met Lys Tyr Leu Gln Thr Asp Leu Asp Asn Ala Phe
            260                 265                 270

Trp Thr Gln His Lys Phe Leu Thr Pro Phe Val Gly Val Ala Leu Glu
        275                 280                 285

Tyr Gly Ser Lys Tyr Lys Val Glu Pro Trp Ala Lys Ser Trp Asn Arg
    290                 295                 300

Trp Val Tyr Glu Asp Trp Ala Gly Ile Trp Leu Gly Arg Leu Gln Gln
305                 310                 315                 320

Phe Gly Val Lys Thr Pro Lys Cys Leu Pro Asp Ala Lys Lys Asp Ala
                325                 330                 335

Val Trp Ala His His Asp Leu Ala Leu Leu Ala Leu Ala Leu Trp Pro
            340                 345                 350

Leu Thr Gly Ile Arg Met Glu Leu Pro Asp Ser Leu Ala Met Glu Trp
        355                 360                 365

Phe Glu Ala Asn Tyr Pro Gly Trp Tyr Asn His Tyr Gly Lys Ile Tyr
    370                 375                 380

Glu Glu Trp Arg Ala Ala Gly Phe Glu Asp Pro Lys Ser Gly Phe Cys
385                 390                 395                 400

Gly Ala Leu Trp Leu Met Glu Arg Gly His Gly Ile Phe Val Asp His
                405                 410                 415

Ala Ser Gly Leu Pro Phe Cys Pro Ser Leu Ala Lys Ser Ser Ile Lys
            420                 425                 430

Pro Arg Phe Thr Glu Tyr Asn Gly Lys Arg Tyr Ala Phe Ala Glu Pro
        435                 440                 445

Tyr Gly Glu Arg Gln Trp Leu Leu Glu Pro Glu Arg Tyr Glu Phe Gln
    450                 455                 460

Asn Phe Phe Glu Gln Phe Glu Gly Trp Glu Leu Ser Asp Leu Val Lys
465                 470                 475                 480

Ala Ala Gly Gly Val Arg Ser Asp Gly Lys Thr Leu Ile Ala Gln Pro
                485                 490                 495

His Leu Arg Asp Thr Asp Met Trp Thr Leu Asp Asp Leu Lys Arg Ile
            500                 505                 510

Asn Leu Thr Ile Pro Asp Pro Met Lys Ile Leu Asn Trp Gln Pro Val
        515                 520                 525

Ala Gln
    530


<210> SEQ ID NO 16
<211> LENGTH: 276
<212> TYPE: PRT
<213> ORGANISM: Methylacidiphilum infernorum

<400> SEQUENCE: 16

Met Ala Gln Ala Thr Thr Gln Thr Val Ser Ile Ser Ile Pro Glu Arg
1               5                   10                  15

Ser Ala Phe Ser Trp Lys Asn Leu Trp Ile Ala Leu Ala Ile Ile Thr
            20                  25                  30

Ala Phe Glu Leu Ala Val Asn Val Tyr Glu Arg Thr Phe Ala Ile Ala
        35                  40                  45

Lys Gly Leu Asp Tyr Phe Thr Pro Gln Tyr Gln Thr Tyr Trp Met Ser
    50                  55                  60
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```
Ile Leu Tyr Thr Glu Leu Ile Leu Glu Pro Thr Thr Leu Ile Ala Leu
65                  70                  75                  80

Cys Ser Trp Leu Trp Val Thr Arg Asp Arg Ala Met Glu Asn Leu Ala
                85                  90                  95

Pro Ala Glu Glu Leu Lys Arg Tyr Trp Asn Leu Gly Leu Phe Val Val
            100                 105                 110

Val Tyr Thr Val Leu Leu Tyr Trp Gly Ala Ser Tyr Tyr Thr Glu Gln
        115                 120                 125

Asp Gly Thr Trp His Gln Thr Val Ile Arg Asp Thr Asp Phe Thr Pro
    130                 135                 140

Ser His Ile Ile Glu Phe Tyr Gln Ser Tyr Pro Ile Tyr Ile Val Ala
145                 150                 155                 160

Gly Val Gly Ser Met Val Tyr Ala Met Thr Arg Leu Pro Gln Tyr Ala
                165                 170                 175

Lys Ala Phe Ser Val Pro Tyr Ala Val Leu Val Gly Ser Pro Leu Met
            180                 185                 190

Ile Phe Pro Asn Val Gly Leu Asn Glu Phe Gly His Thr Arg Trp Phe
        195                 200                 205

Met Glu Glu Leu Phe Val Ala Pro Leu His Trp Gly Phe Val Met Phe
    210                 215                 220

Gly Trp Gly Ala Leu Ala Ile Leu Gly Thr Trp Leu Gln Ile Cys Pro
225                 230                 235                 240

Arg Val Val Glu Leu Ile Lys Gln Val Tyr Tyr Gly Lys Pro Ala Thr
                245                 250                 255

Ala Pro Ala Val Val Leu Asn Asp Pro Glu Lys Ala Ala Asp Pro Ala
            260                 265                 270

Leu Cys Glu Ile
        275
```

<210> SEQ ID NO 17

<400> SEQUENCE: 17

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<210> SEQ ID NO 18

<400> SEQUENCE: 18

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<210> SEQ ID NO 19

<400> SEQUENCE: 19

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<210> SEQ ID NO 20

<400> SEQUENCE: 20

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<210> SEQ ID NO 21

<400> SEQUENCE: 21

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<210> SEQ ID NO 22
<211> LENGTH: 345
<212> TYPE: PRT
<213> ORGANISM: Corynebacterium glutamicum

<400> SEQUENCE: 22

```
Met Thr Thr Ala Ala Pro Gln Glu Phe Thr Ala Ala Val Val Glu Lys
1               5                   10                  15

Phe Gly His Asp Val Thr Val Lys Asp Ile Asp Leu Pro Lys Pro Gly
            20                  25                  30

Pro His Gln Ala Leu Val Lys Val Leu Thr Ser Gly Ile Cys His Thr
        35                  40                  45

Asp Leu His Ala Leu Glu Gly Asp Trp Pro Val Lys Pro Glu Pro Pro
    50                  55                  60

Phe Val Pro Gly His Glu Gly Val Gly Glu Val Val Glu Leu Gly Pro
65                  70                  75                  80

Gly Glu His Asp Val Lys Val Gly Asp Ile Val Gly Asn Ala Trp Leu
                85                  90                  95

Trp Ser Ala Cys Gly Thr Cys Glu Tyr Cys Ile Thr Gly Arg Glu Thr
            100                 105                 110

Gln Cys Asn Glu Ala Glu Tyr Gly Gly Tyr Thr Gln Asn Gly Ser Phe
        115                 120                 125

Gly Gln Tyr Met Leu Val Asp Thr Arg Tyr Ala Ala Arg Ile Pro Asp
    130                 135                 140

Gly Val Asp Tyr Leu Glu Ala Ala Pro Ile Leu Cys Ala Gly Val Thr
145                 150                 155                 160

Val Tyr Lys Ala Leu Lys Val Ser Glu Thr Arg Pro Gly Gln Phe Met
                165                 170                 175

Val Ile Ser Gly Val Gly Gly Leu Gly His Ile Ala Val Gln Tyr Ala
            180                 185                 190

Ala Ala Met Gly Met Arg Val Ile Ala Val Asp Ile Ala Asp Asp Lys
        195                 200                 205

Leu Glu Leu Ala Arg Lys His Gly Ala Glu Phe Thr Val Asn Ala Arg
    210                 215                 220

Asn Glu Asp Ser Gly Glu Ala Val Gln Lys Tyr Thr Asn Gly Gly Ala
225                 230                 235                 240

His Gly Val Leu Val Thr Ala Val His Glu Ala Ala Phe Gly Gln Ala
                245                 250                 255

Leu Asp Met Ala Arg Arg Ala Gly Thr Ile Val Phe Asn Gly Leu Pro
            260                 265                 270

Pro Gly Glu Phe Pro Ala Ser Val Phe Asn Ile Val Phe Lys Gly Leu
        275                 280                 285

Thr Ile Arg Gly Ser Leu Val Gly Thr Arg Gln Asp Leu Ala Glu Ala
    290                 295                 300

Leu Asp Phe Phe Ala Arg Gly Leu Ile Lys Pro Thr Val Ser Glu Cys
305                 310                 315                 320

Ser Leu Asp Glu Val Asn Gly Val Leu Asp Arg Met Arg Asn Gly Lys
                325                 330                 335

Ile Asp Gly Arg Val Ala Ile Arg Phe
            340                 345
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<210> SEQ ID NO 23
<211> LENGTH: 506
<212> TYPE: PRT
<213> ORGANISM: Corynebacterium glutamicum

<400> SEQUENCE: 23

```
Met Thr Val Tyr Ala Asn Pro Gly Thr Glu Gly Ser Ile Val Asn Tyr
1               5                   10                  15

Glu Lys Arg Tyr Glu Asn Tyr Ile Gly Gly Lys Trp Val Pro Pro Val
            20                  25                  30

Glu Gly Gln Tyr Leu Glu Asn Ile Ser Pro Val Thr Gly Glu Val Phe
        35                  40                  45

Cys Glu Val Ala Arg Gly Thr Ala Ala Asp Val Glu Leu Ala Leu Asp
    50                  55                  60

Ala Ala His Ala Ala Ala Asp Ala Trp Gly Lys Thr Ser Val Ala Glu
65                  70                  75                  80

Arg Ala Leu Ile Leu His Arg Ile Ala Asp Arg Met Glu Glu His Leu
                85                  90                  95

Glu Glu Ile Ala Val Ala Glu Thr Trp Glu Asn Gly Lys Ala Val Arg
            100                 105                 110

Glu Thr Leu Ala Ala Asp Ile Pro Leu Ala Ile Asp His Phe Arg Tyr
        115                 120                 125

Phe Ala Gly Ala Ile Arg Ala Gln Glu Asp Arg Ser Ser Gln Ile Asp
    130                 135                 140

His Asn Thr Val Ala Tyr His Phe Asn Glu Pro Ile Gly Val Val Gly
145                 150                 155                 160

Gln Ile Ile Pro Trp Asn Phe Pro Ile Leu Met Ala Thr Trp Lys Leu
                165                 170                 175

Ala Pro Ala Leu Ala Ala Gly Asn Ala Ile Val Met Lys Pro Ala Glu
            180                 185                 190

Gln Thr Pro Ala Ser Ile Leu Tyr Leu Ile Asn Ile Ile Gly Asp Leu
        195                 200                 205

Ile Pro Glu Gly Val Leu Asn Ile Val Asn Gly Leu Gly Gly Glu Ala
    210                 215                 220

Gly Ala Ala Leu Ser Gly Ser Asn Arg Ile Gly Lys Ile Ala Phe Thr
225                 230                 235                 240

Gly Ser Thr Glu Val Gly Lys Leu Ile Asn Arg Ala Ala Ser Asp Lys
                245                 250                 255

Ile Ile Pro Val Thr Leu Glu Leu Gly Gly Lys Ser Pro Ser Ile Phe
            260                 265                 270

Phe Ser Asp Val Leu Ser Gln Asp Asp Ala Phe Ala Glu Lys Ala Val
        275                 280                 285

Glu Gly Phe Ala Met Phe Ala Leu Asn Gln Gly Glu Val Cys Thr Cys
    290                 295                 300

Pro Ser Arg Ala Leu Val His Glu Ser Ile Ala Asp Glu Phe Leu Glu
305                 310                 315                 320

Leu Gly Val Lys Arg Val Gln Asn Ile Lys Leu Gly Asn Pro Leu Asp
                325                 330                 335

Thr Glu Thr Met Met Gly Ala Gln Ala Ser Gln Glu Gln Met Asp Lys
            340                 345                 350

Ile Ser Ser Tyr Leu Lys Ile Gly Pro Glu Glu Gly Ala Gln Thr Leu
        355                 360                 365
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Thr Gly Gly Lys Val Asn Lys Val Asp Gly Met Glu Asn Gly Tyr Tyr
    370                 375                 380

Ile Glu Pro Thr Val Phe Arg Gly Thr Asn Asp Met Arg Ile Phe Arg
385                 390                 395                 400

Glu Glu Ile Phe Gly Pro Val Leu Ser Val Ala Thr Phe Ser Asp Phe
                405                 410                 415

Asp Glu Ala Ile Arg Ile Ala Asn Asp Thr Asn Tyr Gly Leu Gly Ala
            420                 425                 430

Gly Val Trp Ser Arg Asp Gln Asn Thr Ile Tyr Arg Ala Gly Arg Ala
        435                 440                 445

Ile Gln Ala Gly Arg Val Trp Val Asn Gln Tyr His Asn Tyr Pro Ala
    450                 455                 460

His Ser Ala Phe Gly Gly Tyr Lys Glu Ser Gly Ile Gly Arg Glu Asn
465                 470                 475                 480

His Leu Met Met Leu Asn His Tyr Gln Gln Thr Lys Asn Leu Leu Val
                485                 490                 495

Ser Tyr Asp Pro Asn Pro Thr Gly Leu Phe
            500                 505


<210> SEQ ID NO 24

<400> SEQUENCE: 24

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<210> SEQ ID NO 25

<400> SEQUENCE: 25

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<210> SEQ ID NO 26

<400> SEQUENCE: 26

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<210> SEQ ID NO 27

<400> SEQUENCE: 27

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<210> SEQ ID NO 28

<400> SEQUENCE: 28

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<210> SEQ ID NO 29
<211> LENGTH: 671
<212> TYPE: PRT
<213> ORGANISM: Chloroflexus aurantiacus

<400> SEQUENCE: 29

Met Ser Ala Thr Thr Gly Ala Arg Ser Ala Ser Val Gly Trp Ala Glu
1               5                   10                  15

Ser Leu Ile Gly Leu His Leu Gly Lys Val Ala Leu Ile Thr Gly Gly
            20                  25                  30
```

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```
Ser Ala Gly Ile Gly Gly Gln Ile Gly Arg Leu Leu Ala Leu Ser Gly
        35                  40                  45

Ala Arg Val Met Leu Ala Ala Arg Asp Arg His Lys Leu Glu Gln Met
    50                  55                  60

Gln Ala Met Ile Gln Ser Glu Leu Ala Glu Val Gly Tyr Thr Asp Val
65                  70                  75                  80

Glu Asp Arg Val His Ile Ala Pro Gly Cys Asp Val Ser Ser Glu Ala
                85                  90                  95

Gln Leu Ala Asp Leu Val Glu Arg Thr Leu Ser Ala Phe Gly Thr Val
            100                 105                 110

Asp Tyr Leu Ile Asn Asn Ala Gly Ile Ala Gly Val Glu Glu Met Val
        115                 120                 125

Ile Asp Met Pro Val Glu Gly Trp Arg His Thr Leu Phe Ala Asn Leu
    130                 135                 140

Ile Ser Asn Tyr Ser Leu Met Arg Lys Leu Ala Pro Leu Met Lys Lys
145                 150                 155                 160

Gln Gly Ser Gly Tyr Ile Leu Asn Val Ser Ser Tyr Phe Gly Gly Glu
                165                 170                 175

Lys Asp Ala Ala Ile Pro Tyr Pro Asn Arg Ala Asp Tyr Ala Val Ser
            180                 185                 190

Lys Ala Gly Gln Arg Ala Met Ala Glu Val Phe Ala Arg Phe Leu Gly
        195                 200                 205

Pro Glu Ile Gln Ile Asn Ala Ile Ala Pro Gly Pro Val Glu Gly Asp
    210                 215                 220

Arg Leu Arg Gly Thr Gly Glu Arg Pro Gly Leu Phe Ala Arg Arg Ala
225                 230                 235                 240

Arg Leu Ile Leu Glu Asn Lys Arg Leu Asn Glu Leu His Ala Ala Leu
                245                 250                 255

Ile Ala Ala Ala Arg Thr Asp Glu Arg Ser Met His Glu Leu Val Glu
            260                 265                 270

Leu Leu Leu Pro Asn Asp Val Ala Ala Leu Glu Gln Asn Pro Ala Ala
        275                 280                 285

Pro Thr Ala Leu Arg Glu Leu Ala Arg Arg Phe Arg Ser Glu Gly Asp
    290                 295                 300

Pro Ala Ala Ser Ser Ser Ser Ala Leu Leu Asn Arg Ser Ile Ala Ala
305                 310                 315                 320

Lys Leu Leu Ala Arg Leu His Asn Gly Gly Tyr Val Leu Pro Ala Asp
                325                 330                 335

Ile Phe Ala Asn Leu Pro Asn Pro Pro Asp Pro Phe Phe Thr Arg Ala
            340                 345                 350

Gln Ile Asp Arg Glu Ala Arg Lys Val Arg Asp Gly Ile Met Gly Met
        355                 360                 365

Leu Tyr Leu Gln Arg Met Pro Thr Glu Phe Asp Val Ala Met Ala Thr
    370                 375                 380

Val Tyr Tyr Leu Ala Asp Arg Val Val Ser Gly Glu Thr Phe His Pro
385                 390                 395                 400

Ser Gly Gly Leu Arg Tyr Glu Arg Thr Pro Thr Gly Gly Glu Leu Phe
                405                 410                 415

Gly Leu Pro Ser Pro Glu Arg Leu Ala Glu Leu Val Gly Ser Thr Val
            420                 425                 430

Tyr Leu Ile Gly Glu His Leu Thr Glu His Leu Asn Leu Leu Ala Arg
```

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```
        435                 440                 445

Ala Tyr Leu Glu Arg Tyr Gly Ala Arg Gln Val Val Met Ile Val Glu
    450                 455                 460

Thr Glu Thr Gly Ala Glu Thr Met Arg Arg Leu Leu His Asp His Val
465                 470                 475                 480

Glu Ala Gly Arg Leu Met Thr Ile Val Ala Gly Asp Gln Ile Glu Ala
                485                 490                 495

Ala Ile Asp Gln Ala Ile Thr Arg Tyr Gly Arg Pro Gly Pro Val Val
            500                 505                 510

Cys Thr Pro Phe Arg Pro Leu Pro Thr Val Pro Leu Val Gly Arg Lys
        515                 520                 525

Asp Ser Asp Trp Ser Thr Val Leu Ser Glu Ala Glu Phe Ala Glu Leu
    530                 535                 540

Cys Glu His Gln Leu Thr His His Phe Arg Val Ala Arg Trp Ile Ala
545                 550                 555                 560

Leu Ser Asp Gly Ala Arg Leu Ala Leu Val Thr Pro Glu Thr Thr Ala
                565                 570                 575

Thr Ser Thr Thr Glu Gln Phe Ala Leu Ala Asn Phe Ile Lys Thr Thr
            580                 585                 590

Leu His Ala Phe Thr Ala Thr Ile Gly Val Glu Ser Glu Arg Thr Ala
        595                 600                 605

Gln Arg Ile Leu Ile Asn Gln Val Asp Leu Thr Arg Arg Ala Arg Ala
    610                 615                 620

Glu Glu Pro Arg Asp Pro His Glu Arg Gln Gln Glu Leu Glu Arg Phe
625                 630                 635                 640

Ile Glu Ala Val Leu Leu Val Thr Ala Pro Leu Pro Pro Glu Ala Asp
                645                 650                 655

Thr Arg Tyr Ala Gly Arg Ile His Arg Gly Arg Ala Ile Thr Val
            660                 665                 670


<210> SEQ ID NO 30
<211> LENGTH: 550
<212> TYPE: PRT
<213> ORGANISM: Chloroflexus aurantiacus

<400> SEQUENCE: 30

Met Ser Gly Thr Gly Arg Leu Ala Gly Lys Ile Ala Leu Ile Thr Gly
1               5                   10                  15

Gly Ala Gly Asn Ile Gly Ser Glu Leu Thr Arg Arg Phe Leu Ala Glu
            20                  25                  30

Gly Ala Thr Val Ile Ile Ser Gly Arg Asn Arg Ala Lys Leu Thr Ala
        35                  40                  45

Leu Ala Glu Arg Met Gln Ala Glu Ala Gly Val Pro Ala Lys Arg Ile
    50                  55                  60

Asp Leu Glu Val Met Asp Gly Ser Asp Pro Val Ala Val Arg Ala Gly
65                  70                  75                  80

Ile Glu Ala Ile Val Ala Arg His Gly Gln Ile Asp Ile Leu Val Asn
                85                  90                  95

Asn Ala Gly Ser Ala Gly Ala Gln Arg Arg Leu Ala Glu Ile Pro Leu
            100                 105                 110

Thr Glu Ala Glu Leu Gly Pro Gly Ala Glu Glu Thr Leu His Ala Ser
        115                 120                 125
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```
Ile Ala Asn Leu Leu Gly Met Gly Trp His Leu Met Arg Ile Ala Ala
    130                 135                 140

Pro His Met Pro Val Gly Ser Ala Val Ile Asn Val Ser Thr Ile Phe
145                 150                 155                 160

Ser Arg Ala Glu Tyr Tyr Gly Arg Ile Pro Tyr Val Thr Pro Lys Ala
                165                 170                 175

Ala Leu Asn Ala Leu Ser Gln Leu Ala Ala Arg Glu Leu Gly Ala Arg
            180                 185                 190

Gly Ile Arg Val Asn Thr Ile Phe Pro Gly Pro Ile Glu Ser Asp Arg
        195                 200                 205

Ile Arg Thr Val Phe Gln Arg Met Asp Gln Leu Lys Gly Arg Pro Glu
    210                 215                 220

Gly Asp Thr Ala His His Phe Leu Asn Thr Met Arg Leu Cys Arg Ala
225                 230                 235                 240

Asn Asp Gln Gly Ala Leu Glu Arg Arg Phe Pro Ser Val Gly Asp Val
                245                 250                 255

Ala Asp Ala Ala Val Phe Leu Ala Ser Ala Glu Ser Ala Ala Leu Ser
            260                 265                 270

Gly Glu Thr Ile Glu Val Thr His Gly Met Glu Leu Pro Ala Cys Ser
        275                 280                 285

Glu Thr Ser Leu Leu Ala Arg Thr Asp Leu Arg Thr Ile Asp Ala Ser
    290                 295                 300

Gly Arg Thr Thr Leu Ile Cys Ala Gly Asp Gln Ile Glu Glu Val Met
305                 310                 315                 320

Ala Leu Thr Gly Met Leu Arg Thr Cys Gly Ser Glu Val Ile Ile Gly
                325                 330                 335

Phe Arg Ser Ala Ala Ala Leu Ala Gln Phe Glu Gln Ala Val Asn Glu
            340                 345                 350

Ser Arg Arg Leu Ala Gly Ala Asp Phe Thr Pro Pro Ile Ala Leu Pro
        355                 360                 365

Leu Asp Pro Arg Asp Pro Ala Thr Ile Asp Ala Val Phe Asp Trp Gly
    370                 375                 380

Ala Gly Glu Asn Thr Gly Gly Ile His Ala Ala Val Ile Leu Pro Ala
385                 390                 395                 400

Thr Ser His Glu Pro Ala Pro Cys Val Ile Glu Val Asp Asp Glu Arg
                405                 410                 415

Val Leu Asn Phe Leu Ala Asp Glu Ile Thr Gly Thr Ile Val Ile Ala
            420                 425                 430

Ser Arg Leu Ala Arg Tyr Trp Gln Ser Gln Arg Leu Thr Pro Gly Ala
        435                 440                 445

Arg Ala Arg Gly Pro Arg Val Ile Phe Leu Ser Asn Gly Ala Asp Gln
    450                 455                 460

Asn Gly Asn Val Tyr Gly Arg Ile Gln Ser Ala Ala Ile Gly Gln Leu
465                 470                 475                 480

Ile Arg Val Trp Arg His Glu Ala Glu Leu Asp Tyr Gln Arg Ala Ser
                485                 490                 495

Ala Ala Gly Asp His Val Leu Pro Pro Val Trp Ala Asn Gln Ile Val
            500                 505                 510

Arg Phe Ala Asn Arg Ser Leu Glu Gly Leu Glu Phe Ala Cys Ala Trp
        515                 520                 525

Thr Ala Gln Leu Leu His Ser Gln Arg His Ile Asn Glu Ile Thr Leu
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530 535 540

Asn Ile Pro Ala Asn Ile
545 550

<210> SEQ ID NO 31

<400> SEQUENCE: 31

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<210> SEQ ID NO 32

<400> SEQUENCE: 32

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<210> SEQ ID NO 33

<400> SEQUENCE: 33

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<210> SEQ ID NO 34

<400> SEQUENCE: 34

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<210> SEQ ID NO 35

<400> SEQUENCE: 35

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<210> SEQ ID NO 36
<211> LENGTH: 97
<212> TYPE: PRT
<213> ORGANISM: Escherichia coli

<400> SEQUENCE: 36

Met Asn Ile Arg Pro Leu His Asp Arg Val Ile Val Lys Arg Lys Glu
1 5 10 15

Val Glu Thr Lys Ser Ala Gly Gly Ile Val Leu Thr Gly Ser Ala Ala
20 25 30

Ala Lys Ser Thr Arg Gly Glu Val Leu Ala Val Gly Asn Gly Arg Ile
35 40 45

Leu Glu Asn Gly Glu Val Lys Pro Leu Asp Val Lys Val Gly Asp Ile
50 55 60

Val Ile Phe Asn Asp Gly Tyr Gly Val Lys Ser Glu Lys Ile Asp Asn
65 70 75 80

Glu Glu Val Leu Ile Met Ser Glu Ser Asp Ile Leu Ala Ile Val Glu
85 90 95

Ala

<210> SEQ ID NO 37
<211> LENGTH: 542
<212> TYPE: PRT
<213> ORGANISM: Escherichia coli

<400> SEQUENCE: 37

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```
Met Ala Ala Lys Asp Val Lys Phe Gly Asn Asp Ala Arg Val Lys Met
1               5                   10                  15

Leu Arg Gly Val Asn Val Leu Ala Asp Ala Val Lys Val Thr Leu Gly
            20                  25                  30

Pro Lys Gly Arg Asn Val Val Leu Asp Lys Ser Phe Gly Ala Pro Thr
        35                  40                  45

Ile Thr Lys Asp Gly Val Ser Val Ala Arg Glu Ile Glu Leu Glu Asp
    50                  55                  60

Lys Phe Glu Asn Met Gly Ala Gln Met Val Lys Glu Val Ala Ser Lys
65                  70                  75                  80

Ala Asn Asp Ala Ala Gly Asp Gly Thr Thr Thr Ala Thr Val Leu Ala
                85                  90                  95

Gln Ala Ile Ile Thr Glu Gly Leu Lys Ala Val Ala Ala Gly Met Asn
            100                 105                 110

Pro Met Asp Leu Lys Arg Gly Ile Asp Lys Ala Val Thr Ala Ala Val
        115                 120                 125

Glu Glu Leu Lys Ala Leu Ser Val Pro Cys Ser Asp Ser Lys Ala Ile
    130                 135                 140

Ala Gln Val Gly Thr Ile Ser Ala Asn Ser Asp Glu Thr Val Gly Lys
145                 150                 155                 160

Leu Ile Ala Glu Ala Met Asp Lys Val Gly Lys Glu Gly Val Ile Thr
                165                 170                 175

Val Glu Asp Gly Thr Gly Leu Gln Asp Glu Leu Asp Val Val Glu Gly
            180                 185                 190

Met Gln Phe Asp Arg Gly Tyr Leu Ser Pro Tyr Phe Ile Asn Lys Pro
        195                 200                 205

Glu Thr Gly Ala Val Glu Leu Glu Ser Pro Phe Ile Leu Leu Ala Asp
    210                 215                 220

Lys Lys Ile Ser Asn Ile Arg Glu Met Leu Pro Val Leu Glu Ala Val
225                 230                 235                 240

Ala Lys Ala Gly Lys Pro Leu Leu Ile Ile Ala Glu Asp Val Glu Gly
                245                 250                 255

Glu Ala Leu Ala Thr Leu Val Val Asn Thr Met Arg Gly Ile Val Lys
            260                 265                 270

Val Ala Ala Val Lys Ala Pro Gly Phe Gly Asp Arg Arg Lys Ala Met
        275                 280                 285

Leu Gln Asp Ile Ala Thr Leu Thr Gly Gly Thr Val Ile Ser Glu Glu
    290                 295                 300

Ile Gly Met Glu Leu Glu Lys Ala Thr Leu Glu Asp Leu Gly Gln Ala
305                 310                 315                 320

Lys Arg Val Val Ile Asn Lys Asp Thr Thr Thr Ile Ile Asp Gly Val
                325                 330                 335

Gly Glu Glu Ala Ala Ile Gln Gly Arg Val Ala Gln Ile Arg Gln Gln
            340                 345                 350

Ile Glu Glu Ala Thr Ser Asp Tyr Asp Arg Glu Lys Leu Gln Glu Arg
        355                 360                 365

Val Ala Lys Leu Ala Gly Gly Val Ala Val Ile Lys Val Gly Ala Ala
    370                 375                 380

Thr Glu Val Glu Met Lys Glu Lys Lys Ala Arg Val Glu Asp Ala Leu
385                 390                 395                 400

His Ala Thr Arg Ala Ala Val Glu Glu Gly Val Val Ala Gly Gly Gly
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                405                 410                 415

Val Ala Leu Ile Arg Val Ala Ser Lys Leu Ala Asp Leu Arg Gly Gln
            420                 425                 430

Asn Glu Asp Gln Asn Val Gly Ile Lys Val Ala Leu Arg Ala Met Glu
        435                 440                 445

Ala Pro Leu Arg Gln Ile Val Leu Asn Cys Gly Glu Glu Pro Ser Val
    450                 455                 460

Val Ala Asn Thr Val Lys Gly Gly Asp Gly Asn Tyr Gly Tyr Asn Ala
465                 470                 475                 480

Ala Thr Glu Glu Tyr Gly Asn Met Ile Asp Met Gly Ile Leu Asp Pro
                485                 490                 495

Thr Lys Val Thr Arg Ser Ala Leu Gln Tyr Ala Ala Ser Val Ala Gly
            500                 505                 510

Leu Met Ile Thr Thr Glu Cys Met Val Thr Asp Leu Pro Lys Asn Asp
        515                 520                 525

Ala Ala Asp Leu Gly Ala Ala Gly Gly Met Gly Gly Met Met
    530                 535                 540


<210> SEQ ID NO 38
<211> LENGTH: 96
<212> TYPE: PRT
<213> ORGANISM: Methylococcus capsulatus

<400> SEQUENCE: 38

Val Lys Ile Arg Pro Leu His Asp Arg Val Ile Ile Lys Arg Leu Glu
1               5                   10                  15

Glu Glu Arg Thr Ser Ala Gly Gly Ile Val Ile Pro Asp Ser Ala Ala
            20                  25                  30

Glu Lys Pro Met Arg Gly Glu Ile Leu Ala Val Gly Asn Gly Lys Val
        35                  40                  45

Leu Asp Asn Gly Glu Val Arg Ala Leu Gln Val Lys Val Gly Asp Lys
    50                  55                  60

Val Leu Phe Gly Lys Tyr Ala Gly Thr Glu Val Lys Val Asp Gly Glu
65                  70                  75                  80

Asp Val Val Val Met Arg Glu Asp Asp Ile Leu Ala Val Leu Glu Ser
                85                  90                  95


<210> SEQ ID NO 39
<211> LENGTH: 559
<212> TYPE: PRT
<213> ORGANISM: Methylococcus capsulatus

<400> SEQUENCE: 39

Met Ala Lys Glu Val Val Tyr Arg Gly Ser Ala Arg Gln Arg Met Met
1               5                   10                  15

Gln Gly Ile Glu Ile Leu Ala Arg Ala Ala Ile Pro Thr Leu Gly Ala
            20                  25                  30

Thr Gly Pro Ser Val Met Ile Gln His Arg Ala Asp Gly Leu Pro Pro
        35                  40                  45

Ile Ser Thr Arg Asp Gly Val Thr Val Ala Asn Ser Ile Val Leu Lys
    50                  55                  60

Asp Arg Val Ala Asn Leu Gly Ala Arg Leu Leu Arg Asp Val Ala Gly
65                  70                  75                  80

Thr Met Ser Arg Glu Ala Gly Asp Gly Thr Thr Thr Ala Ile Val Leu
```

-continued

```
                85                  90                  95

Ala Arg His Ile Ala Arg Glu Met Phe Lys Ser Leu Ala Val Gly Ala
            100                 105                 110

Asp Pro Ile Ala Leu Lys Arg Gly Ile Asp Arg Ala Val Ala Arg Val
        115                 120                 125

Ser Glu Asp Ile Gly Ala Arg Ala Trp Arg Gly Asp Lys Glu Ser Val
    130                 135                 140

Ile Leu Gly Val Ala Ala Val Ala Thr Lys Gly Glu Pro Gly Val Gly
145                 150                 155                 160

Arg Leu Leu Leu Glu Ala Leu Asp Ala Val Gly Val His Gly Ala Val
                165                 170                 175

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Pro Gly Thr Glu Ala Ser Arg Ala Arg Leu Glu Gly Leu Arg Leu Glu
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<210> SEQ ID NO 45
<211> LENGTH: 6364
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 45

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<210> SEQ ID NO 46
<211> LENGTH: 7032
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 46

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1. A synthetic microorganism comprising or consisting of at least one exogenous polynucleotide that encodes one or more polypeptides comprising or consisting of an ethane monooxygenase, an ethylotrophy pathway, and/or a malonyl-CoA-product pathway.

2. The synthetic microorganism according to claim **1**, wherein the synthetic microorganism further comprises a microorganism with the at least one exogenous polynucleotide inserted or integrated into the genome of the microorganism.

3. The synthetic microorganism of claim **1**, wherein the ethane monooxygenase comprises or consists of an ethane monooxygenase having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID NOs: 1-16.

4. The synthetic microorganism of claim **1**, wherein the ethylotrophy pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID NOs 22-23.

5. The synthetic microorganism of claim **1**, wherein the malonyl-CoA-product pathway comprises or consists of one or more enzymes having an amino acid sequence that is more than about 70%, more than about 75%, more than about 80%, more than about 85%, more than about 90%, or more than about 95% identical or identical to any of SEQ ID NOs: 29-30.

6. The synthetic microorganism of claim **1**, wherein the synthetic microorganism generates a product comprising or consisting of 3-hydroxypropionic acid.

7. The synthetic microorganism of claim **1**, wherein the synthetic microorganism generates a product comprising or consisting of a polymer of 3-hydroxypropionic acid.

8. The synthetic microorganism of claim **1**, wherein the synthetic microorganism generates one or more products comprising or consisting of fatty acids or fatty acid derivatives.

9. The synthetic microorganism of claim **1**, wherein the synthetic microorganism generates a product comprising or consisting of polymer(s) of 3-hydroxy-fatty acids.

10. The synthetic microorganism of claim **1**, wherein the synthetic microorganism generates a product comprising or consisting of malonic acid.

11. The synthetic microorganism of claim **1**, further comprising one or more exogenous polynucleotides which overexpress acetyl-CoA carboxylase.

12. The synthetic microorganism of claim **1**, wherein the microorganism is derived from *Escherichia coli*.

13. A method of culturing the synthetic microorganism of claim **1** for a suitable period of time and under conditions sufficient for conversion of ethane into the malonyl-CoA-derived product.

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